

WJEC GCE Biology

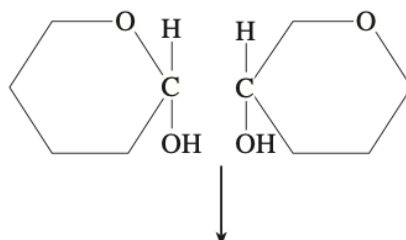
AS UNIT 1 PAST PAPER BY TOPIC (LEGACY)

IGC HK Exam



1.1

6. (a) (i) Complete the diagram to show the reaction between two molecules of glucose and the products formed. [3]



- (ii) Name the organic molecule formed. [1]

.....

- (iii) Name the inorganic molecule released during the reaction. [1]

.....

- (iv) Name the type of reaction. [1]

.....

6. (a) (i) Showing, 1 O and 2 H s removed.

Elimination of water, stated.

Molecules joined by oxygen bridge. [3]

- (ii) Maltose (not: disaccharide). [1]

- (iii) Water. [1]

- (iv) Condensation. [1]

- (b) One species of bacterium *Streptococcus mutans* produces the enzyme glucosyl transferase that catalyses the polymerisation of glucose molecules. Polymers of glucose form part of the plaque on teeth which leads to dental disease.

(i) What is polymerisation? [1]

- (b) (i) Joining together sub units / monomers /repeating units/ residues
(to make a larger molecule)
(not: joining molecules into a chain/ specific example)

[1]

1. Below is a list of chemicals which may be found in cells.

A	magnesium	F	calcium
B	maltose	G	phosphate
C	amino acid	H	water
D	cellulose	I	sucrose
E	iron		

Each of the following statements applies to one or more of these chemicals.
After each statement, write in the box the letter or letters which correspond(s) to the chemical(s) being described. Each letter may be used once, more than once, or not at all.

- | | |
|--|--------------------------|
| (a) Found in bones and teeth. | ☐ |
| (b) A non-reducing sugar | ☐ |
| (c) Found in chlorophyll in leaves. | ☐ |
| (d) A polysaccharide found in plant cell walls. | ☐ |
| (e) Found attached to lipids in the plasma membrane. | ☐ |
| (f) A large quantity of energy is needed to raise its temperature. | ☐ |

(Total 6 marks)

1. (a) F / calcium or G / phosphate
(b) I / sucrose
(c) A / magnesium
(d) D / cellulose
(e) G / phosphate
(f) H / water
(mark first answer)

[1 mark each]

[Total 6 marks]

2. (a) The level of protein structure is decided by the folding of the molecule and the types of bonds present.

Tick (✓) the boxes in the table to show all the bonds present at each level of protein structure. [3]

Level of protein structure	Types of bonds			
	peptide	hydrogen	disulphide	ionic
Primary				
Secondary				
Tertiary				

- (b) (i) How does the quaternary structure of a protein differ from the tertiary structure? [1]

- (ii) Give an example of a quaternary protein. [1]

(Total 5 marks)

2. (a)

Level of protein structure	Types of bonds			
	peptide	hydrogen	disulphide	ionic
Primary	/			
Secondary	/	/		
Tertiary	/	/	/	/

[1 mark per row across]

[3]

- (b) (i) More than one polypeptide chain present /

two or more polypeptide chains present.
(not: ref. to protein/tertiary)

[1]

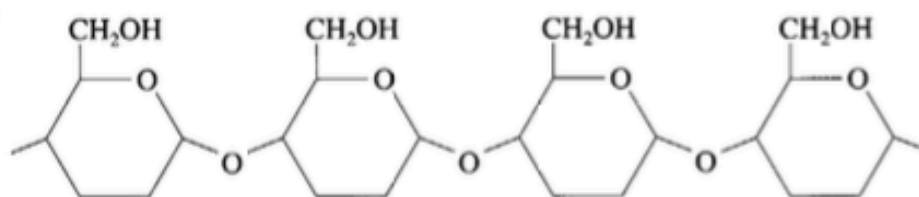
- (ii) haemoglobin / collagen / insulin/antibodies

[1]

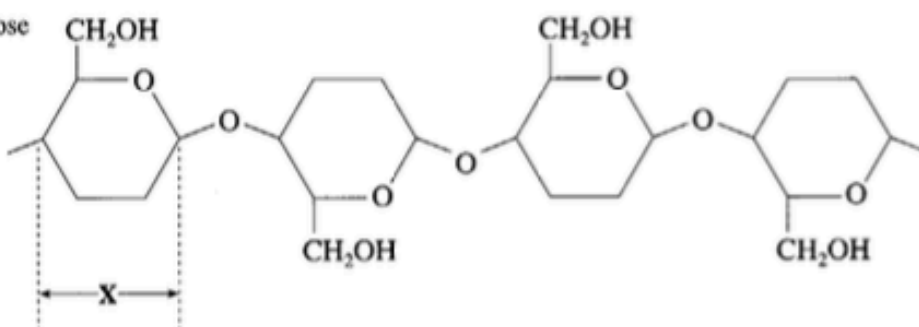
[Total 5 marks]

3. The diagrams show part of a molecule of starch (A) and part of a molecule of cellulose (B). The hexagonal shapes represent hexose sugars.

A. starch



B. cellulose



- (a) Name monosaccharide X and its form. [1]

- (b) Name the bond formed between two hexose sugars. [1]

- (c) State two structural differences between starch and cellulose. [2]

- (d) Starch is a compact storage polysaccharide. Cellulose has a structural role in plant cell walls. Describe how cellulose units are arranged in a complete molecule and how this arrangement gives cellulose a high tensile strength. [2]

(Total 6 marks)

3. (a) β glucose [1]
- (b) glycosidic [1]
- (c) starch has α glucose molecules cellulose has β glucose;
 starch two polysaccharides (or named), cellulose one;
 starch has (two polysaccharides) one of which is branched whereas
 cellulose is unbranched;
 starch has 1-4 and 1-6 linkages, cellulose 1-4 only;
 amylose / starch coiled, cellulose (cross linked) in straight chains;
 starch consists of single chains cellulose has many parallel chains
 linked with hydrogen bonds.
 Starch, glucose all same way 'up', rotated (90°) in cellulose.
 (must be a comparison) [Any 2] [2]

1

3. (d) long chains cross linked by hydrogen bonds;
 adjacent glucose molecules are rotated by 180° ;
 hydrogen bonds form between hydroxyl groups of adjacent parallel chains;
 microfibrils. [2]
 (not: fibres/fibrils) [Any 2]

[Total 6 marks]

- Or** (b) Explain how the properties of water are important to plants and animals. Use examples where appropriate. [10]
8. (b) A water is a polar molecule / dipolar / allows chemical reactions to take place in solution; [1]
- B explanation of dipolar H +ve, O –ve (allows diagram); [1]
- C explanation of hydrogen bonding; [1]
- D water is an universal solvent / dissolves polar and ionic substances or examples (**not:** many substances dissolve in water); [1]
- E since chemicals dissolve in water it acts as a transport medium in blood or phloem / xylem; [1]
- F water molecules show cohesion qual. / tall columns of water can be drawn up xylem vessels in tall trees / reference to cohesion tension theory and transpiration/ surface tension importance to living organisms e.g. walk on surface [1]
- G gases such as O₂ and CO₂ dissolve in water, available for respiration / photosynthesis; [1]
- H high latent heat of vaporisation; [1]
- I explanation of importance, cooling body when sweating;
- J water has a high specific heat / large amount of heat energy is needed to raise the temperature of water / high thermal capacity; [1]
- K explanation – heats up slowly and cools down slowly; [1]
- L importance of this ability to maintain constant temperature / this prevents large fluctuations in the temperature of water / important in keeping the temperature of aquatic habitats stable / that organisms do not have to endure extremes of temperature; [1]
- M water is transparent allowing light to pass through enabling aquatic plants to photosynthesise effectively; [1]
- O Water in its solid form (ice) is less dense than water and so floats on the surface; [1]
- P Ice forms an insulating layer and allows organisms to survive beneath it / preventing further heat loss / hibernation in ponds [1]
- Q Water is a reactant in photosynthesis / hydrolysis

[Total 10 marks]

WJEC 2010

-
- Chemical structure of the repeating unit of chondroitin-6-sulfate (CS-6S) disaccharide. The structure shows a disaccharide unit consisting of N-acetylmannosamine (NAM) and N-acetylglucosamine (NAG) linked by a β -1,3-glycosidic bond. The NAM unit has a methyl group at C2 and a carboxylic acid group at C6. The NAG unit has a methyl group at C2 and a carboxylic acid group at C6. The structure is shown in a chair conformation with various substituents labeled.

-
-
-
-

- (c) Catalase is an enzyme which breaks down hydrogen peroxide into water and oxygen. Hydrogen peroxide is a highly toxic waste product of metabolism. An investigation was carried out to determine the relative amounts of catalase in samples of potato, liver and apple. The samples were ground to a pulp and added to hydrogen peroxide in measuring cylinders. The table shows the height of the resulting bubbles in the cylinders.

<i>Sample</i>	<i>Height of bubbles / cm³</i>
Potato	4
Liver	9
Apple	1

- (i) State **two** variables that should be controlled during this investigation. [2]

.....

.....

- (ii) From the table, liver contains the most catalase. Suggest an explanation for this result. [2]

.....

.....

.....

(Total 14 marks)

6. (a) (i) nitrogen containing part; [1]
- (ii) arrow pointing to glycosidic bond; [1]
- (iii) hydrolysis; [1]
- (iv) hydroxyl groups point outwards;
link with neighbouring chains;
via hydrogen bonding;
to form microfibrils;
strong structure because of large number of hydrogen bonds;
chains associate in groups / fibres formed;
beta glucose units. [3 max]
ref. alternating rotation
- (b) (i) tertiary; [1]
- (ii) links between different parts of polypeptide chains;
produces a specific shape for the molecule / lysozyme;
reference to active site;
complementary to substrate;
allows enzyme – substrate complexes to form; [3 max]
- (c) (i) mass/volume of tissue/sample; (not: amount/size)
concentration of hydrogen peroxide;
same time intervals between measurements;
equal volumes of hydrogen peroxide used;
pH;
temperature. [2 max]
- (ii) most metabolically active;
produces most hydrogen peroxide;
needs to be broken down because of toxicity; [2 max]

[Total 14 marks]

(b) Describe the structure and functions of lipids in plants and animals.

7. (b)
- A. Energy storage; [1]
 - B. Release more energy per unit mass than carbohydrates; [1]
 - C. Makes seeds lighter / smaller for dispersal / energy store for hibernation; [1]
 - D. Phospholipid component of cell membrane; [1]
 - E. Controls entry of molecules into cell; [1]
 - F. Insulation; [1]
 - G. Protection of (delicate) organs or e.g.; [1]
 - H. Buoyancy for aquatic animals or e.g.; [1]
 - I. consist of the elements carbon hydrogen and oxygen; [1]
 - J. Glycerol plus three fatty acids; [1]
 - K. Joined by condensation reactions; [1]
 - L. Via ester bonds; [1]
 - M. saturated and unsaturated fatty acids; [1]
 - N. phosphate group present in phospholipids [1]
 - O. water proofing+ wax/oils;hormones+ steroids;
myelin sheath + insulation; AVP

Any 5 structure i.e. points B, D, E, I to N and 5 others

[Total 10 marks]

6. (a) Name the cell organelle which is the site of energy release in a mammal. [1]

(b) The table shows figures from the respiration of carbohydrate and lipid in a mammal.

<i>Energy source</i>	<i>Energy released / kJg⁻¹ food</i>	<i>Metabolic water produced / g g⁻¹ food</i>	<i>Oxygen consumed / dm³ g⁻¹ food</i>
Carbohydrate	17.2	0.56	0.83
Lipid	38.9	1.07	2.02

- (i) With reference to the figures in the table, state **one advantage** and **one disadvantage** of storing lipid rather than carbohydrate. [2]

advantage

disadvantage

- (ii) Suggest **two other** reasons why mammals store lipids. [2]

1.

2.

(c) The potato contains stored starch and protein.

- (i) State **two** uses for the protein in the growing potato plant. [2]

1.

2.

- (ii) The starch and protein must be hydrolysed before being used by the potato plant. Explain what is meant by the term *hydrolysed*. [2]

.....

.....

.....

- (iii) Name the products resulting from the **complete** hydrolysis of: [2]

Starch

Protein

- | | | | |
|---|-----|--|---------|
| 6 | (a) | Mitochondrion. | 1 |
| | (b) | (i) Advantage:
*Higher energy yield <u>per unit mass</u> /higher yield <u>per g</u> .
Disadvantage:
More oxygen required for respiration. | 2 |
| | | (ii) Heat/thermal/electrical insulation; (not: insulation unequal.)
*Better source of metabolic water;
Buoyancy;
Protection against knocks (not: protection unequal.) | (2 max) |
| | | (points marked * are interchangeable and could be credited in either (i) or (ii)
but not credited in both) | |
| | (c) | (i) New tissue manufacture/growth qualified/repair;
Enzyme manufacture. | 2 |
| | | (ii) Breaking a bond; (not: molecule broken down)
Insertion of a molecule of water/chemical addition of water
(not: adding water) | 2 |
| | | (iii) Glucose; (not: beta glucose)
Amino acids. | 2 |

11 MARKS

WJEC 2011

1. Inorganic ions are needed by living organisms. Complete the table below to give a function for **each** of the four ions. [4]

<i>Ion</i>	<i>Function</i>
Magnesium	
Iron	
Phosphate	
Calcium	

(Total 4 marks)

- Q.1** needed for / found in / used in / component of chlorophyll;
 (allow: middle lamella / enzyme co-factors)
 component haemoglobin; enzyme Co factors
 component nucleic acids/DNA/RNA/ATP/ (plasma) membrane/ phospholipids
 hardens bone / nucleotide;
 hardens/deposited in bones/teeth/ossification/synaptic transmission
 enzyme co-factors/middle lamella (not: strengthen bones)

2. Complete the following passage by inserting the correct terms in the spaces provided. [6]

Cellulose is a fibrous molecule. It is a carbohydrate and is the main component of the cell walls of plants. Cellulose consists of chains of glucose molecules which are joined together by 1-4 bonds. Each adjacent glucose molecule is rotated by 180° resulting in a chain. Chains are held together by hydrogen bonds forming groups of chains known as microfibrils.

(Total 6 marks)

Q.2 cell wall;
 beta/β;
 glycosidic;
 180;
 hydrogen;
 microfibrils; (not: microfibres)

5. (a) When a triglyceride molecule is broken down name:

(i) the products formed; [2]

.....

(ii) the type of bond broken and describe the process. [3]

.....

.....

.....

(b) Describe **two** functions of lipids in plants. [2]

.....

.....

(Total 7 marks)

Q.5 (a) (i) glycerol;
 (3) fatty acids;

(ii) ester;
 hydrolysis;
 chemical insertion of water/water added to bond

(iii) energy storage / respiratory substrate/source of energy
 waxy cuticle/leaf waterproofing;
 membrane structure;

(Total 10 marks)

(a) Describe the properties of water and its functions in living organisms. [10]

|

- Q.8** (a)
- A. polar molecule/dipole;
 - B. uneven distribution of charges/ H^+O^- ;
 - C. forms hydrogen bonds (between molecules);
 - D. dissolves ionic/polar substances; (not: ref glucose/solvent u
 - E. used for transportation of molecules;
 - F. high latent heat of evaporation / vapourisation / large amount of energy needed to make water evaporate
 - G. has a role in cooling body;
 - H. high specific heat / large amount of heat energy needed to r
 - I. helps maintain stable/constant environmental temperature; (Internal)
 - J. (transparent) to allow light through for photosynthesis
- K. molecules of water stick together / (high) cohesion (not: adhesion)
- L. allows movement through xylem/ adhesion (not: ref capillarity)
- M. surface tension allows insects to walk on water or example
- N. reactant in photosynthesis/hydrolysis or description (not: used in Photosynthesis)
- O. ice less dense than water so floats on surface therefore insulation of pond life when ice forms / correct ref to buoyancy qual
- P. chemical reactions occur in solution

(Points F and H only in correct context of explanation)

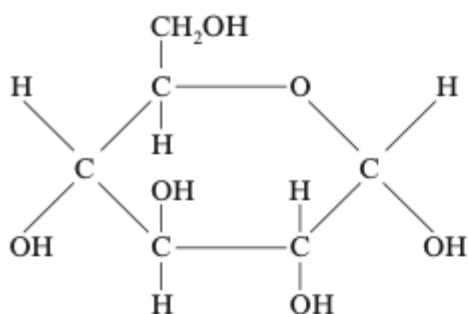
[Total 10 marks]

5. (a) (i) amino acid;
triglyceride; (not: lipid/triglycerol)
- (ii) nitrogen/sulphur; (not: chemical symbols)
- (b) condensation;
peptide;
- (c) (i) add Biuret to test solution; (not: if ref. to boiling)
blue changing to mauve/purple colour is positive result;
- (ii) little colour change/mauve colour may be masked;

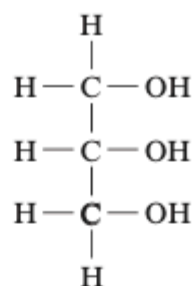
(Total

WJEC 2012

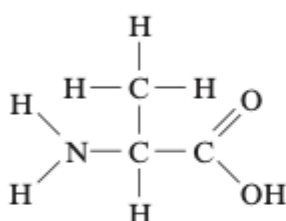
2. The following diagrams represent the structure of four biologically important compounds.



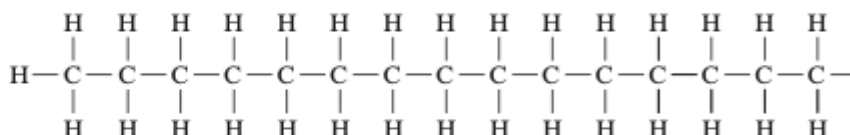
A



B



C



D

- (a) A chemical element found in a molecule of compound **C** is not found in molecules of the other three compounds. **Name** this element. [1]

.....

- (b) A reducing sugar in solution can be detected in the laboratory.

- (i) Describe the biochemical test you would use to show that the solution contained a reducing sugar. [2]

.....

- (ii) Which of the compounds **A** to **D** will give a positive result with this biochemical test? [1]

.....

- (c) Which of the compounds **A** to **D** has molecules that will join together by peptide bonds? [1]

.....

(d) (i) Which of the compounds **A** to **D** is a fatty acid? [1]

.....

(ii) State how the structure of a saturated fatty acid differs from the structure of an unsaturated fatty acid. [2]

.....

.....

.....

.....

(Total 8 marks)

2. (a) nitrogen (not: N)

(b) (i) heat/boil with {Benedict's/ Fehlings A + B} solution; NOT boil
with acid

colour change from blue to {green/yellow/orange/brick red/
brown}

(ii) A

(c) C

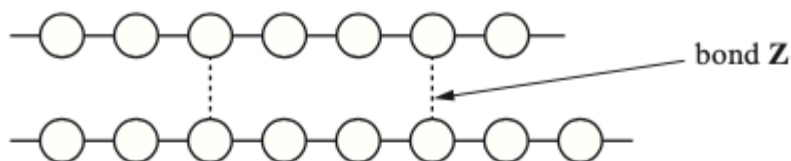
(d) (i) D

(ii) saturated- no double bonds/ all carbon atoms have/attached
to two hydrogens ;

Fewer hydrogen atoms (or converse)

Must have comparison for each

3. The diagram represents part of a cellulose molecule.



- (a) (i) Name bond **Z** as shown on the diagram. [1]

.....

- (ii) Explain the importance of bond **Z** in the role of cellulose in plant cell walls. [2]

.....
.....
.....

- (b) (i) Name the chemical reaction by which monomers join together to form cellulose. [1]

.....

- (ii) Chitin has many chemical and structural similarities to cellulose.
In chitin what additional compound replaces one of the -OH groups in each of its monosaccharides? [1]

.....

- (iii) State the structural role of chitin in insects and describe its properties that make it suitable for this role. [1]

.....
.....



(c) Other polysaccharides have a storage function.
Name a storage polysaccharide found in:

(i) animal cells;

[1]

.....

(ii) plant cells.

[1]

.....

(Total 8 marks)

3. (a) (i) hydrogen/H

(ii) {Holds/binds} {cellulose/glucose} {chains/molecules}
together/ forms microfibrils;
strengthens (the wall)/ (cellulose fibres are) strong/ rigid/
gives structural stability/
can resist turgor/ osmotic pressure/ prevents plant cells
bursting.

(b) (i) condensation/ polymerisation

(ii) (Has) amino acid (added)/glucosamine
(to form a mucopolysaccharide)/ amine/ NH_2

(iii) (exo)skeleton – strong/waterproof/ light/ rigidity/ tough
NOT exoskeleton gives protection

(c) (i) glycogen

(ii) starch (accept amylose/ amylopectin)

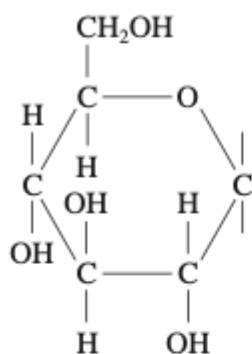
(b) Describe the structure and role of proteins in living organisms.

8. (b)
- A primary structure, {sequence/ order} of amino acids in its polypeptide chain
 - B linked by peptide bonds
 - C secondary structure consists of – α helix/ pleated sheet
 - D hydrogen bonds
 - E tertiary structure described - 3D folding/ irregular/ further folding
 - F as shown by globular proteins
 - G disulphide bridges/ ionic/ hydrogen/ hydrophobic (any two)
 - H Quaternary structure described- combination of two or more polypeptide chains
 - I Some proteins have non-protein groups/ prosthetic groups
 - J enzymes – function or description of
 - K antibodies/hormones/ plasmaproteins with function
 - L haemoglobin – (carries/ transport) of oxygen
 - M fibrous proteins + example connective tissue/ keratin/ collagen
 - N Function of fibrous protein - strength
 - O carriers in active transport/ facilitated diffusion
/fibrinogen in blood clotting /histones/ ribosomal proteins

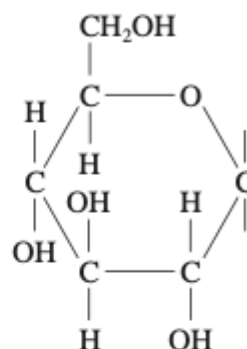
(Any 10 out of 15 points)

2. Carbohydrate molecules contain the chemical elements carbon, hydrogen and oxygen only.

- (a) The diagrams below show structural formulae of two isomers of glucose. Complete the drawings to distinguish between the α and β isomers. [1]



α glucose



β glucose

- (b) (i) Starch and cellulose are both polymers of glucose, but they are formed from different isomers. State the isomer which is found in: [1]

Cellulose Starch

- (ii) Explain how the structures of starch and cellulose are related to their functions as storage and structural molecules respectively. [4]

.....

.....

.....

.....

.....

.....

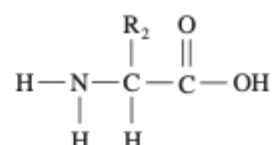
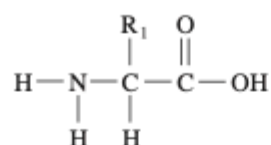
.....

.....

(Total 6 marks)

2.	(a)	(i)	α glucose OH on C1 down, H up + β glucose OH on C1 up, H down; Allow HO (both for 1 mark).	1
	(b)	(i)	Cellulose –Beta Starch – alpha; (both for 1 mark). Allow symbols.	1
		(ii)	Starch: any 2 correct reference to amylose and/or amylopectin; glycosidic bonds (α 1-4); molecules coil/branch (in amylopectin); NOT compact NOT: amylopectin – coiled or amylase branched easy to add/remove { <u>glucose / maltose</u> } units;	2
			Cellulose: any 2 alternate units rotate / head up, head down / 180° rotation; straight chain only / no branches; NOT parallel hydrogen bonds between / reference to cross linking; gives strength or stability / forming microfibrils;	2
Question total				6

5. The diagram below shows two molecules which are sub-units of proteins.



(a) (i) Complete the diagram above to show how a reaction takes place to join the two molecules. [3]

(ii) Name the type of reaction involved. [1]

.....

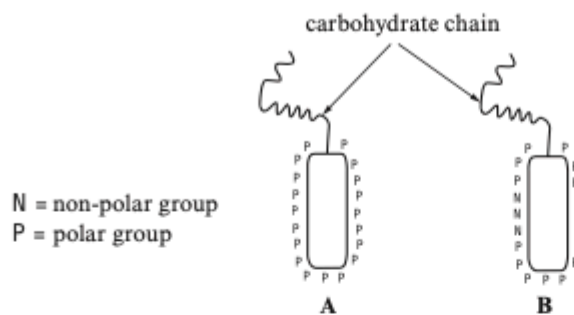
(iii) Name the type of bond formed. [1]

.....

(b) (i) Why is the model of the structure of biological membranes described as 'fluid mosaic'? [2]

.....

The diagrams below represent two glycoprotein molecules found in the plasma membranes of mammalian cells.



- (ii) Which of the molecules **A** or **B** will form an **intrinsic** protein in the plasma membrane? [1]

Molecule

- (iii) Draw a **labelled** diagram of the plasma membrane using the diagrams above to show the correct positioning of glycoproteins **A** and **B**. [2]

- (iv) Give **one** function of the carbohydrate chains on the glycoproteins. [1]

.....

(c) Some diseases are caused by abnormal proteins called prions. Some prions have a higher proportion of β pleated sheet in place of the normal α helix structure.

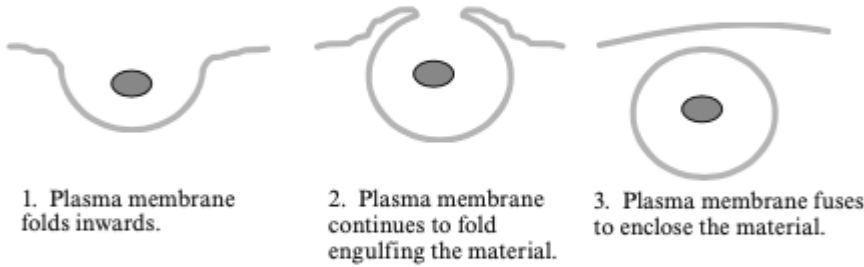
(i) What level of protein structure is described by the terms α helix and β pleated sheet? [1]

.....

(ii) Which organelles are involved in synthesising proteins? [1]

.....

(d) The following diagram shows one way that prions may pass into cells.



(i) Name the process shown in the diagram above. [1]

.....

(ii) Name **two other** ways in which substances might pass into the cell. [2]

.....
.....

(Total 16 marks)

5.	(a)	(i)	OH and H removal shown on diagram; formation of water (H ₂ O) shown; dipeptide correctly drawn with C joined to N;	3
		(ii)	Condensation;	1
		(iii)	Peptide; NOT dipeptide;	1
	(b)	(i)	Mosaic: <u>Proteins</u> are scattered (in lipid layer); Fluid: molecules / components / (phospho)lipids / proteins are free to move around;	2
		(ii)	B;	1
		(iii)	Drawing shows a lipid bilayer with A and B in the correct places, B intrinsic (through the middle) A extrinsic (on top or bottom, outside phosphate heads); Need not use N and P, but must be clear which is A and B any 1 correct label from phospholipid / hydrophobic / hydrophilic / cholesterol / phosphate (head) / lipid or fatty acid (tails);	1 1
		(iv)	<u>Cell</u> {recognition / interaction / identification / cell to cell recognition / adhesion / signalling} / receptor qualified e.g. {hormone receptor / antigens};	1
	(c)	(i)	Secondary;	1
		(ii)	Ribosomes / rough endoplasmic reticulum; Accept nucleus; NOT golgi body / nucleolus.	1
	(d)	(i)	Endocytosis (accept phagocytosis / pinocytosis); NOT exocytosis.	1
		(ii)	Any 2: Diffusion / osmosis; Facilitated diffusion; Active transport;	2

WJEC 2013

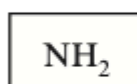
4. Red blood cells are involved with the transport of oxygen around the body. Red blood cells lack internal organelles and their cytoplasm contains haemoglobin. Haemoglobin is a protein that consists of four polypeptide chains linked together.

(a) State the level of protein structure shown by haemoglobin.

[1]

.....

(b) The diagram below shows one of the polypeptide chains from haemoglobin.



- (i) On the diagram above, use an arrow to **clearly label** an alpha –helix. [1]
- (ii) Complete the diagram above by writing in the empty box, the molecular group that would be present at the end of the polypeptide chain. [1]
- (iii) Name **two** types of bonds that would be present to maintain the 3D shape of this polypeptide chain. [1]

.....

.....

- (c) The plasma membrane contains proteins and phospholipids. Describe **two** ways in which the structure of phospholipids differ from triglycerides. [2]

4. (a) Quaternary/ 4°; 1

(b) (i) (Labelled) arrow in correct position; 1

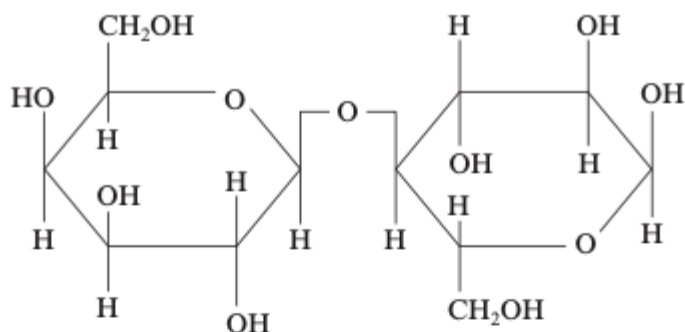
(ii) COOH/ carboxyl/ carboxylic acid; 1

(iii) Disulphide {bond/ bridges} / ionic bonds / hydrogen / 1
hydrophobic interactions / Van der Waals; (Any 2)
NOT peptide / S-S (covalent – neutral)

(c) **Mark points must be comparative** Max 2

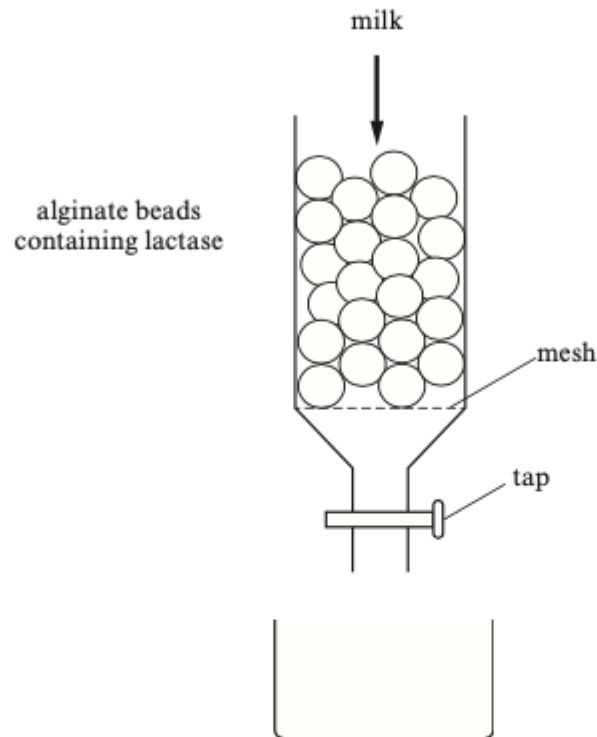
phospholipid	triglyceride
2 fatty acids	3 fatty acids;
phosphate (head)	do not contain a phosphate (head)
polar/hydrophilic head and non-polar/hydrophobic tails	non-polar/hydrophobic;

6. Lactose is a disaccharide found in milk. The diagram below shows the structure of lactose.



- (a) Lactose can be broken down into its constituent monosaccharides.
- Complete the diagram above to show how lactose is broken down. [2]
 - State the type of reaction involved in the breakdown of lactose. [1]
.....
 - Name the bond that is broken during this reaction. [1]
.....
 - Name the molecules produced when lactose is broken down. [1]
.....

- (b) The enzyme lactase can be used to break down lactose. In an experiment lactase was immobilised inside alginate beads and placed in a column, as shown in the diagram below. Fresh milk was then poured into the column and left for one minute before being allowed to drain into the beaker below. As the milk passes through the column the lactose in the milk is broken down.



- (i) What is meant by the term immobilised enzyme?

[1]

.....

.....

- (ii) Describe **two** advantages of using immobilised enzymes in this way.

[2]

.....

.....

.....

.....

- (c) (i) The products produced from the breakdown of lactose are reducing sugars. Describe how you could test for the presence of a reducing sugar. [2]

.....

.....

.....

.....

- (ii) The products produced could also be detected by a biosensor. What is meant by the term biosensor? [1]

.....

.....

- (iii) What would be the main advantage of using the biosensor to detect the products? [1]

.....

.....

.....

- (d) Some bacteria which are found in milk can convert sugars within the milk to lactic acid. Over time the number of these bacteria increase and this eventually causes milk to go sour. The experiment above was repeated with milk that had been left for seven days. State and explain the effect this would have on the concentration of reducing sugars detected. [4]

.....

.....

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.....

.....

(Total 16 marks)

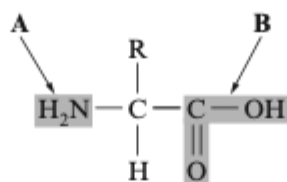
- | | | | |
|----|-----|--|-------|
| 6. | (a) | (i) Molecule of water (drawn with arrow towards the O atom of the glycosidic bond); NOT water going out
Monosaccharides drawn with –OH groups in correct position on C1 and C4 (involved in bond); | 2 |
| | | (ii) Hydrolysis; NOT hydrolysatation (ignore reference to acid) | 1 |
| | | (iii) Glycosidic; | 1 |
| | | (iv) Glucose <u>and</u> galactose; ignore alpha/ beta | 1 |
| | (b) | (i) An <u>enzyme</u> that has been fixed to an <u>inert</u> {matrix/support/ substance}; | 1 |
| | | (ii) The enzyme can easily be recovered/ reused;
The product is free from contamination;
Enzyme is {stable at / tolerates/ withstand} higher temperatures/denatures at a higher temperature/ functions over a wide range of pH;
NOT wider range of temperature alone
Several enzymes with differing optima can be used at the same time;
More control over the reaction/enzymes easily added or removed/ can be used in a continuous process; | Max 2 |

- (c) (i) Heat with Benedict's solution/reagent; 2
NOT warm/ water bath/ ref to acid
Blue to {red/ orange/ green/ yellow/ brown};
- (ii) Instrument/equipment that can detect a specific 1
molecule/metabolite (in a mixture of molecules/bodily fluid).
- (iii) Any one from: 1
The biosensor would give quantitative data/
it would detect {a particular product/glucose/galactose}/
Can detect even at {very low concentrations/ small volumes};
- (d) 1. (The concentration of reducing sugars) would decrease; Max 4
2. {Lactose/ substrate} concentration is lower (in the sour
milk);
3. Lactic acid lowers the pH;
4. Enzyme would be inactivated/denatured;
5. Hydrogen/ ionic bonds (maintaining the 3D shape)
would break;
6. This will change the shape/charge of the active site (of
lactase);
7. Fewer enzyme-substrate complexes would be
formed/fewer successful collisions;
8. Benedicts would remain {blue/ change to {orange/
yellow/ green/ brown}/ negative}

Question 6 Total

[16]

1. (a) The diagram below shows the structure of an amino acid.



- (i) Name the two shaded groups **A** and **B** shown on the diagram above. [2]

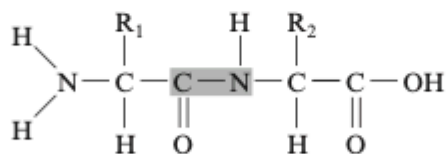
A

B

- (ii) What is represented by letter **R** in the diagram above? [1]

.....

- (b) The diagram below shows two amino acids joined together.



Using the diagram shown above:

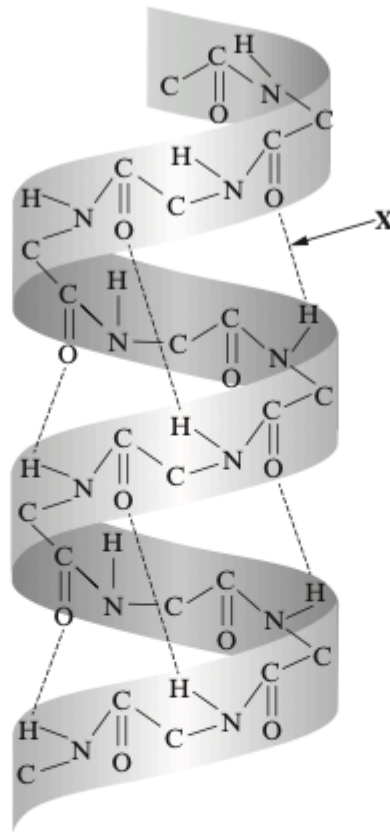
- (i) Name the type of molecule shown. [1]

.....

- (ii) Name the bond highlighted in diagram (b) above. [1]

.....

(c) The diagram below shows part of a protein molecule.



(i) Name bond X shown on the diagram above.

[1]

(ii) What name is given to the shape of this molecule?

[1]

(iii) What level of protein structure does the diagram above show?

[1]

(Total 8 marks)

1.	(a)	(i)	A <u>amino/amine</u> ;	2
			B <u>carboxyl</u> ;	
		(ii)	variable group/side chain OR description of; NOT element/ hydrocarbon chain/ R group	1
	(b)	(i)	Dipeptide; NOT polypeptide	1
		(ii)	peptide (bond);	1
	(c)	(i)	hydrogen bonds; NOT H bond	1
		(ii)	Alpha/ α helix; NOT double helix	1
		(iii)	secondary/ 2° (structure) NOT second	1
			Question 1 total	[8]

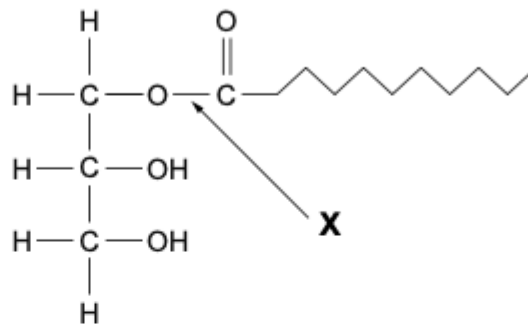
(a) Give an account of the structure and function of carbohydrates.

[10]

8.	(a)	A. Monosaccharides / single sugars plus 2 suitable examples; B. Diagram of hexose/glucose; C. Alpha and beta forms of glucose shown; (can be description) D. Pentoses/deoxyribose/ribose and presence in DNA/RNA; E. Trioses in photosynthesis/respiration/metabolic pathways; F. Disaccharides plus 2 suitable examples; G. Correct formation of glycosidic bond (stated or diagrams, labelled); H. 2 suitable examples of where disaccharides are found (milk sugar/germinating seeds/transport in plant stems); I. Starch in plant cells for storage of <u>glucose</u>; NOT energy J. Correct reference to starch structure (alpha glucose/amylose & amylopectin/1 -4 and 1 – 6 linkages/amylose spiral chain/amylopectin branched); K. Glycogen in <u>animal</u> cells for <u>glucose</u> storage ; L. Glycogen has branched chains; M. Cellulose in <u>plant</u> cell walls/structural polysaccharide; N. Correct reference to cellulose structure (beta glucose/microfibrils/ chains held together by H – bonds/alternate 180° glucose); O. Correct reference to chitin (amino groups/ use in {exoskeleton/ fungal cell walls})	
----	-----	--	--

WJEC 2014

3. (a) The diagram below shows a monoglyceride.



During the digestion of monoglycerides, the bond labelled **X** is broken down by the enzyme lipase.

- (i) Name the bond labelled **X** in the diagram above. [1]

.....

- (ii) State the **type** of reaction involved in the breakdown of the monoglyceride. [1]

.....

- (iii) In the space below **draw** and **name** the products of the breakdown of the monoglyceride. [2]

(iv) Explain why triglycerides are not considered to be polymers.

[1]

.....

.....

.....

- (b) (i) Two fatty acids which are common in triglycerides are stearic acid and oleic acid. Stearic acid has the chemical formula of $C_{17}H_{35}COOH$ and oleic acid has the chemical formula of $C_{17}H_{33}COOH$.
What type of fatty acid is oleic acid? Give a reason for your answer. [2]

.....

.....

.....

- (ii) Apart from energy storage, state **two** functions of triglycerides in a mammal. [2]

.....

.....

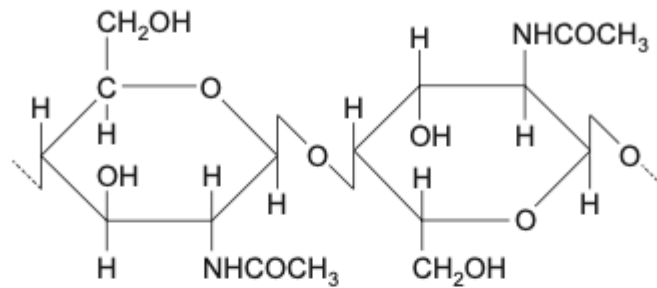
.....

3	(a)	(i) Ester;	1
		(ii) Hydrolysis;	1
		(iii) Glycerol and fatty acid drawn correctly; Glycerol and fatty acid named;	2
		(iv) Glycerol and fatty acids have different structures / OWTTE; (not just reference to monomers)	1
	(b)	(i) (Oleic acid is) unsaturated; It contains at least one C=C double bond (in the hydrocarbon chain) / is not fully saturated with hydrogen (atoms); NOT hydrogen bonds/ fewer hydrogens	2
		(ii) Any 2 protection of internal organs against impact; <u>thermal</u> insulation; buoyancy; waterproofing skin/fur; source of metabolic water;	Max 2
		Question 3 total	[9]

(a) Using examples to illustrate your answer, describe how the structure of polysaccharides are related to their functions.

- 8 (a) A polysaccharides {are polymers/ formed during condensation reactions};
- B (monomers are) joined by glycosidic bonds;
- C starch is made up from alpha glucose;
- D starch is composed of amylose and amylopectin / contains both 1,4 & 1,6 bonds;
- E glycogen is made from (alpha) glucose;
- F {Starch/glycogen} are insoluble and therefore osmotically inert/ OWTTE;
- G {Starch/glycogen} are storage molecules because {glucose can be added or removed easily / they have a compact structure};
- H cellulose is composed of beta glucose;
- I alternate glucose molecules are rotated by 180° / head up head down structure;
- J this form long straight chains (of beta glucose)/ only contains 1-4 bonds;
- K {hydrogen bonds / cross links} form between the chains;
- L forming microfibrils;
- M cellulose provides {strength/rigidity} to plant cell walls / cellulose prevents osmotic lysis in plant cells;
- N in chitin some OH groups are replaced with amino acids / amine groups / glucose amine;
- O chitin provides strength to fungal cell walls / (arthropod) exoskeletons;

(b) The exoskeleton of insects contains the molecule shown below.



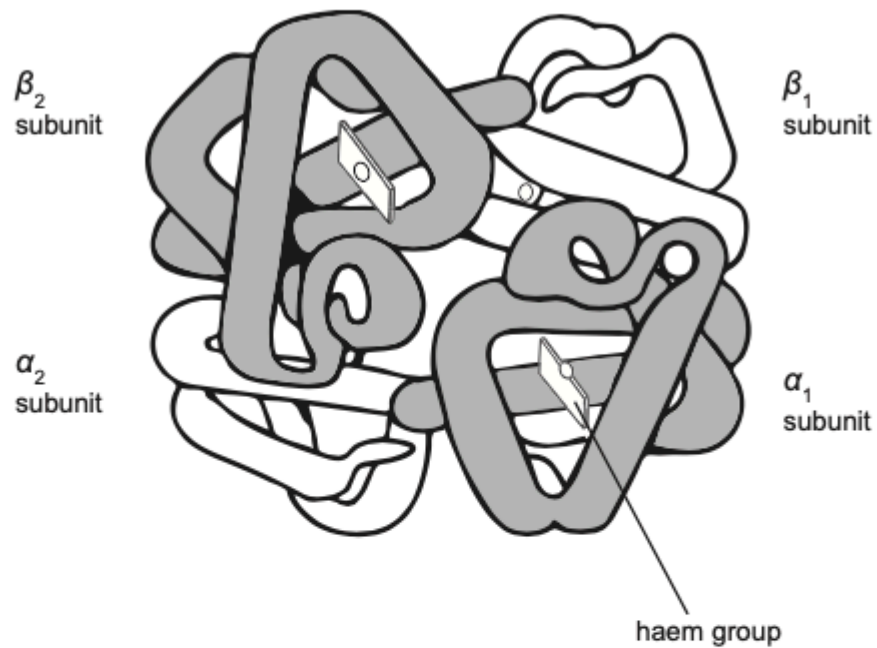
(i) Name the **group** of biological molecules to which the molecule shown above belongs. [1]

(ii) Explain how the molecule shown gives strength to the exoskeleton. [2]

(iii) Some OH groups in the molecule above have been replaced with NHCOCH₃ groups. Name **one other** group of biological molecules that contain nitrogen. [1]

- | | | |
|-----|---|-------|
| (b) | (i) Carbohydrates;
Accept polysaccharides | 1 |
| | (ii) Any two from
Alternating molecules rotated through 180° form <u>straight</u>
chains;
{Cross links/hydrogen bonds/ H bonds} form between chains;
forming <u>microfibrils</u> ; | Max 2 |
| | (iii) Proteins/amino acids/nucleic acids/ nucleotides/
{organic/nitrogenous} bases;
NOT DNA/RNA | 1 |

3. The diagram below shows a molecule of haemoglobin.



- (a) State the inorganic ion present in the haem group. [1]
- (b) Using the diagram above, explain why this molecule is regarded as having a quaternary structure. [2]

(c) Describe the biochemical test that could be performed to test for a protein. [2]

.....

.....

.....

.....

(d) Suggest how the concentration of a specific protein could be measured in a sample of urine. [1]

.....

.....

- | | | | |
|----------|-----|---|----------|
| 3 | (a) | Iron / Fe ²⁺ ; | 1 |
| | (b) | {Four polypeptide chains / two alpha and two beta subunits};
in tertiary form are {combined/joined}; | 2 |
| | (c) | Add { <u>biuret</u> (reagent) / copper sulphate <u>and</u> sodium hydroxide};
Reject boil/heat
Colour changes from <u>blue</u> to { <u>purple/lilac/violet</u> }; | 2 |
| | (d) | <u>Biosensor</u> ; | 1 |

Question 3 Total [6]

- (a) Using examples, describe the functions of carbohydrates and lipids in living organisms. [10]

8 (a) Carbohydrates

- A Glucose for respiration;
- B Starch for storage of {glucose/energy} in plants;
- C Cellulose for structural support in plant cell walls/ chitin in {insect exoskeleton/ fungi};
- D Glycogen for storage of {glucose/energy} in animals;
- E {Glycogen/starch} insoluble so no osmotic effect;
- F Disaccharides or named + function (e.g. sucrose for transport in plants);

Lipids

- G Saturated fatty acids for storage in animals/ unsaturated fatty acids for storage in {seeds/plants};
- H Thermal insulation/buoyancy;
- I Waxes for waterproofing in leaves;
- J Good source of energy, twice as many as carbohydrates or value 38 kJ per g;
- K Correct ref to protection of organ from physical damage (e.g. kidney);
- L Electrical insulation in neurons (ref to myelin);
- M Source of metabolic water from respiration of lipids;

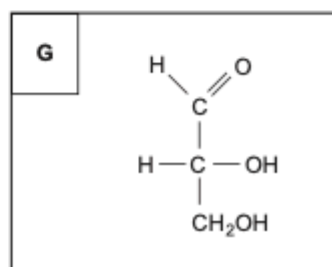
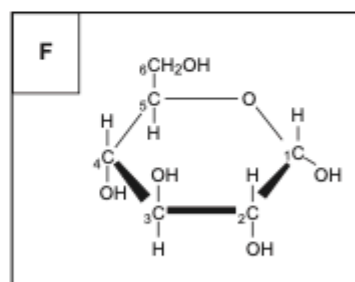
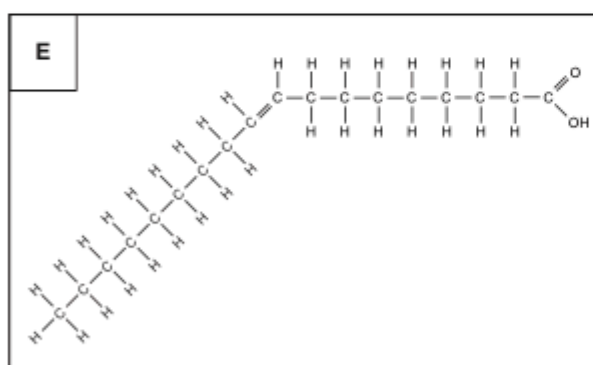
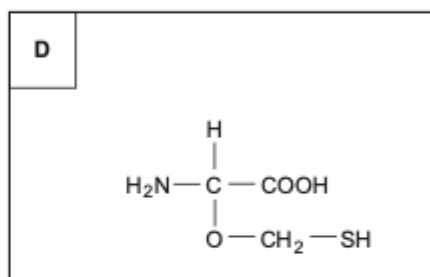
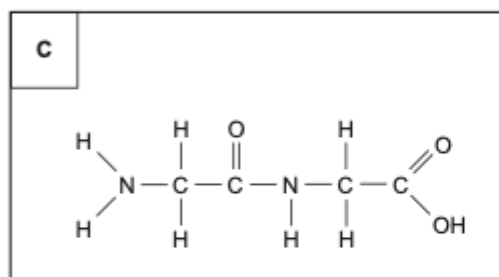
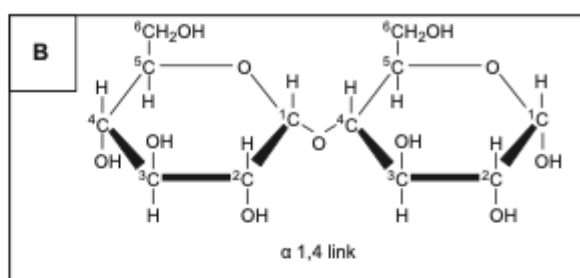
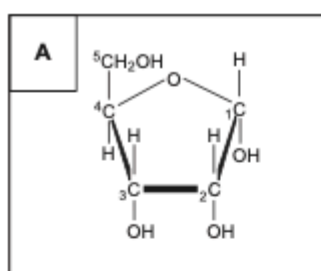
Used to make other molecules

(CHO / glucose / lipids needed to make)

- N+ Any two for one mark each from:
 - O Chlorophyll *with* magnesium / phospholipids *with* phosphate/ {DNA/RNA/ATP} *with* nitrogen and phosphate / amino acids *with* nitrogen/ glycoprotein *with* protein;

WJEC 2015

1. The following diagrams represent the structure of some common organic molecules.



- (a) Using letters **A-G**, complete the table below. You may use each letter once, more than once or not at all. [7]

Statement	Letter(s)
is a monosaccharide	
is a dipeptide	
would be found in nucleic acids	
contains C=C bonds	
contains a glycosidic bond	
is a triose sugar	

- (b) (i) Describe a biochemical test for the presence of glucose in a solution. [2]

.....

.....

.....

.....

.....

- (ii) Explain why a positive result is seen with glucose but not for sucrose. [1]

.....

.....

.....

.....

1	(a)	Statement	Letter(s)
		Is a monosaccharide	A,F,G ;
		Any 2 from 3 correct	
		Is a dipeptide	C ;
		Would be found in nucleic acids	A ;
		contain C=C bonds	E ;
	(b)	Contains a glycosidic bond	B ;
		Is a triose sugar	G ;
		(i) Add Benedicts/Fehlings (reagent) and { <u>heat / boil</u> };	
		Colour would change from <u>blue</u> to {red / orange / green / brown};	
		Blue precipitate = neutral	
		(ii) Glucose is a reducing sugar / reduces copper II sulphate (to copper I oxide) / sucrose is a non reducing sugar;	
		Question 1 total	

WJEC 2016

1. Chemical elements are joined to form biological compounds in cells and tissues.

(a) Using an example from the selection below, complete the following table. You can use the same example once, more than once or not at all. [5]

phosphorus sodium chloride blood leaf
starch amino acid phosphate

	Example
organic molecule	
tissue	
ion	
polymer	
element	

(b) (i) Explain, with reference to its structure, why a phospholipid is both hydrophobic and hydrophilic. [2]

.....

.....

.....

(ii) Describe *one* **cellular** function of each of the following molecules. [2]

Triglyceride

.....

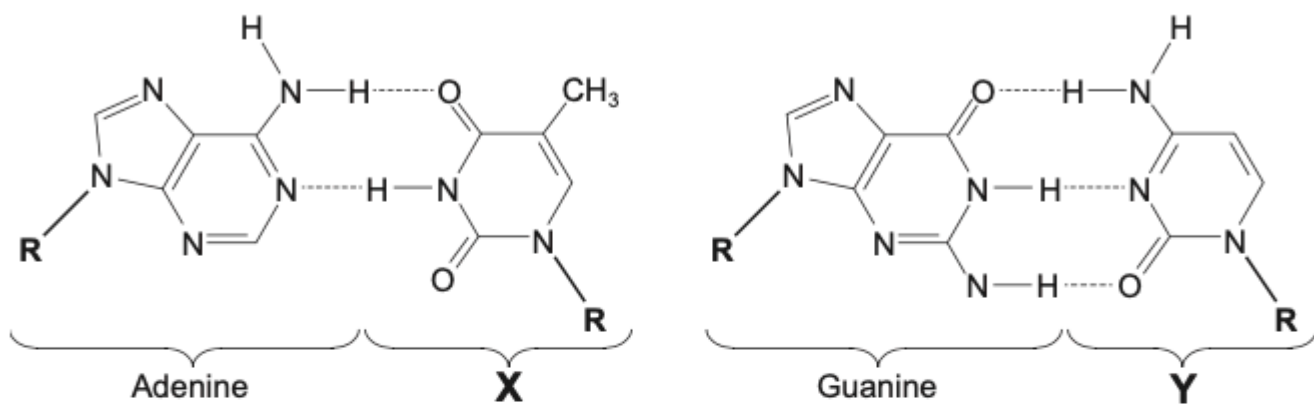
.....

Phospholipid

.....

.....

(c) The diagrams below show the base pairing in DNA.



(i) Explain why DNA is described as being a polymer.

[1]

(ii) Name the group of organic bases to which X and Y belong.

[1]

(iii) R represents a sugar-phosphate group. State the type of reaction that occurs to form a bond between the sugar-phosphate group and the organic base.

[1]

1	(a)		<i>organic molecule</i>	Starch / amino acid ;	5
			<i>tissue</i>	Blood ;	
			<i>ion</i>	Phosphate ;	
			<i>polymer</i>	Starch ;	
			<i>element</i>	Phosphorus ;	
	(b)	(i)	Hydrophobic {fatty acids / tail} are non-polar; Hydrophilic {phosphate / head} is polar ; Accept: ref to charged		2
		(ii)	<i>triglyceride</i> Source / store of energy / metabolic water ; Accept: ref to steroid/hormone/glycolipid synthesis Reject: ref to waterproofing/insulation/protection <i>phospholipid</i> (component of cell) membranes / control of molecules entering/leaving cell ;		2
	(c)	(i)	{Chain / linked / bonded} of repeating / many {nucleotides/monomers }; Accept: description of monomer Reject: chain of polynucleotides / incorrect monomer.		1
		(ii)	Pyrimidine ;		1
		(iii)	Condensation ;		1
			Question 1 total		[12]

3. (a) Explain the formation of hydrogen bonds between water molecules.
You may use annotated diagrams in your answer.

[3]

.....

.....

.....

.....

.....

.....

(b) The following table shows some properties of water and their significance to life. [5]

Complete the table below.

Property	Biological Significance
ice less dense than water	
	has a role in cooling the body through sweating
cohesion between water molecules	
	minimises temperature fluctuation in aquatic habitats
dissolves ionic substances	

3

(a)

1) polar molecule / dipole ;

2) $H^{\delta+}O^{\delta-}$ / hydrogen slightly positive and oxygen slightly negative charges ;

3) forms bonds between hydrogen and oxygen on adjacent molecules ;

(b)

Property	Biological Significance
<i>ice less dense than water</i>	- so floats on surface, to provides a habitat to live on / - insulation of water so life can survive when ice forms ;
high latent heat of evaporation / vaporisation ;	<i>has a role in cooling body</i>
<i>Cohesion between water molecules</i>	- allows movement through xylem Accept: transpiration stream (surface tension) allows insects to walk / live on water surface or example ;
high (specific) heat capacity ;	<i>minimises temperature fluctuation in aquatic habitats</i>
<i>dissolves ionic substances;</i>	used for {transport/ movement } of molecules / {metabolic /chemical} reactions occur in solution ;

5

Question 3 Total

[8]

1.2

WJEC 2009

1. (a) The following table compares plant and animal cells. Complete the table by placing a tick (✓) in the appropriate box if the structure is present. Place a cross (✗) in the box if the structure is not present.

<i>Structure</i>	<i>Plant Cell</i>	<i>Animal Cell</i>
Centrioles		
Mitochondria		
Chloroplasts		

- (b) (i) Draw **and label** a diagram to show a section through a typical mitochondrion.

- (ii) What is the function of mitochondria in cells?

.....

- (iii) Name a type of cell in which you would expect to find large numbers of mitochondria.

.....

- (iv) Explain why the cell you have chosen contains large numbers of mitochondria.

.....

.....

(Total 8 marks)

1. (a)

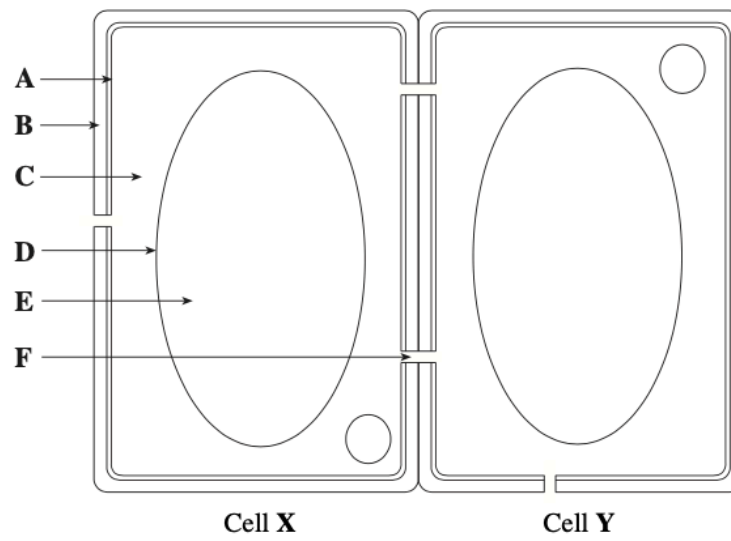
Structure	Plant cell	Animal Cell
Centrioles	X	/
Mitochondria	/	/
Chloroplasts	/	X

(1 for each column)

- (b) (i) Correct diagram showing double membrane with Inner membrane folded.
Any 2 labels from matrix, crista, intermembrane space, double membrane, stalked particles, DNA, ribosomes.
- (ii) Aerobic respiration / ATP production (not: respiration)
- (iii) Muscle cell / liver cell/sperm/pancreas/epithelial cell from small intestine/neurone/companion cell.
- (iv) High requirement for energy for contraction/sperm movement / for chemical activity/high metabolic activity/active transport. (not: exercise)

[Total 8 Ma

5. The drawings below show two adjacent plant cells, placed in a 0.6 Molar glucose solution.



- (a) Name the parts labelled A-F.

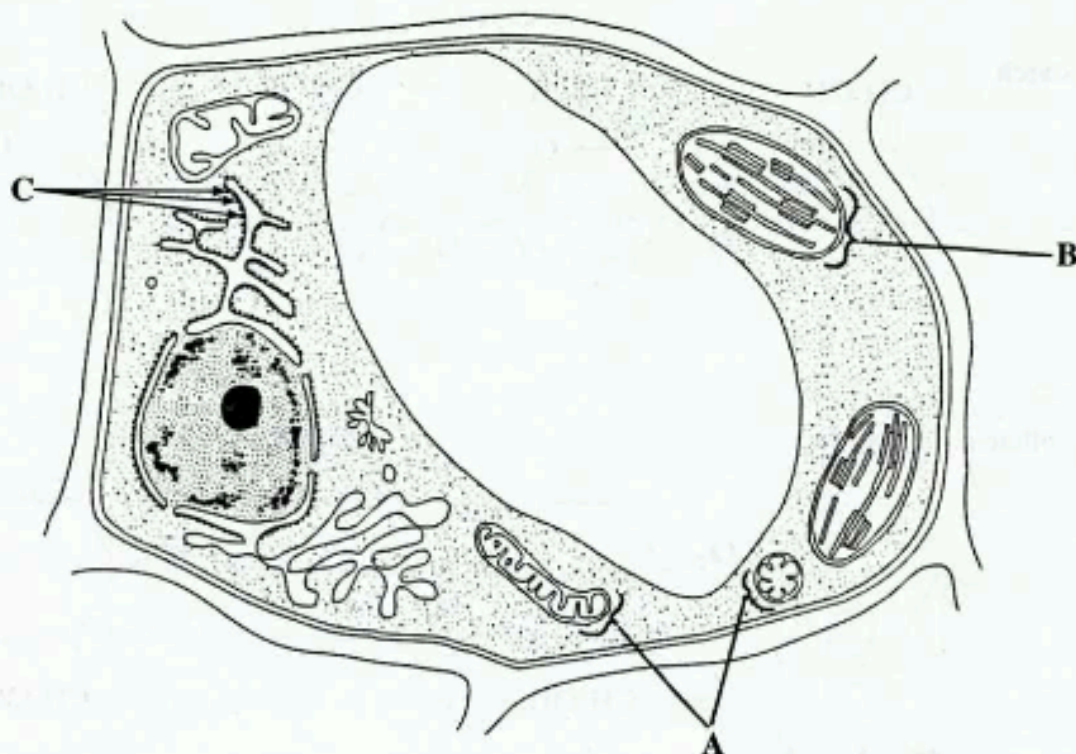
[2]

- A
- B
- C
- D
- E
- F

- 5.** **(a)** **A.** Cell / plasma membrane **B.** Cell wall
- C.** Cytoplasm **D.** Tonoplast / vacuolar membrane
- E.** Vacuole **F.** Plasmodesma(ta)

(2 for all correct 1 if 1 mistake)

4. The diagram shows the ultra-structure of a plant cell.



- (a) Complete the table by naming and stating the function of each of the organelles labelled A, B and C. [6]

Organelle	Name	Function
A		
B		
C		

- (b) The structures labelled A are identical. Explain why they differ in appearance in the diagram. [1]

.....

.....

- (c) State **two** ways in which an animal cell would differ from a plant cell. [2]

.....

.....

(Total 9 marks)

4. (a)

Structure	Name	Function
A	Mitochondrion	(Cell) respiration / ATP production
B	Chloroplast	Photosynthesis / light trapping
C	Ribosome	Protein synthesis

Independent marks

[6]

(b) they have been cut in different planes

[1]

(c) animal cells do not have chloroplast; cell wall; (large) (central) permanent vacuole; centrioles; plasmodesmata

(not: chlorophyll) No need for comparative statements. [Any 2]

[2]

[Total 9 marks]

WJEC 2010

2. The table below describes structural features and functions of some of the organelles found in eukaryotic cells. Complete the table. [6]

<i>Organelle</i>	<i>Structural feature</i>	<i>Function of feature</i>
Nucleus	Nucleolus	
	Inner membrane folded into cristae	
	Vesicle containing hydrolytic enzymes	Cell lysis
Golgi body		

(Total 6 marks)

2.

	Structural feature	Function of feature
Organelle		ribosome manufacture / synthesis rRNA;
Mitochondrion;		<u>Increase surface area</u> for enzyme attachment /ATP synthesis/ oxidative phosphorylation;
Lysosome;		
	stack of cisternae / flattened (membrane bound) sacs	Protein assembly /conjugation of proteins / secretion / lysosome formation/ produces vesicles/packaging or modification molecules/ stores and transports lipids/stores and transports lipids

[Total 6 marks]

1. The table shows three features that may be found in bacteria and viruses. If the feature is present put a tick (✓) in the box, if it is not present put a cross (✗) in the box. [3]

<i>Feature</i>	<i>Bacterium</i>	<i>Virus</i>
Possess nucleic acid		
Surrounded by a protein coat		
Ribosomes in cytoplasm		

(Total 3 marks)

1

Feature	Bacterium	Virus
Possess nucleic acid	✓	✓
Surrounded by a protein coat	X	✓
Ribosomes in cytoplasm	✓	X

(not: hybrid ticks ✗)

(i) Fibrous;

2. (a) The diagram represents a section of a collagen molecule.



- (i) Name the group of proteins to which collagen belongs. [1]

- (ii) Describe the structure of a collagen molecule. [3]

- (iii) State **one** function of collagen. [1]

- (b) Haemoglobin belongs to another group of proteins called globular proteins. The diagram shows a molecule of haemoglobin.



Structure of haemoglobin

- (i) Describe **three** differences between collagen and haemoglobin that you can see in the diagrams. [3]

1.

2.

3.

3.

- (ii) Name **one other** type of globular protein. [1]

.....

(Total 9 marks)

2 (a)

(ii) Polypeptide chains; (not: proteins) (3)

Three chains; (not: strands)

(Three) alpha helices;

Tightly/closely bound;

Held together by hydrogen bonds:

(iii) Structural/relevant example e.g. tendons or named tissue strengthened.
(not: strength or name of tissue unequal./tensile strength)

(b) (i) Four chains vs. three; (3)

Iron/prosthetic/haem group vs. none;

Compact vs. non-compact/long fibres vs spherical;

3 polypeptide chains the same vs 2 different polypeptide chains;

Secondary structure vs. quaternary structure: (not: more complex)

(ii) Hormones/enzyme/ antibodies/plasma proteins. (not: specific examples)

9 M.

WJEC 2011

6. The diagram below shows an organelle found in a liver cell.



(a) (i) Name the organelle. [1]

.....

(ii) State the function of the organelle. [1]

.....

(iii) Name the structures labelled A and B in the diagram. [2]

A

B

(b) Explain why liver cells have large numbers of these organelles present. [2]

.....

(Total 6 marks)

Q.6 (a) (i) mitochondrion; [1]

(ii) aerobic respiration / production / manufacture of ATP;
 (not: make ATP for respiration) [1]

(ii) A = cristae;
 B = matrix; [2]

(b) metabolically active/ many chemical reactions or specified eg active transport
 large amount of ATP produced/required; [2]

(Total 6 Marks)

1. The table below shows some features of prokaryotic and eukaryotic cells. Complete the table to show the differences between the two types of cell.

<i>Feature</i>	<i>Prokaryotic</i>	<i>Eukaryotic</i>
Mitochondria		
Arrangement of DNA		DNA forms chromosomes
Position of DNA	Free in cytoplasm	
Composition of cell wall if present		
Size of ribosomes		

[5]

(Total 5 Marks)

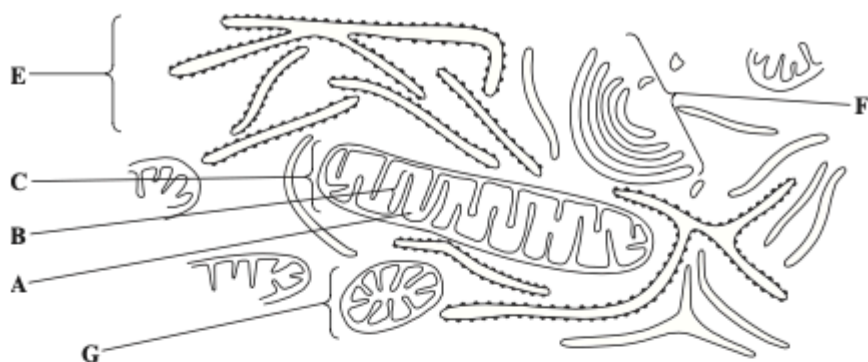
1.

Feature	Prokaryotic	Eukaryotic
mitochondria	Absent (not: no organelles)	present;
Arrangement of DNA	circular/no chromosomes; (not: loop)	DNA forms chromosomes
Position of DNA	Free in cytoplasm	in nucleus/ bound by membrane;
Composition of cell wall if present	murein peptidoglycan (not: not cellulose)	cellulose/chitin;
Size of ribosomes	Small/70S; (not: other figures)	Large/80S;

1 mark per row

(Total 5 marks)

6. The diagram below shows part of a cell that secretes a hormone into the bloodstream.



- (a) Name structures **A** and **B**. [2]

A

B

- (b) Explain the functions of structures **E** and **F**. [4]

E

.....

F

.....

- (c) Suggest why this type of cell is likely to contain large numbers of structure **C**. [2]

.....

.....

.....

- (d) Labels **C** and **G** show the same type of organelle. Explain why they differ in appearance. [1]

.....

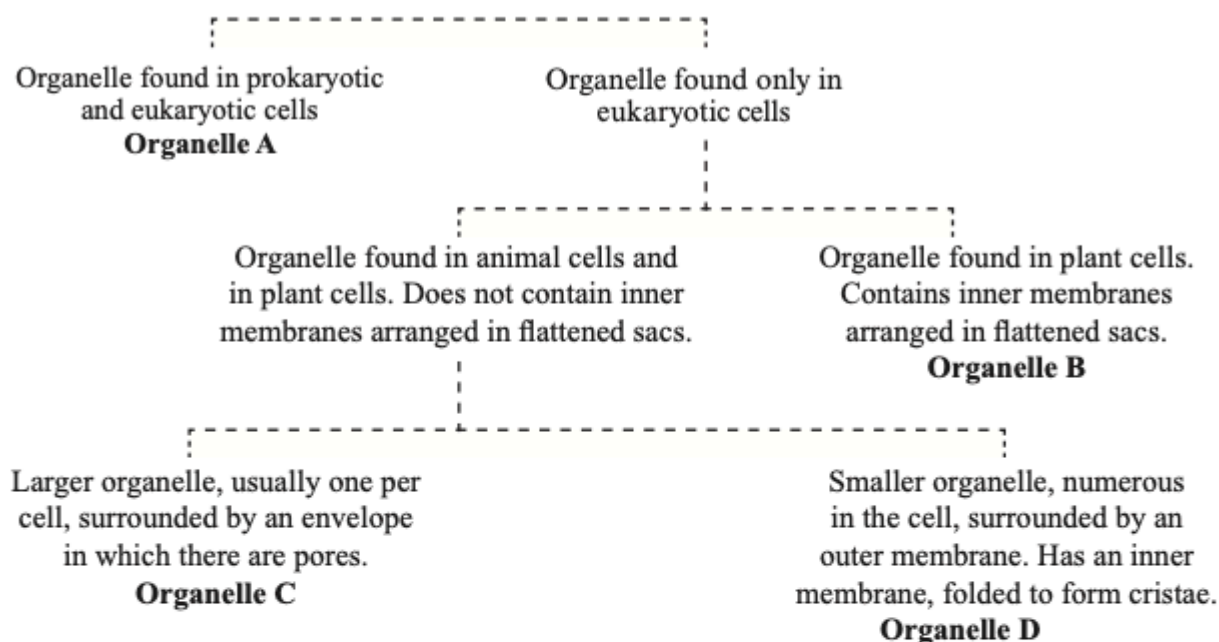
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(Total 9 Marks)

6. (a) A = matrix;
B = crista/internal membrane; **2**
- (b) E is the site of protein synthesis;
Polypeptide chains build up at ribosome;
transports polypeptides/proteins;
ribosomes read genetic code (allow: receive mRNA); **2 max**
- F buds off vesicles/package proteins into vesicles;
these contain molecules for secretion;
transport protein molecules to cell surface/membrane;
synthesis of glycoproteins/modification of proteins; **2 max**
- (c) secretory cell involved in active processes/metabolically active;
ATP/energy dependent;
ATP manufactured by C;
hormone synthesis requires ATP; **2 max**
- (d) cut in different plane/AW; **1**
- (Total 9 marks)**

WJEC 2012

1. The diagram shows how some organelles may be distinguished from each other.



(a) (i) Name organelle **D**. [1]

(ii) Describe the function of organelle **D**. [2]

(iii) Name a cell that contains **large** numbers of organelle **D**. [1]

(b) Which of the organelles **A**, **B**, **C** or **D** is a ribosome? [1]

.....

(c) What is the function of the pores in organelle **C**? [1]

.....

.....

(Total 6 marks)

- | | | |
|------------|--|---|
| 1. (a) (i) | Mitochondrion/ mitochondria | 1 |
| (ii) | Respiration/ aerobic respiration | 1 |
| | stores {energy/ ATP}/ release energy/ {synthesis/produce | 1 |
| | ATP}/ release energy / ATP for respiration =2 marks NOT | |
| | production of energy | |
| (iii) | muscle / liver/ epithelial cells of small intestine / cells of | 1 |
| | proximal convoluted tubule/ neurones/ companion cells/ | |
| | sperm/ secretory cells NOT muscle tissue/ the liver/ cardiac | |
| | tissue | |
| (b) | A | 1 |
| (c) | allows transport of { <u>messenger/ mRNA</u> }/ nucleotides/ | 1 |
| | ribosomes NOT out and in | |

1. (a) Name the following:

- (i) A device containing an enzyme that can be used to detect a specific compound in a fluid. [1]

- (ii) An aggregation of similar cells carrying out the same function. [1]

(b) Give **one** structural difference between each of the following:

- (i) prokaryotic and eukaryotic cells; [1]

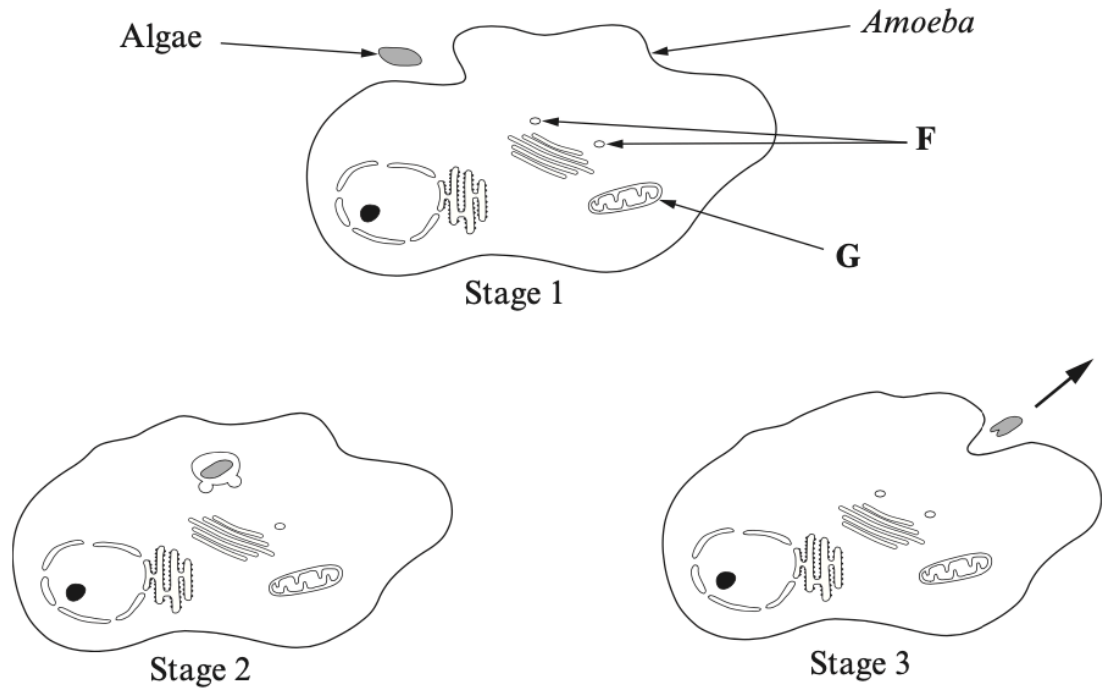
- (ii) chloroplasts and mitochondria. [1]

(Total 4 marks)

1.	(a)	(i)	Biosensor;	1
		(ii)	Tissue;	1
(b)	(i)	Prokaryotic has no nucleus vs eukaryotic has a nucleus / eukaryotic has membrane bound organelles vs prokaryotic no membrane bound organelles (Accept named membrane bound organelle) / prokaryotes smaller ribosomes (70S) vs Eukaryotes larger (80S) / DNA circular v DNA in chromosomes or strands [must refer to both terms]; Reject reference to cell wall; Reject reference to size; Reject reference to plasmid;		1
	(ii)	Chloroplast contain chlorophyll vs mitochondria have no chlorophyll (accept photosynthetic pigments) / grana vs no grana / stroma vs matrix / cristae vs no cristae / thylakoid vs no thylakoids / cristae vs grana / infolding of membrane in mitochondria not in chloroplasts [must refer to both structures];		1
Question total				4

WJEC 2013

3. *Amoeba proteus* is a single celled eukaryotic organism that can be found living in shallow freshwater ponds and streams. *Amoeba proteus* feeds on algae and other unicellular organisms. The diagrams below show the sequence of events during feeding.



- (a) (i) Name and describe the process that has occurred between stages 1 and 2 on the diagram opposite. [2]

.....

.....

.....

.....

- (ii) Structures **F** on the diagram opposite are involved in the digestion of the *Amoeba*'s food.
Name the organelle where Structures **F** are formed. [1]

.....

- (iii) State the name of the process occurring at stage 3 on the diagram opposite. [1]

.....

- (b) (i) What is the function of the organelle on the diagram opposite labelled **G**? [1]

.....

.....

- (ii) Suggest a reason why this organelle is required by *Amoeba* during feeding. [1]

.....

.....

.....

- (c) Describe **three** ways in which the structure of a prokaryotic cell would differ from that of *Amoeba*. [3]

.....

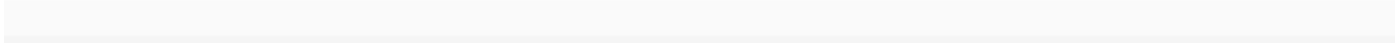
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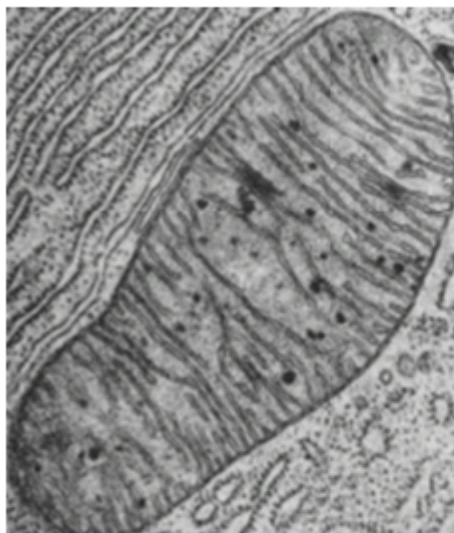


- | | | | |
|----|-----|--|-------|
| 3. | (a) | (i) Phagocytosis/ <u>endocytosis</u> ;
the (cell) <u>membrane</u> {invaginates/infolds/ surrounds/ wraps
around/ engulfs} (to form a vesicle (allow vacuole))around the
{food particle/ algae}; | 2 |
| | | (ii) Golgi { <u>Body/apparatus</u> }; | 1 |
| | | (iii) Exocytosis; | 1 |
| | (b) | (i) (Site of aerobic) respiration / production of ATP;
NOT production of energy alone | 1 |
| | | (ii) Carry out {endo/exo/ phago}cytosis / synthesis of digestive
enzymes/ movement/ form lysosomes;
Reject active transport unqualified
NOT digestion/ feeding | 1 |
| | (c) | 1.No nucleus/nuclear membrane/ DNA free in cytoplasm;
2. No membrane bound organelles / named example/ possess
mesosome;
3. A loop of DNA / circular DNA/ ORA DNA {linear/ on
chromosome/ associated with histone};
4. Smaller/ <u>70S</u> ribosomes;
5. Cell wall; Reject reference to cellulose
6. Capsule/ flagellum/ plasmid;
NOT reference to size (can be neutral) | Max 3 |

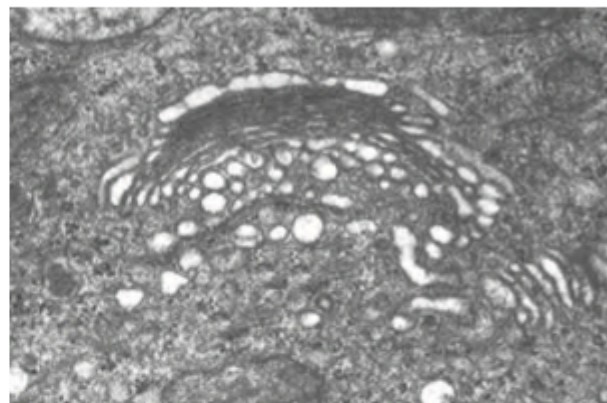
Question 3 Total

[9]

- 3 (a) The electron micrographs below show organelles in eukaryote cells.



A



B

- (i) Identify the organelles in photographs **A** and **B** and state their function. [2]

A

Function

.....

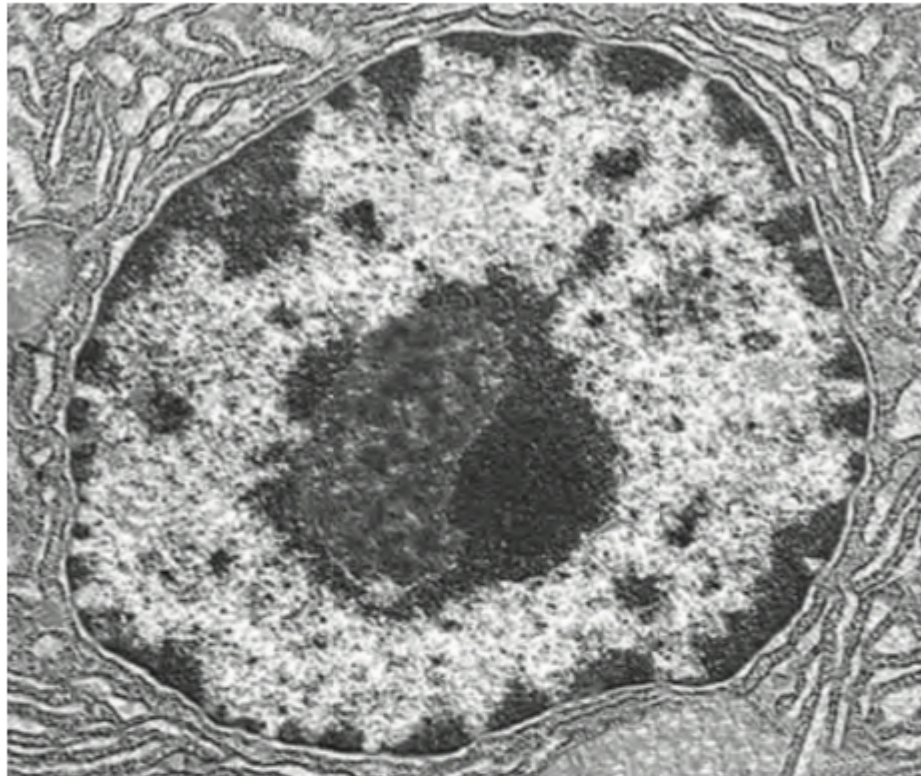
B

Function

.....

- (ii) Name a tissue which contains large numbers of the organelle shown in **A**. [1]

.....



C

- (b) Photograph **C** above shows a nucleus. State **two** features of a nucleus that can be seen in this electron micrograph and their function. [2]

Feature 1

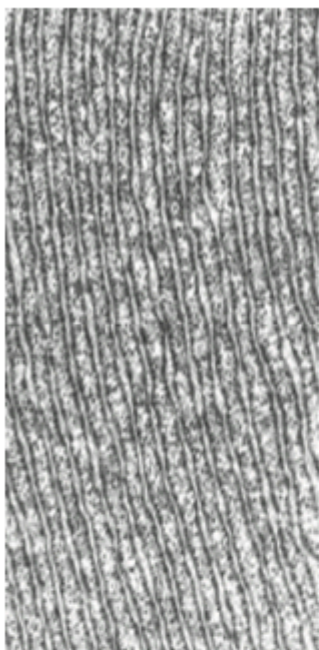
Function

.....

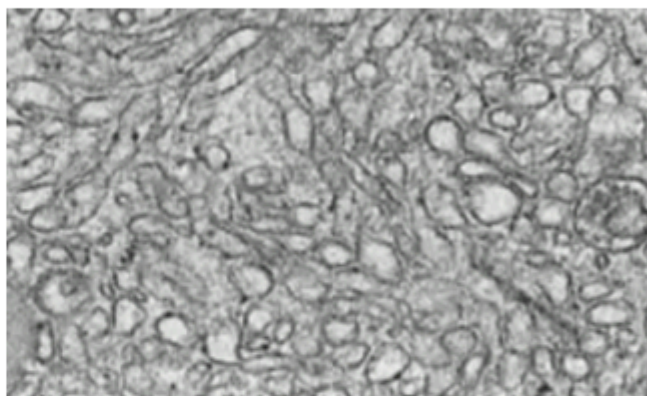
Feature 2

Function

.....



D



E

- (c) Photographs **D** and **E** above show two different types of endoplasmic reticulum. State **two** visible differences between **D** and **E**. [2]

.....

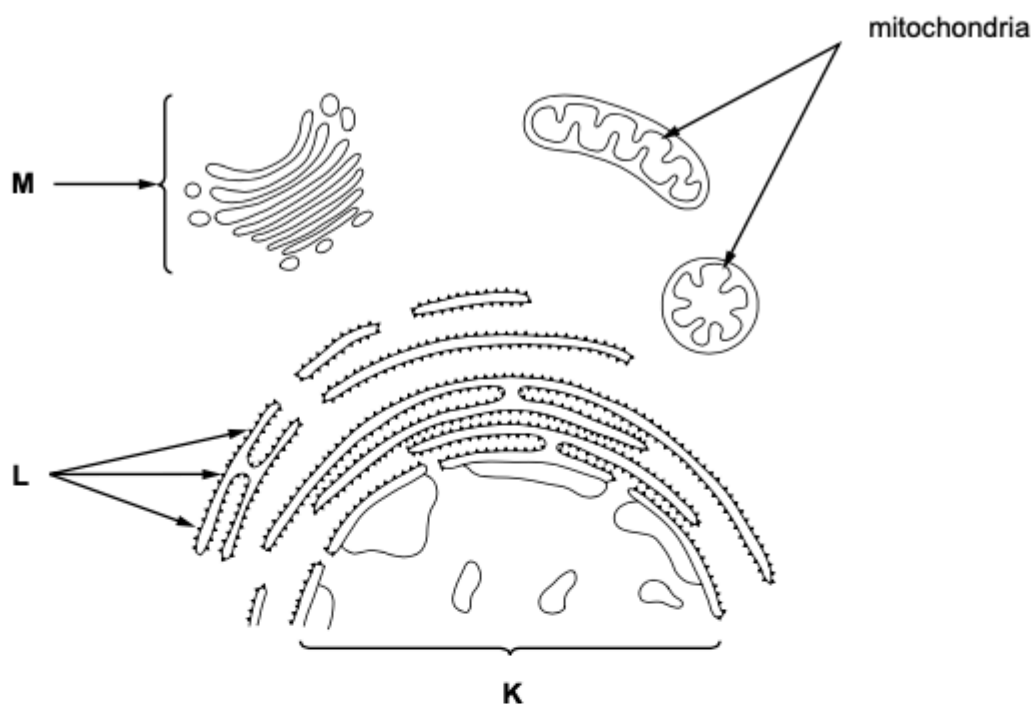
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.....

(Total 7 marks)

3.	(a)	(i)	A Mitochondrion/ mitochondria Plus ATP synthesis/aerobic respiration; NOT produce/ create energy B Golgi Body/ complex/ apparatus NOT golgi alone Plus one of • Modification of {proteins/lipids}/ Addition of sugar chains/ produces glycoprotein • {Transport/storage} of {lipids/digestive enzymes} • Synthesis of {(secretory) vesicles/lysosomes}/ packaging proteins; NOT transport(ation) of proteins/ synthesis of proteins	2
		(ii)	Liver/muscle/nervous tissue/ meristem;	1
	(b)		Nuclear pores + Allows {mRNA/ribosomal RNA/ribosomes} to <u>pass out/through</u> of nucleus; NOT substances Nucleolus + Synthesis of ribosome (components); (Double) nuclear membrane/nuclear envelope + Separates the DNA from the rest of the cellular contents/ holds DNA/ chromosomes; Chromatin+ condenses to form chromosomes/ {involved in/ code for} protein synthesis; Matched pair = 1 mark	2
	(c)		D presence of ribosomes + no ribosomes on E; D {<u>membranes/ cisternae</u>} in parallel/regular lines/ more organised + {open network of <u>membranes/ cisternae</u>}/ less organised/ or description in E;	2
			Question 3 Total	[7]

5. The diagram below shows part of a generalised animal cell.



(a) Complete the table below.

[6]

Organelle	Name	Function
K		
L		
M		

(b) (i) Explain why the mitochondria labelled in the diagram above appear different from one another.

[1]

- (ii) Nearly all eukaryotic cells possess mitochondria. Mitochondria are similar in size to prokaryotic cells and have features in common with them. This led to the biologist, Lynn Margulis, proposing that mitochondria evolved from ancient prokaryotes. The theory of endosymbiosis proposes that these ancient prokaryotes were engulfed by other bacterial cells and both benefited from the relationship – this led to the evolution of eukaryotic cells.

Using your knowledge, state which **two** structures found in prokaryotic cells are also found in mitochondria. [2]

.....
.....

- (iii) Describe **two** differences between mitochondria and prokaryotic cells such as bacteria. [2]

.....
.....
.....
.....

5 (a)

Organelle	Name	Function
K	nucleus;	contains <u>DNA</u> which {codes for / controls} <u>protein synthesis</u> ;
L	ribosomes ;	synthesise proteins;
M	Golgi apparatus/body;	packaging of proteins (for secretion from the cell) / (chemically) modifies proteins / produces glycoproteins / produces lysosomes;

6

(b) (i) They have been cut in different plane/ angle;

1

(ii) (Loop of) DNA;
(70S) ribosomes;

Max 2

Both possess plasma membranes; NOT double membrane

(iii) Mitochondria: (statements should be comparative)

Max 2

Has a double membrane;

No cell wall;

No capsule;

No flagellum/ pili;

No mesosome;

No plasmids;

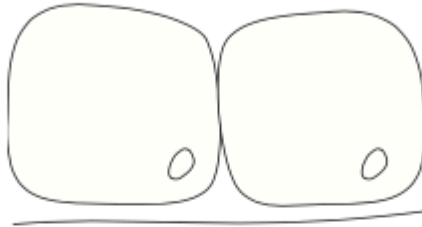
Question 5 Total

[11]

1. (a) Samples of epithelial tissue were examined using a light microscope. Drawings of cells from these tissues are shown below. Identify the type of epithelial tissue shown, and suggest from where in the body the samples were taken.

(i)

[2]

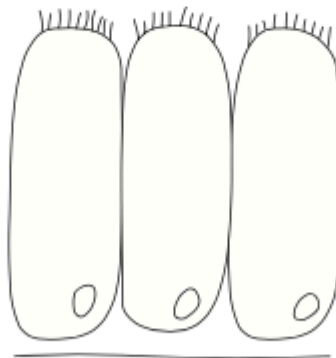


Tissue type

Where found in body

(ii)

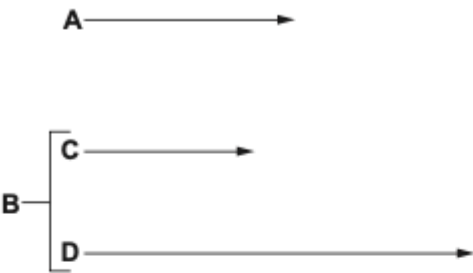
[2]



Tissue type

Where found in body

The electron micrograph below shows part of a typical animal cell.



(b) Complete the table below by naming the structures and organelles shown in the electron micrograph above, and describing their functions. [4]

Letter	Organelle / Structure	Function
B	nucleus	
C	nuclear pores	
D		

- (c) The nucleus has pores in the envelope that surrounds it, whereas organelle **A** does not. Describe **one other** difference between the membranes that surround organelle **A** and those that surround the nucleus. [1]

.....

.....

.....

- (d) Describe **two** differences between the ribosomes found in animal cells and those found in prokaryotic cells. [2]

.....

.....

.....

.....

1 (a) (i) Cuboidal; 2
Kidney tubule;
Accept kidney/ liver/named gland/ureter/ovary/glands

(ii) Ciliated;NOT cilia 2
Trachea / oviduct/fallopian tube/ bronchi/bronchioles;

(b) 4

B	<i>nucleus</i>	contains <u>DNA</u> which {codes for/ controls} <u>protein synthesis</u> / transcription/ <u>DNA</u> {synthesis/replication};
C	<i>nuclear pores</i>	{Transport/movement} of {mRNA/ nucleotides/rRNA}; Accept ribosomes NOT transport of mRNA in
D	Nucleolus;	Produces {rRNA/ribosomes/tRNA}; NOT produces RNA unqualified

(c) 1

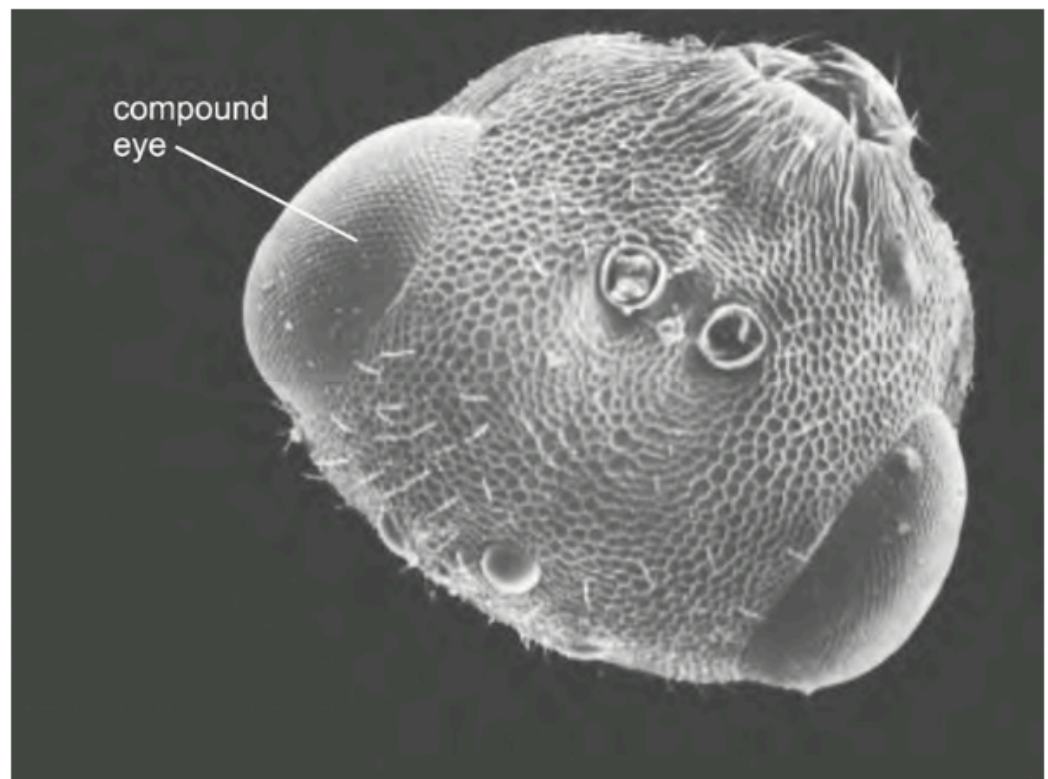
Organelle A	Nucleus
<u>Inner</u> membrane is folded /	No folding of <u>inner</u> membrane /
Has cristae	no cristae;
No ribosomes attached	ribosomes attached;

must be comparative

(d) 2
Ribosomes are not attached to {membranes/ ER} in
prokaryotes (some) are in animal cells;
Ribosomes are {larger/80S} in animal cells than prokaryotes /
70S;
must be comparative

Question 1 Total [11]

2. The photograph below shows the head of a parasitic wasp, *Chlorocytus* species, taken using a scanning electron microscope.



- (a) The photograph shows two compound eyes, which are regarded as organs. State what is meant by the term *organ*.

.....

.....

.....

.....

.....

- 2 (a) (An organ) is an {aggregation/collection} of several tissues;
To carry out a {specific/particular} {function/task/job} (for the
whole organism);

- (b) Describe the structure and function of the rough endoplasmic reticulum, Golgi body and lysosomes. [10]

8 (b) **Rough Endoplasmic Reticulum** Max 10

- A Flattened sacs/cisternae (or from diagram);
- B Continuous with nuclear membrane (or from diagram);
- C With attached ribosomes (must be clearly labelled on diagram);
- D Site of {protein synthesis/translation}/transport system;

Golgi

- E Golgi consists of a {series/system/group/stack} of (dynamic) flattened sacs (diagram must show at least 3);
- F Function in packaging proteins (for secretion);
- G Vesicles containing proteins from RER fuse with Golgi membrane and contents are shed into Golgi sacs/ coalescence of vesicles;
- H (Contents are built into more complex molecules such as) enzymes/glycoproteins;
- I Other Golgi function, e.g. carbohydrate secretion/ transporting or storing lipids;
- J {Vesicles containing product/lysosomes} are budded off;
- K Ref. to exocytosis of contents; NOT in context of lysosomes

Lysosomes

- L Lysosomes contain digestive enzymes/lysozyme;
- M Function is to {break down worn out organelles/digest foreign material/ cause autolysis/ intracellular breakdown};
- N {Lysosomes/vesicles} fuse with membrane of digestive vacuoles;
- O Enclosed by phagocytosis; NOT in context of lysosomes

Award Max 8 if only 2 organelles described

Points A,B,C and E can be accepted from clear diagram

Question 8 Total [10]

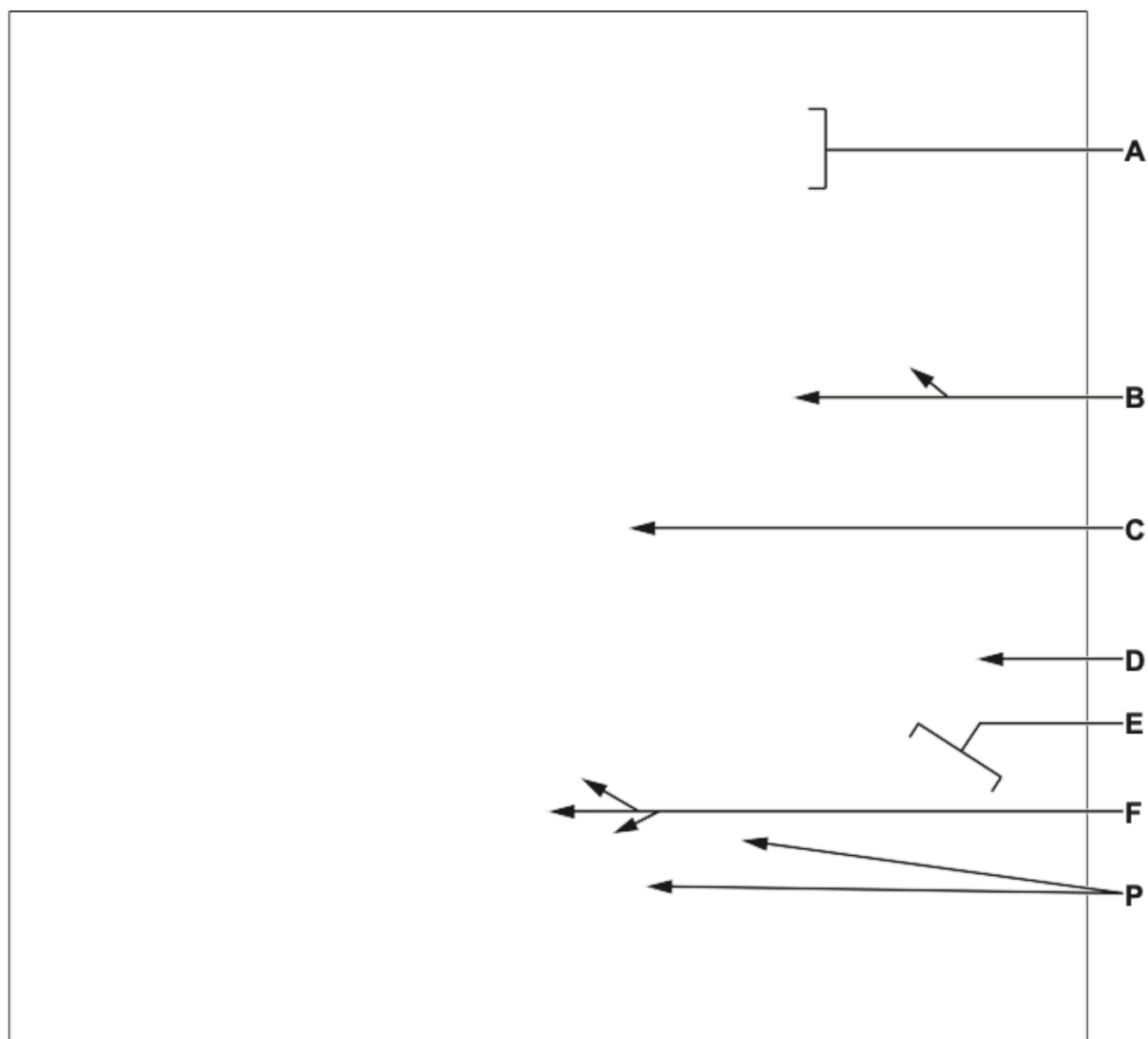
WJEC 2015

- (b) Describe and explain how a mitochondrion **and** a chloroplast are adapted to their functions. Explain how they are similar to a prokaryotic cell. [10]

7	(b)	Organelle essay Mitochondrion A <u>folded</u> inner membrane / cristae; B increases surface area; C for (attachment of) {enzymes / ATP synthetase} / location of {electron transport chain / stalked particles}; D site of (aerobic) respiration / ATP production; E good diagram / description of location of any four from - cristae, matrix, stalked particles, inter membrane space, inner / outer membrane, (circular) DNA, ribosomes ; Chloroplast F thylakoids contain {photosynthetic pigments / chlorophyll}; G thylakoids stacked into grana; H function is photosynthesis; I Starch grains store {photosynthetic products / glucose} J good diagram / description of location of any four from - stroma, {thylakoid / grana}, ribosomes, (circular) DNA, starch grain, lipid droplet, lamellae, inner / outer membrane; <u>similar to prokaryotic cell</u> K contain ribosomes; L similar size; M (circular) DNA; N for self replication; O comparison mesosome with cristae; Question 7 total	[10]
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WJEC 2016

2. The electron micrograph shows part of a plant cell.



(a) Name the structures labelled **A** to **F**.

[6]

- A**
- B**
- C**
- D**
- E**
- F**

- (b) Name a structure, not shown on the diagram, which connects the cytoplasm of adjacent plant cells. [1]

.....

- (c) (i) Suggest a substance, found at point **P** on the electron micrograph, that was processed in the Golgi bodies. [1]

.....

- (ii) Following processing by the Golgi body, describe how this substance was secreted from the cell. [3]

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.....

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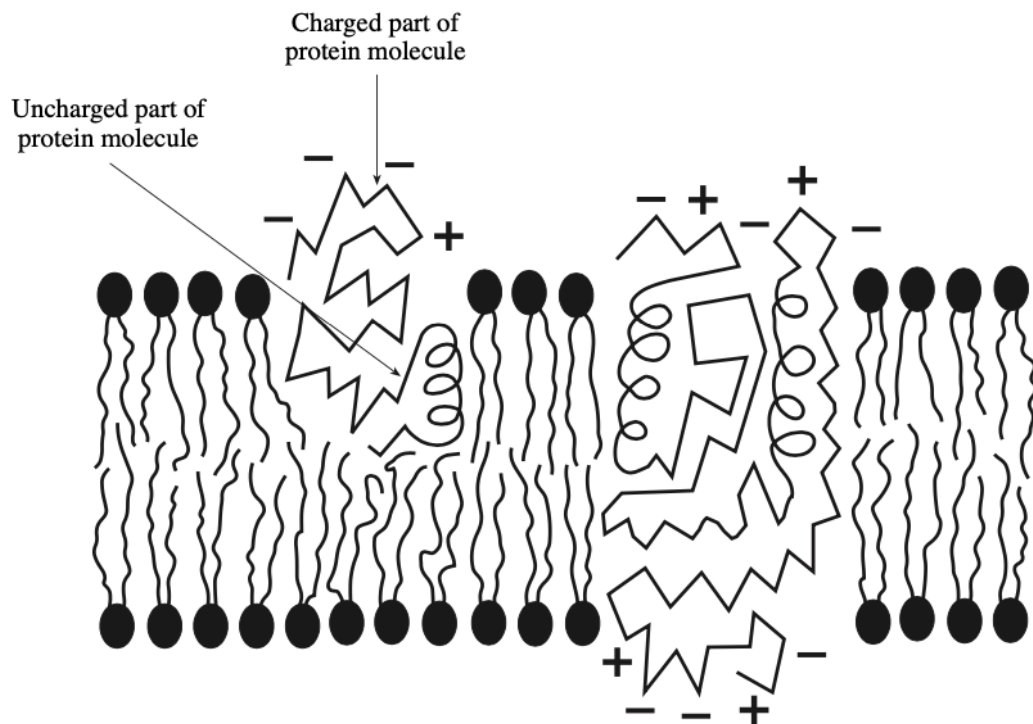
.....

2	(a)	A = Golgi body / Golgi {apparatus/ cisternae} ; B = mitochondrion / mitochondria ; C = endoplasmic reticulum; NOT ER /RER / SER D = {cell / plasma} membrane ; E = cell wall ; F = ribosomes ;	6
	(b)	Plasmodesmid / plasmodesmata ;	1
	(c)	(i) Protein / glycoprotein / pectin/ lipid; Accept: enzymes Reject: lysosome	1
		(ii) • <u>membrane</u> of {secretory vesicles / vesicles containing product} , fuses with cell membrane ; • contents released from cell ; (secreted is neutral) • exocytosis ;	3
Question 2 total			[11]

1.3

WJEC 2009

4. The diagram represents a model of a biological membrane. This model has been described as 'a phospholipid sea with protein icebergs'.



- (a) (i) Name the model, proposed by Singer and Nicholson in 1972. [1]

- (ii) Select a single phospholipid molecule in the diagram and label the part which is hydrophilic and the part which is hydrophobic. [1]

- (b) The proteins are drawn to give some indication of their tertiary structure.

Explain the difference between secondary and tertiary structure of protein molecules including reference to the type of bonds involved. [4]

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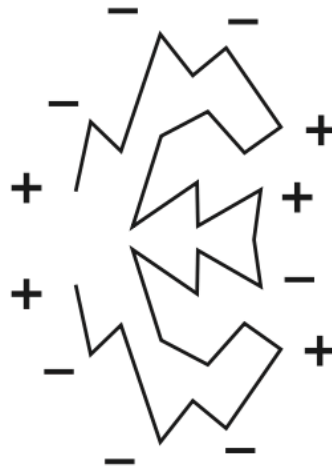
- (c) (i) With reference to the diagram and your answer to part (a) (ii), explain how the distribution of charged and uncharged parts determine the position a protein will take up in a membrane. [2]

.....

.....

.....

- (ii) The diagram below shows another protein.



Suggest how this protein would position itself in the membrane. [1]

.....

.....

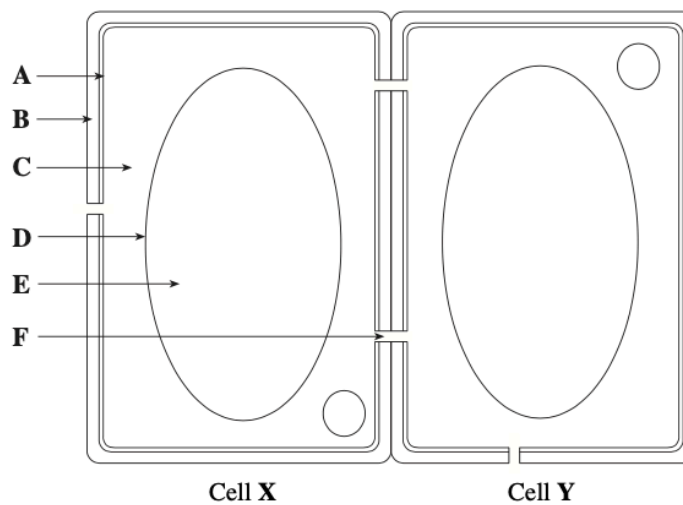
.....

(Total 9 marks)

4. (a) (i) Fluid Mosaic. [1]
- (ii) Head labelled hydrophilic **AND** tail labelled hydrophobic. [1]
- (b) Secondary structure is folding of polypeptide chain / ref. to α helix or β pleated sheet;
held by hydrogen bonds;
tertiary is folding of α helix or secondary structure / correct reference to specific 3D shape;
held by bonds between R groups / name at least 2 from
covalent, disulphide, ionic, salt bridges, hydrophobic, hydrogen, van der Waals.
(Any 4) [4]
- (c) (i) Charged groups will associate with (hydrophilic) heads of lipids / layer; / hydrophobic inside hydrophilic outside.
Uncharged groups will associate with (hydrophobic) tails. [2]
- (ii) Will associate with heads only / attach to outside or inside of the membrane / correct use of extrinsic or would be surface protein.
(not: would not be in the membrane unequal.) [1]

[Total 9 marks]

5. The drawings below show two adjacent plant cells, placed in a 0.6 Molar glucose solution.



- (a) Name the parts labelled **A-F**. [2]

A

B

C

D

E

F

- (b) Name **two** ways by which materials move between cell **X** and cell **Y**. [2]

.....

.....

- (c) Cells **X** and **Y** are at incipient plasmolysis, in a 0.6 molar glucose solution.
Describe **one** change, that would be visible under the microscope, if the cells were placed in a 1 Molar glucose solution. [1]

.....

.....

(d) The water potential of vacuolated cells is represented by the equation:

$$\Psi_{\text{cell}} = \Psi_s + \Psi_p$$

(i) Define the term *water potential*. [1]

.....

.....

.....

(ii) Give the value of Ψ_p in cells at incipient plasmolysis. [1]

.....

(iii) Two different cells **P** and **Q** are adjacent to one another in a plant. Calculate the missing values for each cell and complete the following table. [2]

<i>Cell</i>	Ψ_{cell} <i>kPa</i>	Ψ_s <i>kPa</i>	Ψ_p <i>kPa</i>
P		-1200	+500
Q	-300		+300

(iv) In which direction will water move between these two cells? [1]

.....

(Total 10 marks)

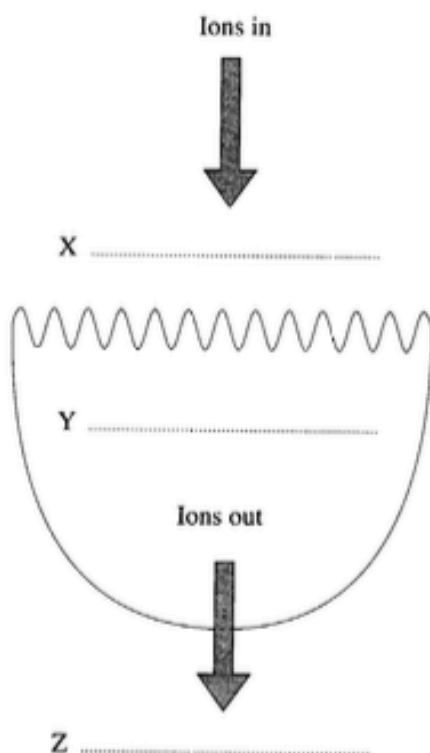
5. (a) A. Cell / plasma membrane B. Cell wall
 C. Cytoplasm D. Tonoplast / vacuolar membrane
 E. Vacuole F. Plasmodesma(ta)

(2 for all correct 1 if 1 mistake)

- (b) Diffusion; osmosis; active transport; facilitated diffusion.
 (Any 2) (not: apoplast/ symplast/ through F) [2]
- (c) Cytoplasm / vacuole shrinks / gaps between wall and cytoplasm.
 (not: plasmolysis/cell shrinks) [1]
- (d) (i) The difference between the free energy of water molecules in a
 system and the free energy of molecules in pure water / the
 tendency for water molecules to leave / move out of a system. [1]
 (not: ref. to equation)
- (ii) Zero. [1]
- (iii) P – 700 kPa.
 Q – 600 kPa. [2]
- (iv) From Q to P/ into P (not: out of Q). [1]
 (independent mark from (iii))

[Total 10 marks]

5. (a) The diagram shows an animal cell.



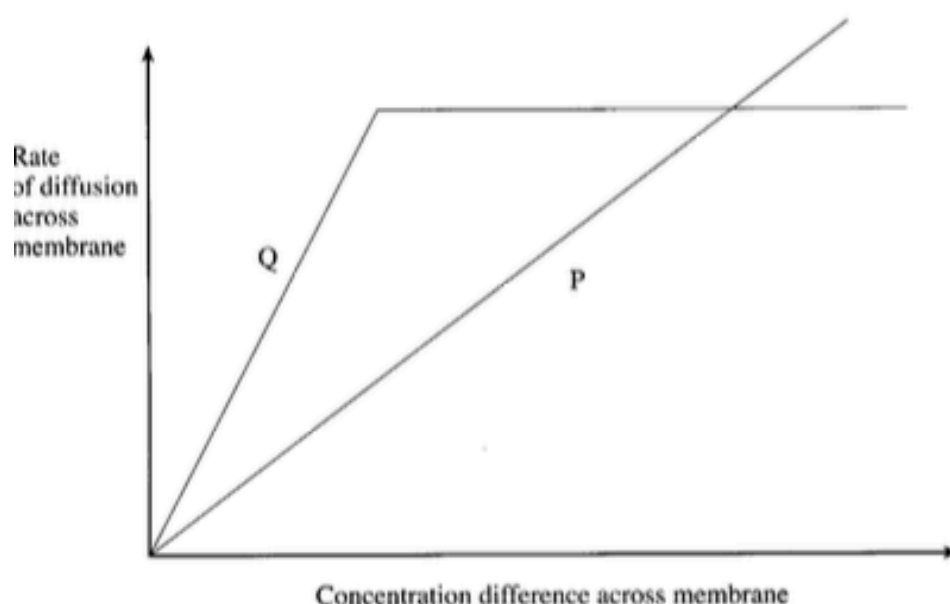
- (i) Sodium ions diffuse into the cell. They then move out of the cell by active transport. Complete the diagram using the words **HIGH** or **LOW** to show the relative concentration of sodium ions at X, Y and Z. [1]
- (ii) Explain how **one** structural feature of the cell shown helps to ensure a rapid rate of diffusion. [1]
-
- (b) (i) A number of factors influence the rate of diffusion. In the table below circle the **letter** which shows the combination of factors which give the most rapid rate of diffusion. [1]

<i>Appearance of membrane</i>	<i>Concentration gradient</i>	<i>Thickness of membrane</i>	
	high to low	thick	A
	low to high	thin	B
	high to low	thin	C
	high to low	thick	D
	low to high	thin	E

- (ii) Using the information in the diagram in part (a), explain how active transport of sodium ions out of the cell helps to ensure a rapid rate of diffusion of sodium ions into the cell. [1]

- (iii) Describe and explain the effect of an increase in temperature on the rate of diffusion. [2]

- (c) The graph shows the relationship between concentration difference across a membrane and the rate of diffusion, for diffusion and facilitated diffusion.



- (i) Identify the two lines.

P

Q

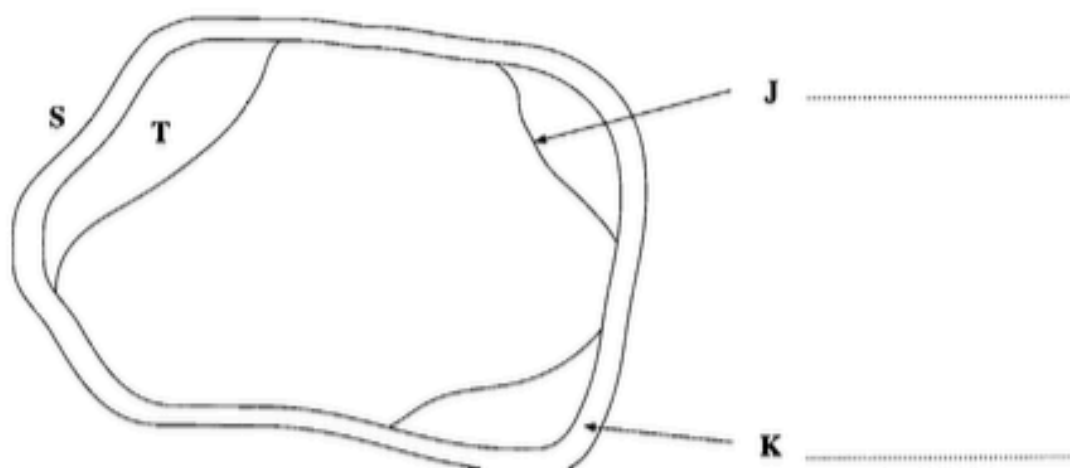
[1]

- (ii) Using the graph and your knowledge of membrane structure explain the difference between P and Q. [2]

(d) Define the term *water potential*.

[1]

(e) A turgid plant cell was placed in a concentrated solution of sucrose. The diagram shows the appearance of the cell after one hour.



(i) Label structures **J** and **K** on the diagram.

[2]

(ii) What evidence on the diagram shows that the water potential of the cell sap must be higher (less negative) than that of the sucrose solution?

[1]

(iii) Use your knowledge of a property of structure **K** to explain why the water potential at **T** must be equal to that at **S**.

[2]

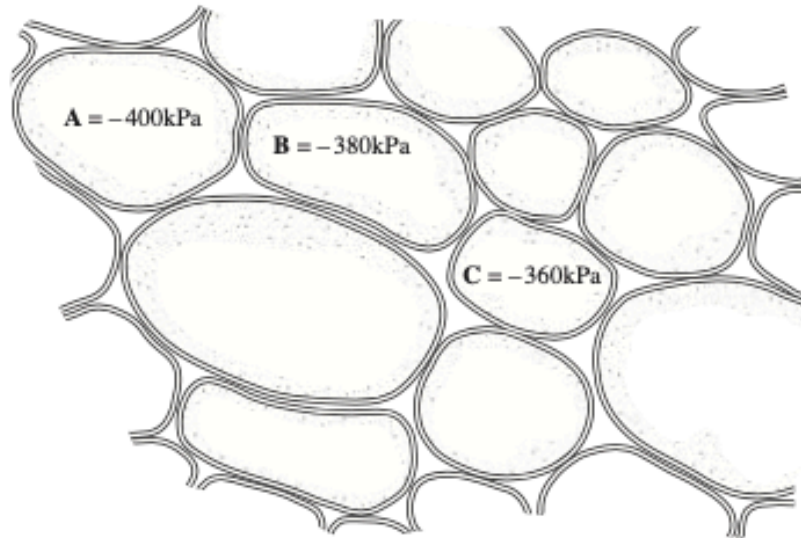
(Total 15 marks)

5. (a) (i) X high
Y low
Z high (all three correct) [1]
- (ii) There are folds in the membrane /
microvilli (which increases the surface area). [1]
- (b) (i) C [1]
- (ii) Moving sodium ions out of the cell will reduce their concentration and create a greater difference in concentration between the inside of the cell and the outside so ensuring a rapid rate of diffusion into the cell. [1]
- (iii) as the temperature is increased kinetic energy/the movement of molecules increases; this results in an increase in rate of diffusion. (allow: diffuse faster) [2]

5. (c) (i) P = diffusion Q = facilitated diffusion (must have both) [1]
- (ii) facilitated diffusion causes more rapid movement of molecules through a membrane than simple diffusion;
it relies on protein carriers / channel proteins;
and the rate is limited by the number of proteins in the membrane i.e. protein carriers 'full';
therefore the curve flattens out; or converse
Diffusion – no carriers involved then just limited by concentration
gradient = 2 [2]
Q flattens out as channel proteins are fully occupied=2
- (d) water potential is the capacity of water to leave or enter a system/cell
Concentration free water molecules in a solution/ ref. kinetic energy [1]
- (e) (i) K cell wall [1]
J cell/plasma membrane [1]
- (ii) cell is plasmolysed / cell membrane has pulled away from the cell wall. Allow incipient plasmolysis, not: cytoplasm has shrunk [1]
- (iii) cell wall is (fully) permeable;
sucrose diffused/moved through to T;
there must be the same (concentration) solution on both sides of the wall; [2]
- [Total 15 marks]**

WJEC 2010

5. The diagram shows cells taken from the stem of a plant. Cells A, B and C are adjacent cells and the figures give the water potential Ψ of each cell.



- (a) (i) Draw arrows on the diagram to show the overall direction of water movement between these **three** cells. [1]

- (ii) Explain your answer in terms of water potential. [2]

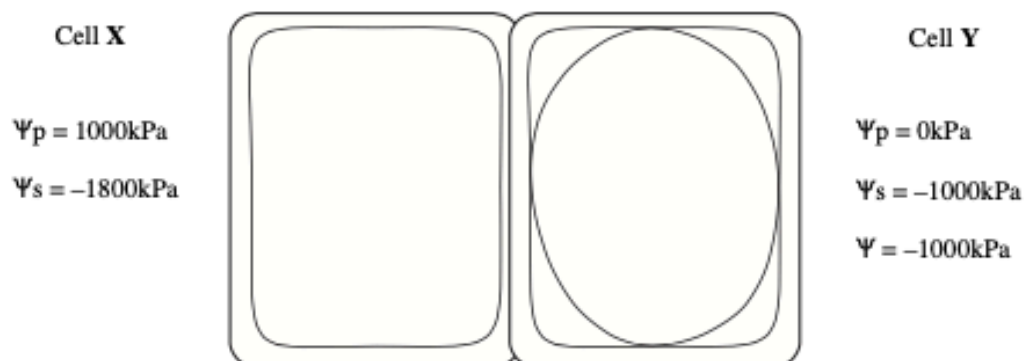
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- (b) The diagram below shows two plant cells, X and Y as seen through a microscope. The figures show the solute potential Ψ_s and the pressure potential Ψ_p for both cells and the water potential Ψ for cell Y.



Water relations in the cells are given by the following equation:

$$\Psi_{\text{cell}} = \Psi_s + \Psi_p$$

- (i) Calculate the water potential, Ψ , of cell X. Show your working. [2]

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- (ii) State the name of the condition shown by cell Y and explain how this condition could have arisen. [3]

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- (c) (i) Cell X has the higher pressure potential Ψ_p . Explain how this pressure potential is built up in cell X. [3]

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- (ii) Suggest the effect on seedlings if all of their cells were in the condition as shown in cell Y. [1]

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(Total 12 marks)

5. (a) (i) C to B to A; [1]
- (ii) water moves down a water potential gradient / high to low;
by osmosis; (not: ref. water concentration)
reference actual figures on diagram; [2]
- (b) (i) $\Psi = +1000 - 1800$;
 $= - 800\text{kPa}$ [2]
- (ii) plasmolysed; [1]
cell in concentrated solution / low water potential;
water passes out;
cytoplasm/vacuole shrinks. [2 max]
(not: cell membrane comes away from wall)

3

- (c) (i) water passes into cell by osmosis;
cytoplasm expands;
cell becomes turgid;
as cytoplasm / contents push against wall;
wall inelastic / resists further expansion. (not: rigid) [3 max]
- (ii) wilts. (not: dies) [1]

[Total 12 marks]

- (a) The cell (plasma) membrane consists of a number of parts.
Describe the structure of the cell membrane and the functions of these parts. [10]

7. (a) A. Singer Nicholson / fluid mosaic model; [1]
- B. Phospholipids / lipid bilayer; [1]
- C. Separate contents from outside / acts as barrier; [1]
- D. Phospholipid allows fat soluble substances through / selective; [1]
- E. Hydrophobic / water hating tails face each other; [1]
- F. Hydrophilic / water loving heads face water / outwards; [1]
- G. Carrier protein; [1]
- H. Used for active transport; [1]
- I. Specific substances transported; [1]
- J. Cholesterol affects fluidity; [1]
- K. Channel/carrier protein for facilitated diffusion; [1]
- L. some are enzymes; [1]
- M. Hydrophilic channels; [1]
- N Glycoprotein / glycolipid; [1]
- O. For cell recognition/signalling/hormonerecognition. [1]

[Total 10 marks]

7. In an experiment 1 cm^3 of blood was added to 10 cm^3 of isotonic saline (salt) solution. To three separate test tubes, 1 cm^3 of the blood was added to equal volumes of each of the following: distilled water, ammonium chloride solution with a water potential of -476 kPa and glycerol solution with a water potential of -896 kPa . The time taken for the red blood cells to burst (haemolysis) is shown in the table.

<i>Solution</i>	<i>Time taken for haemolysis/s</i>	<i>Ψ/kPa</i>
Distilled water	10	0
Ammonium chloride solution	50	-476
Glycerol solution	720	-896

- (a) (i) Suggest **one** problem in performing any investigation using blood. [1]

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- (ii) Explain why haemolysis of the red blood cells occurred quickest when placed in distilled water. [3]

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- (b) A sample of red blood cells were placed in a concentrated solution ($\Psi = -2000\text{ kPa}$) of sodium chloride.
Draw a diagram to show the expected appearance of one of the blood cells after five minutes and explain its appearance. [3]

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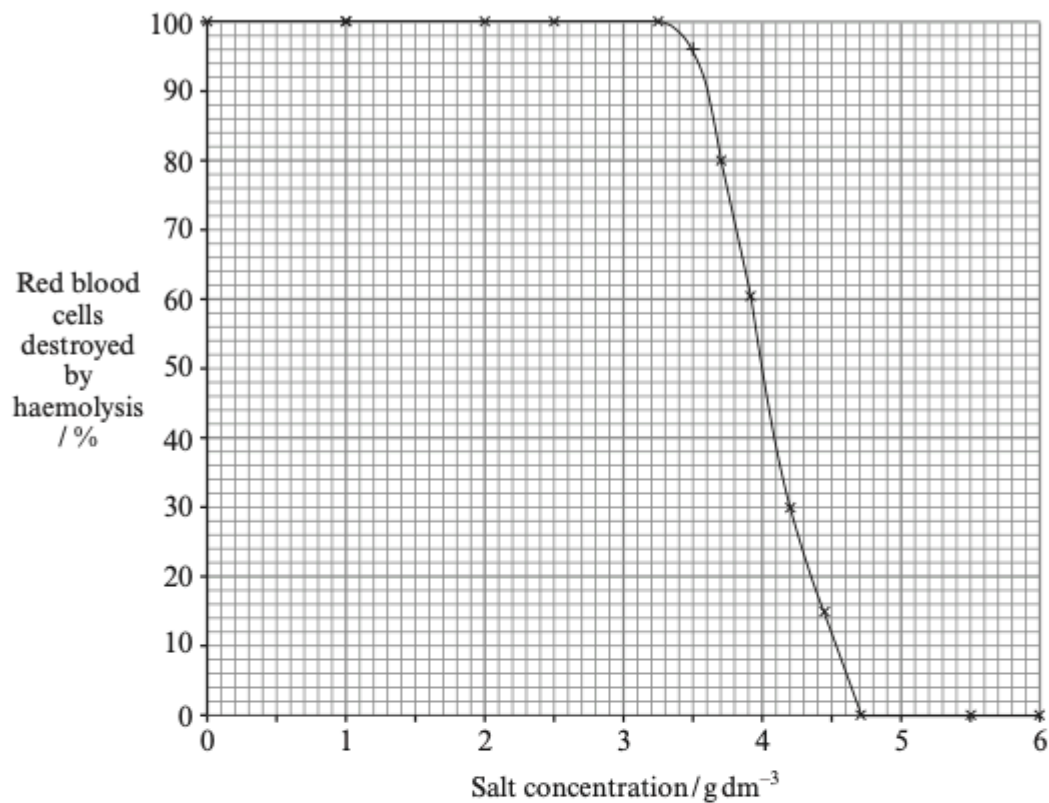
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- 7 (a) (i) Blood clots/infection. 1
- (ii) Water has highest water potential/0 compared with -476/-896kPa; 3
Water passes down water potential gradient/from high to low water potential;
Passes into cell by osmosis.
(not: ref. water concentration)
- (b) Diagram showing crinkled cells; (not: showing plant cell or nucleus) 3
Higher water potential inside cell;
Water passes out of cell (causing shrinkage/distortion).

7 MARKS

WJEC 2011

3. (a) An investigation was carried out in which red blood cells were placed in salt (sodium chloride) solutions of different concentrations. The percentage of cells destroyed by bursting (haemolysis) was recorded and the results shown in the graph below.



- (i) Explain, in terms of water potential, why red blood cells burst when placed in a solution that has a lower concentration than plasma. [3]

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- (ii) State the salt concentration at which the number of cells haemolysed is equal to that not haemolysed. [1]

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- (iii) The graph shows that haemolysis occurs between $3.3 - 4.7 \text{ g dm}^{-3}$ salt concentration. Suggest why there is a range. [2]

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- (b) An investigation was carried out on the uptake of potassium ions by root tissue. The root was cut into four discs of uniform size and each disc was added to an equal volume of a solution containing a fixed potassium ion concentration. The experiment was carried out in different oxygen concentrations and the results are shown in the table below.

Oxygen concentration / arbitrary units	0	4	11	20
Rate of potassium ion uptake / arbitrary units	7	27	92	100

- (i) State, with a reason, **one other** variable that should be kept constant. [2]

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- (ii) Using the information in the table, state with an explanation, the **main** method by which potassium ions are taken into the root. [3]

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- (c) State the rate of uptake you would expect if a drop of cyanide solution had been added to **each** of the four solutions. Explain your answer. [3]

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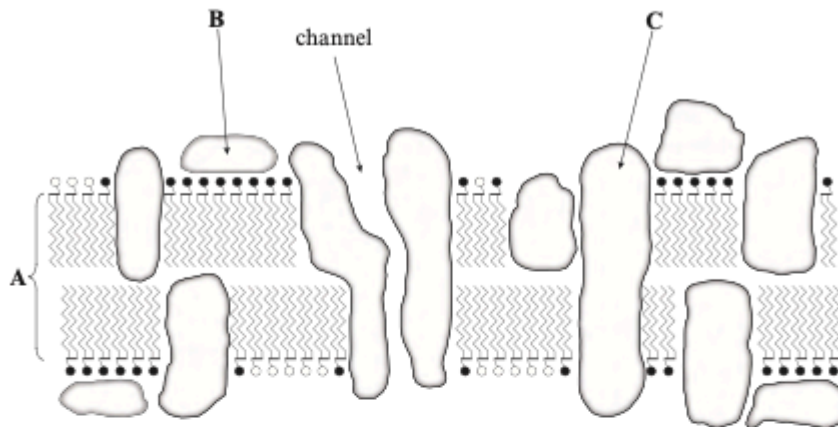
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(Total 14 marks)

- Q.3**
- (a) (i) higher water potential outside rbc/lower inside;
(not: ref solute concentration / ref water concentration)
water moves in by osmosis;
down water potential gradient;
ref. no cell wall to prevent bursting/cell
membrane unable to withstand pressure. [3]
- (ii) 4g dm^{-3} ; [1]
- (iii) different concentration/solute/ water potential of contents;
requires different concentration of external salts/water potential, for
movement of water/ to burst the cell [2]
- (b) (i) temperature/pH;
change enzyme activity/reaction rate/diffusion rate/respiration rate
(not: time/root/ref fair test) [2]
- (ii) active transport; [1]
is energy/ATP dependent;
aerobic respiration/oxygen required, to liberate energy/for ATP prod;
greater oxygen concentration produces greater uptake; [2]
- (c) * all would be 7au; [1]
cyanide inhibits aerobic respiration/ inhibits cytochrome oxidase/
stops/ reduces prevents ATP production;
when no oxygen is present there is still some uptake;
by diffusion;
which is a passive process; [2]
must have *. Plus 2 others

(Total 14 Marks)

3. (a) The diagram below is of a model of a section through a cell surface membrane, as proposed by Singer and Nicholson.



- (i) State the name given to this model and give reasons why it is so-called. [3]

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- (ii) Name the structures labelled A, B and C. [3]

A

B

C

- (iii) Describe the function of the channel shown in the diagram. [1]

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- (b) Some molecules are transported across the membrane by active transport. Explain what is meant by the term *active transport*. [2]

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- (c) Suggest **two** reasons why transport across the membrane is vital to the cell. [2]

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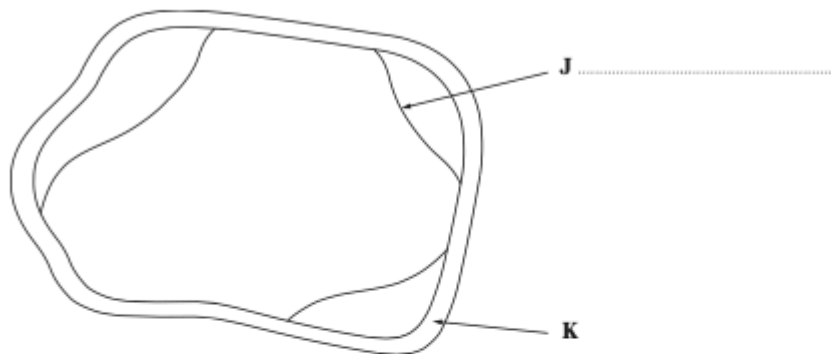
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(Total 11 Marks)

3. (a) (i) fluid mosaic model;
mosaic of protein molecules/irregularly or randomly
arranged;
lipid layer fluid/can move; **3**
- (ii) A = phospholipid bilayer/fatty acid tails; (not: ref.
hydrophobic)
B = extrinsic/surface protein/glycoprotein;
C = transmembrane/carrier/intrinsic protein; **3**
- (iii) allows passage of polar/charged/ionic/hydrophilic
molecules/facilitated diffusion; (allow: ref. water/non lipid
soluble; not: named molecule) **1**
- (b) movement up/against a concentration gradient;
requires energy/ATP; **2**
- (c) maintain water potential;
obtain nutrients/metabolites or named e.g. glucose;
obtain oxygen/remove carbon dioxide;
secrete molecules;
remove toxic substances or named;
(not: waste products) **2 max**
- (Total 11 marks)**

7. A student carried out an investigation on the solute potential of a plant tissue. The tissue was placed in a sucrose solution that had a water potential (Ψ) of -600 kPa and was left for one hour. The diagram below shows one cell after that time. Approximately 50% of the cells showed signs of plasmolysis, the other 50% did not.



(a) Label structure **J** on the diagram. [1]

(b) The student concluded that the solute potential of the cell contents was -600 kPa . Explain why you think the student reached that conclusion. [3]

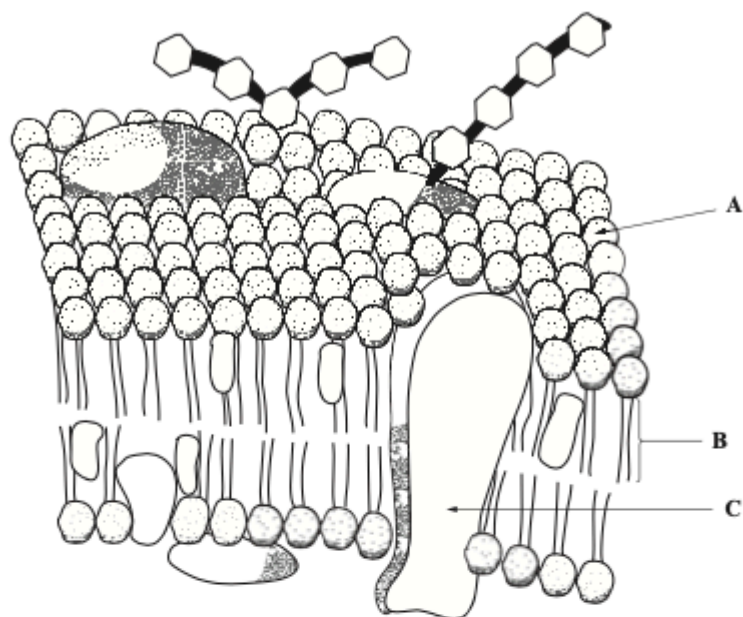
(c) Explain the role of structure **K** in generating pressure potential in the cell. [3]

(Total 7 Marks)

7. (a) cell/plasma membrane; 1
- (b) 50% of cells plasmolysed/point of incipient
plasmolysis/membrane just in contact with wall/at incipient
plasmolysis $\Psi P=0\text{KPa}$;
because cell left in solution for one hour;
equilibrium reached/no net movement of water;
solute potential inside equal to that outside;
outside solution is given as -600kPa ; 3 max
- (c) K is cell wall which is inelastic/won't stretch;
as protoplast/cell contents expand/swell; (not: ref. vacuole
unqualified)
as water passes into cell;
pushes against expanding protoplast/cytoplasm/cell
contents;
pressure potential is generated by resistance of cell wall; 3 max
- (Total 7 marks)**

WJEC 2012

7. The diagram shows the plasma membrane of an animal cell.



- (a) State the names of the structures labelled **A**, **B** and **C**.

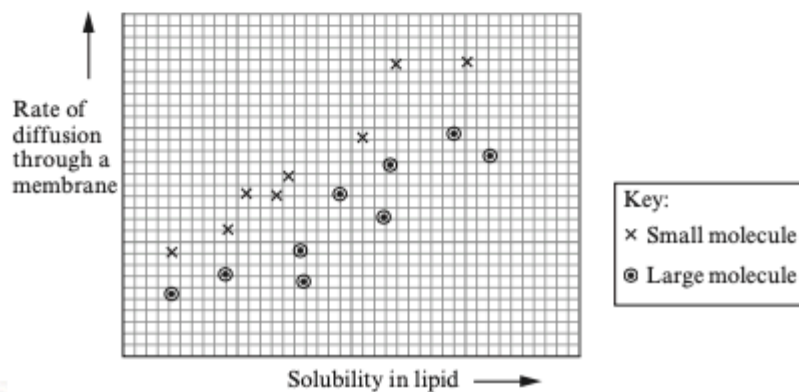
[3]

A

B

C

- (b) The graph shows the effect of molecule size and solubility in lipid on the rate of diffusion of substances through a cell surface membrane.



- (i) State with an explanation how the solubility in lipid affects the rate of diffusion through a membrane. [2]

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- (ii) Describe how molecular size affects the rate of diffusion. Suggest an explanation for your answer. [2]

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- (c) Name **two** factors which affect the rate of facilitated diffusion of a substance through a membrane. [2]

1.

2.

- (d) Vitamins B₁ and K enter cells by crossing the plasma membrane. As vitamin B₁ is water soluble while vitamin K is fat soluble they take different routes across the membrane. Explain how the different routes taken by these vitamins into a cell, is determined by the structure of the plasma membrane. [4]

vitamin B₁

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vitamin K

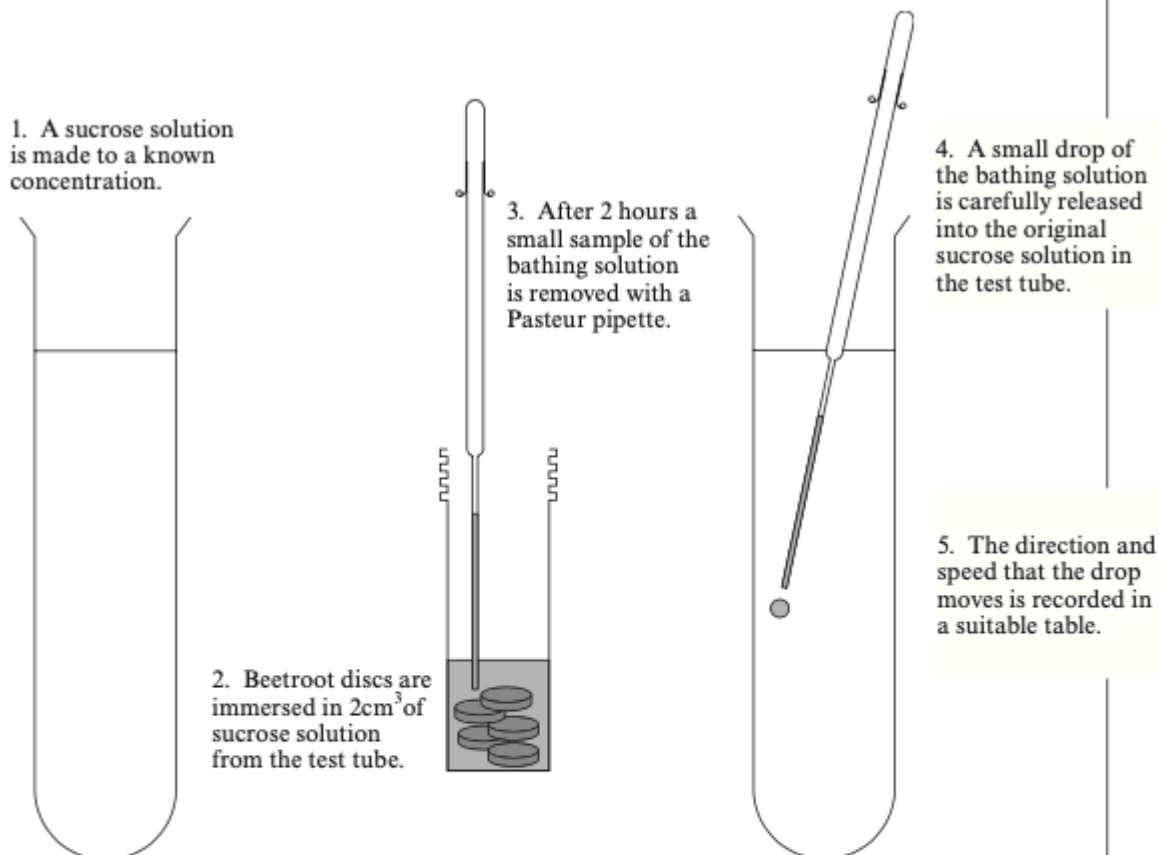
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(Total 13 marks)

7. (a)	A = phospholipid head/hydrophilic head/phosphate/polar group;	1
	B = hydrophobic tails/ fatty acids/ non polar tails: (Not: tails/ lipid layer)	1
	C = transmembrane protein/ carrier protein/ channel protein/ intrinsic protein.	1
(b) (i)	As lipid solubility increases the rate increases; NOT rate of reaction Membrane contains (a double layer) of phospholipids/ Lipid soluble substances can {move/pass/ diffuse} through the membrane (more easily than water soluble substances.) (any two)	2
(ii)	small molecules diffuse faster(or converse);	1
	Higher kinetic energy/ easier to pass between phospholipid molecules.	1
(c)	concentration/ diffusion gradient/ concentration difference; {amount/number} of carriers/ channel proteins/ larger surface area contains more carrier proteins; temperature. (any two)	2
(d)	vitamin B ₁ – polar/ ionic;	2
	Cannot pass through phospholipid layer/ hydrophobic region; Uses <u>protein</u> channels/ carriers/ transport <u>proteins</u> / Hydrophilic (lining to) channels; (any two)	
	Vitamin K – non polar/non ionic; dissolves in phospholipid/ hydrophobic regions; so can pass (directly) through phospholipid/ hydrophobic regions; (any two)	2

6. The diagram below summarises a technique used to measure the water potential of beetroot cells.



- (a) The table shows the results of an experiment carried out by some students.

Concentration of sucrose solution (M)	Direction droplet moved (number of arrows indicates speed of movement)
0.1	↓↓↓
0.2	↓↓
0.3	↓
0.4	↔
0.5	↑
0.6	↑↑
0.7	↑↑↑

- (i) According to these results which concentration of sucrose has the same water potential as the beetroot cells? [1]

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- (ii) Use the table below to find the water potential (Ψ_{cell}) of the beetroot cells.

Concentration of sucrose solution (M)	Solute potential, Ψ_s (kPa)
0.1	-269
0.2	-526
0.3	-790
0.4	-1052
0.5	-1322
0.6	-1596
0.7	-1882

water potential (Ψ_{cell}) of the beetroot cells = [1]

- (b) Explain why the drop of bathing solution rose in the 0.6 M solution. [4]

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- (c) The solute potential (Ψ_s) of the contents of the beetroot cells was known to be -1100kPa . Use the equation below to calculate the pressure potential (Ψ_p) of the beetroot cells when they were at equilibrium in the 0.3M sucrose bathing solution. Show your workings. [2]

$$\Psi_{\text{cell}} = \Psi_s + \Psi_p$$

- (d) (i) In the space below, draw a **labelled** diagram of a single beetroot cell from the 0.7M sucrose bathing solution, to show how it would have appeared under a light microscope. [3]

- (ii) What term is used to describe cells in this condition? [1]

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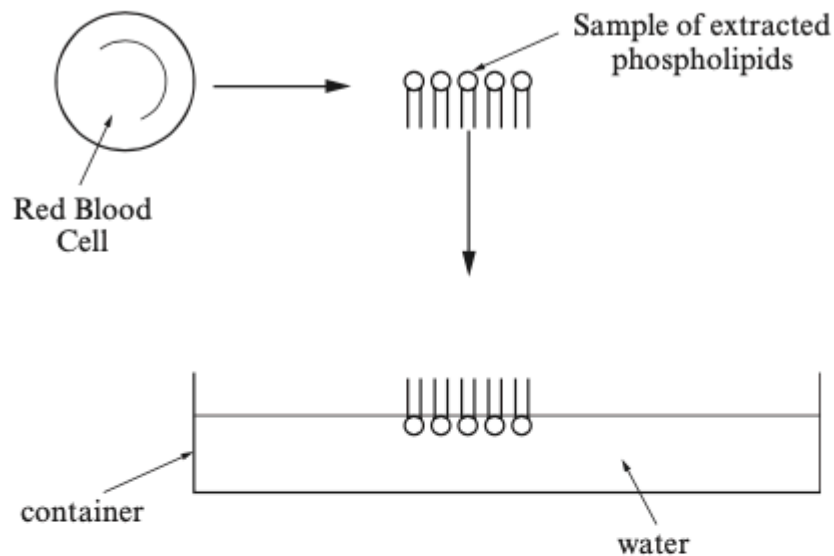
(Total 12 marks)

6. (a) (i) 0.4M; no units no marks. 1
- (ii) -1052 (kPa); 1
allow ECF
- (b) correct reference to osmosis; 4
bathing solution {has a lower water potential / is more concentrated /
is more negative / hypertonic} than the water potential of beetroot
cells / ORA;
water leaves / moved {out of / from} cells / into bathing solution;
bathing solution became less dense / lighter than original sucrose
solution;
REJECT reference to water moving into or out of the drop.
- (c) $-790 = -1100 + \Psi_p$; 2
 $\Psi_p = 310 \text{ kPa}$;
2 marks for correct answer.
- (d) (i) Diagram shows cell plasmolysed (any stage); 1
Mark diagram using labels.
No labels = 0 marks.
Any **2** correct labels from
cell wall; plasma / cell membrane (part or all of which must be away 2
from cell wall); tonoplast or vacuolar membrane; vacuole;
IGNORE incorrect labels.
- (ii) Plasmolysed / plasmolysis; 1

Question total 12

WJEC 2013

- (d) In 1925, two scientists, Gorter and Grendel investigated the arrangement of phospholipids in the plasma membrane. This involved the removal of the phospholipids from the surface membrane of all the red blood cells in 10cm^3 of blood. The phospholipids were then placed on the surface of water and allowed to spread out to form a single layer, called a monofilm.



- (i) Explain fully the arrangement of the phospholipid molecules as shown in the container on the diagram above. [2]

- (ii) The area covered by all the phospholipids in the monofilm was found to be 12.2m^2 . The total surface area of the intact red blood cells had been previously measured. Using your knowledge of membrane structure, what would you expect the total surface area of the red blood cells to be? Explain your answer. [2]

(Total 10 marks)

- (d) (i) {Heads/ phosphates} are {hydrophilic/ polar} and are {attracted to/ in} the water; 2
{Tails/ fatty acids} are {hydrophobic/ non polar} and are {repelled by/ above/ avoid} water;
NOT react/ dissolve with water
- (ii) $6.1(\text{m}^2)$; 2
The phospholipids are {arranged in/ formed} a {bilayer/ double layer} in the membrane;
Ref to phospholipid bilayer alone- insufficient

Question 4 Total

[10]

5. (a) Oxygen (an uncharged molecule) and phosphate ions are both required by plants. They enter the cells through the plasma membrane. Use your knowledge of the structure of the plasma membrane to explain how each of these molecules enter the cell.

(i) Oxygen:

[2]

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(ii) Phosphate ions:

[2]

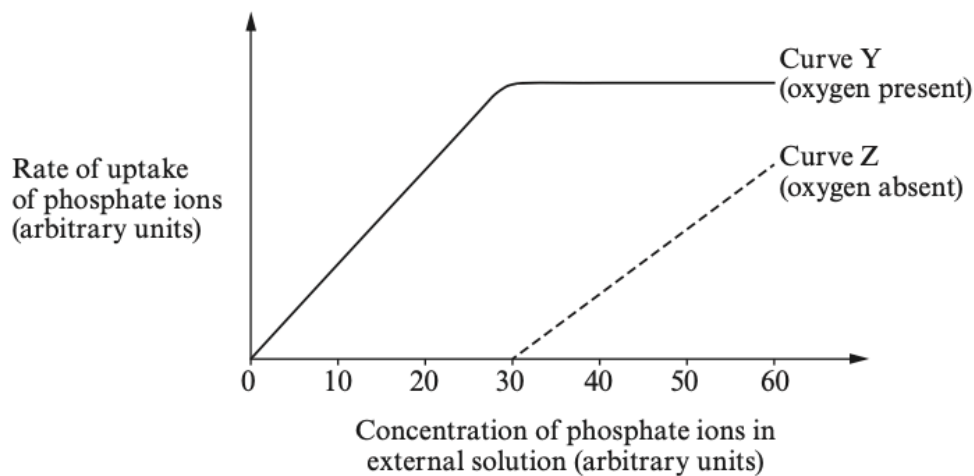
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- (b) Phosphate ions are taken up by specialised cells in the roots called root hair cells. The graph below shows the effect of the external concentration of phosphate ions on the rate of uptake of phosphate ions.



- (i) With reference to curve Y opposite, name the process that the cells use to uptake phosphate ions when the external concentration of phosphate ions is between 0 – 30 arbitrary units. Explain your answer. [3]

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- (ii) Explain the shape of curve Y between concentrations of 30 – 60 arbitrary units. [2]

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- (iii) Explain why the rate of uptake increases on curve Z between concentrations of 30 – 60 arbitrary units. [2]

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- (c) State **one** reason (other than as a component of phospholipids) why the plant needs phosphate ions. [1]

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(Total 12 marks)

- | | | | |
|----|-----|--|------------|
| 5. | (a) | (i) <i>Oxygen</i>
by (simple) diffusion;
through the phospholipid (bilayer); | 2 |
| | | (ii) <i>Phosphate ions</i>
by { <u>facilitated</u> diffusion/active transport};
through {carrier /channel}proteins/ <u>protein pumps (active transport)</u> ; (not channel proteins with active transport)
NOT intrinsic
Pass through hydrophilic pore; (not with active transport) | Max 2 |
| | (b) | (i) Active transport;
(Between 0-30au) the concentration of phosphate ions is lower outside (the root)/higher inside (the root)/ Ions are being taken up against a concentration gradient;
With oxygen present (aerobic) respiration can occur;
Providing {ATP/ energy} (for active transport)/ active transport needs {energy/ ATP}; | 1
Max 2 |
| | | (ii) There are a {limited/fixed} number of {carriers/ proteins/ channels} (for phosphate ions) in the membrane;
(The curve levels off/the rate of uptake becomes constant)
when all of the {carriers/ channels/ proteins} are in use; | 2 |
| | | (iii) (Ions are being taken up by) <u>facilitated</u> diffusion;
Uptake {only begins/ occurs} when the external concentration is <u>higher</u> than the concentration inside the root hair cells/ <u>down</u> a concentration gradient; | 2 |
| | (c) | They are a {component of/required to synthesise} {DNA/ RNA/ ATP/ NAD/ FAD/ NADP/ nucleotides/ nucleic acids}; | 1 |

Question 5 Total

[12]

- (b) Describe and explain the effects of placing animal and plant cells in solutions of differing solute concentration. [10]

(b) Describe the effects of placing animal and plant cells in solutions of differing solute concentration.

- A Osmosis is the (net) movement of water molecules down a water potential gradient/from a higher water potential to a lower water potential;
- B through a partially/selectively permeable membrane;
- C Hypotonic solutions have a higher water potential than the (cytoplasm of the) cells;
- D Water moves into the cells (by osmosis);
- E Animal cells swell /burst/ref osmotic lysis; reject turgid
- F Plant cells the cytoplasm swells up/cell contents/plasma membrane pushes against the cell wall;
- G (plant cells) becomes turgid/ $\psi_p > 0$ /cell wall prevents osmotic lysis;
- H Hypertonic solutions have a lower water potential than the (cytoplasm of the) cells;
- I Water moves out of the cells (by osmosis);
- J Animal cells shrink/crenated; reject flaccid
- K In plant cells the cytoplasm shrinks / the (plasma) membrane is pulled away from the cell wall;
- L Plant cell becomes plasmolysed/ $\psi_p = 0$;
- M Isotonic solutions have the same water potential as the cytoplasm of the cell;
- N (In isotonic solutions) there is no net movement of water molecules;
- O At incipient plasmolysis 50% of the cells in a plant tissue will be turgid and 50% will be plasmolysed;

- 5.
- Doxorubicin (DOX) and idarubicin (IDA) are antibiotics.
 - They are widely used in human cancer treatment.
 - DOX causes rapid changes in red blood cell membranes following injection.
 - These changes are
 - decreased fluidity of the hydrophobic parts of the lipid bilayer
 - the membrane proteins change shape.
 - IDA is considered to be less toxic to cancer patients than DOX.

(a) (i) Explain what is meant by the term 'lipid bilayer'. [1]

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(ii) Name the 'hydrophobic parts' referred to in the information above. [1]

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(iii) State **two** functions of membrane proteins. [2]

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(b) Use the information above to suggest why the changes in red blood cell membranes caused by DOX make it more toxic than IDA. [2]

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(c) These drugs are used in cancer treatment. Explain briefly what is meant by the term cancer. [2]

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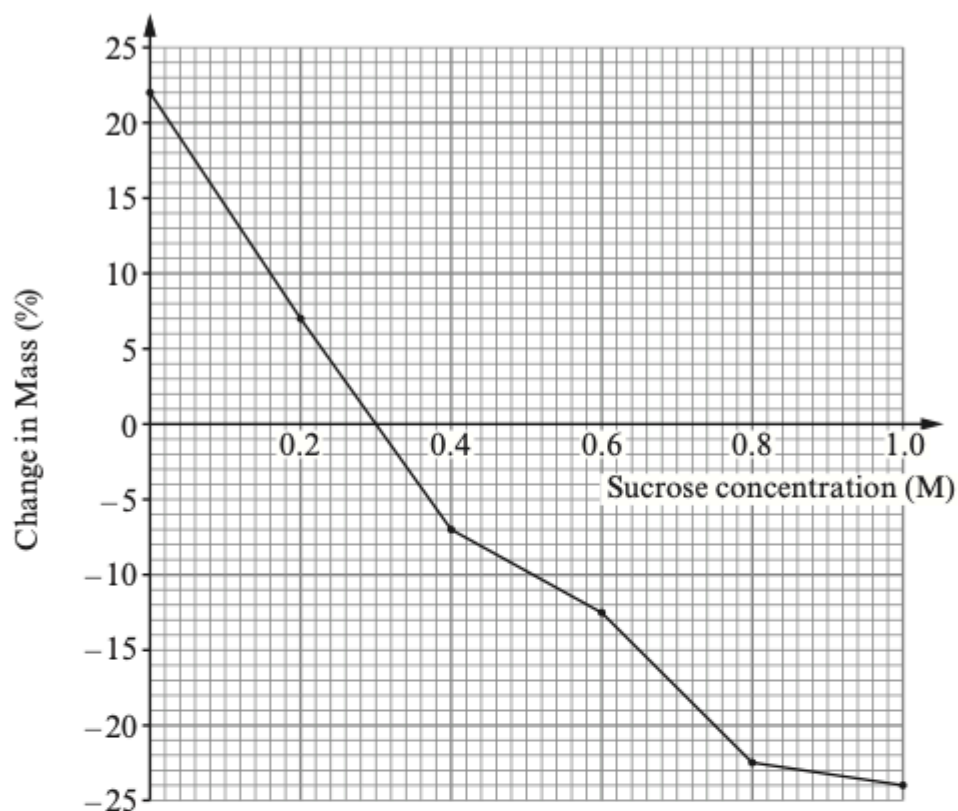
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(Total 8 marks)

5.	(a)	(i)	<u>two layers/ double layer of phospholipids;</u> NOT bilayer	1
		(ii)	<u>fatty acid;</u>	1
		(iii)	Any 2 from: transport/ form hydrophilic pores/ active transport/ channel proteins/ facilitated diffusion; receptors/ cell recognition; enzyme systems;	2 max
	(b)		Decreased fluidity/ rigid membrane - cells/ membranes more easily damaged (as blood flows)/ cannot pass through capillaries so easily;	2
			Membrane proteins change shape / denatured {carriers/ receptors/membrane enzymes} - so {reduced/no} {transport/movement} of molecules;	
	(c)		Any 2 from: {Unrestricted/ uncontrolled} {Cell division/mitosis}; Forming a mass of cells/ tumour; Preventing {normal cells/ organs} from functioning;	2 max
			Question 5 Total	[8]

7. An experiment was carried out to determine the water potential (Ψ_{cell}) of potato. A range of sucrose concentrations were prepared. Potato cylinders were weighed and then one was immersed into each of the solutions. After 2 hours they were blotted dry and reweighed. The percentage change in mass was calculated and the graph below plotted.



- (a) (i) Describe the changes in mass in 0.0M (distilled water) **and** 1.0M sucrose solution. [1]

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- (ii) What term is used to describe the appearance of the cells in 0.0M (distilled water)? [1]

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- (iii) Explain the mass change in the 1.0M sucrose solution in terms of water potential. [3]

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- (iv) What term is given to the solution which causes no change in mass? [1]

.....

Table to convert molarity to solute potential (kPa).

<i>Molarity of sucrose solution (M)</i>	<i>Solute potential kPa</i>
0.05	-130
0.10	-260
0.15	-410
0.20	-540
0.25	-680
0.30	-860
0.35	-970
0.40	-1120
0.45	-1280
0.50	-1450
0.55	-1620
0.60	-1800

- (v) Using the graph opposite and the table above determine the water potential (Ψ_{cell}) of potato tissue and **explain** how you arrived at your answer. [3]

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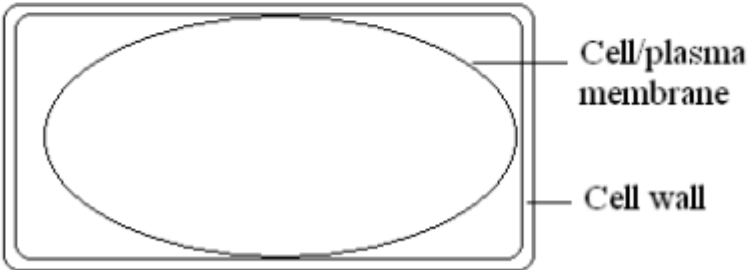
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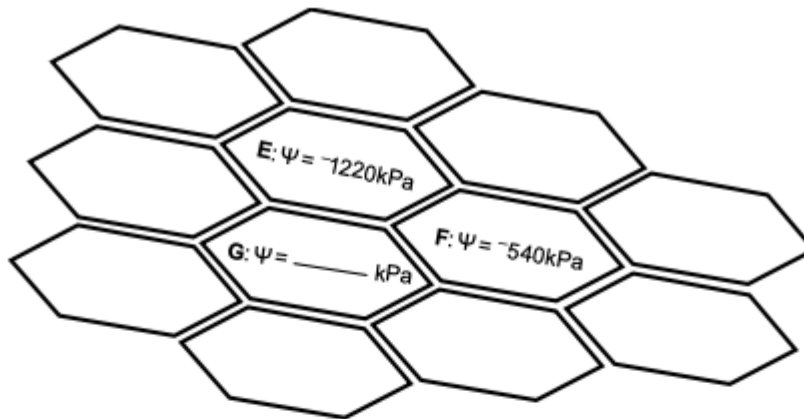
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- (b) Draw a labelled diagram of a cell as it would appear in the 1.0M solution. [2]

(Total 11 marks)

7.	(a)	(i) <u>{0.0M/distilled water} increase in mass and {1.0M/ sucrose solution} decrease in mass;</u> (ii) Turgid; (iii) Water moves out of the {cell/ potato}; By osmosis; The external solution has a {lower water potential than the cell/is hypertonic/ more negative}/ ORA ; Potato becomes flaccid/cells are plasmolysed; (iv) Isotonic; (v) 1. Where the line crosses the {X/ horizontal axis} there is no change in {mass/weight}; 2. So $\Psi_{\text{cell}} = \Psi_{\text{external}}$ solution (can be expressed in words); 3. This is <u>0.3(M)</u> sucrose; (must be linked to point 1 or 2) 4. And converts to -860kPa from the (conversion) table; 5. (So $\Psi_{\text{cell potato}} = -860\text{kPa}$;	1 1 3 max 1 3 max 2 [11]
	(b)	 • 1 mark for correct drawing of a plasmolysed plant cell(at any stage); (cell wall must be double line) • 1mark for correct labelling of a plasmolysed plant cell (plasma membrane pulled away from cell wall – both labelled correctly/ accurately); Question 7 Total	

7. The diagram below shows cells taken from plant tissue. Cells **E**, **F**, and **G** are adjacent cells and the water potential of cells **E** and **F** are given.



The water potential of a plant cell can be determined using the following formula:

$$\psi_{\text{cell}} = \psi_s + \psi_p$$

- (a) (i) The pressure potential of cell **G** is 900 kPa and the solute potential is -1600 kPa . Calculate the water potential of cell **G** and write your answer in the space on the diagram above. [1]

- (ii) I Draw arrows on the diagram above to show the net movement of water molecules between these **three** cells. [1]

- II Explain your answer in terms of water potential. [2]

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- (b) (i) In an experiment, a student immersed plant tissue in salt solutions of different concentrations. The student then observed the plant tissue under the microscope. In one of the solutions the student concluded that the cells within the tissue were at incipient plasmolysis.
What observation had the student made that allowed her to make this conclusion?
[1]

- (ii) The water potential of a cell at incipient plasmolysis was -430kPa .
Using this information state the value of the solute potential of the cell and explain how you arrived at your answer.
[2]

- 7 (a) (i) $-700(\text{kPa})$; 1
- (ii) I arrows drawn from F to G, F to E and from G to E; (allow ecf) 1
- II Water molecules move down a water potential gradient / from a{higher /less negative} water potential to a{lower /more negative} water potential; 2
By osmosis; (in correct context)
- (b) (i) 50% of the cells were plasmolysed; 1
- (ii) -430kPa ; 2
(At incipient plasmolysis) {the pressure potential equals zero/ the solute potential = water potential};

Question 7 Total [7]

- (b) Globular proteins are an important component of plasma membranes. Describe the structure and function of membrane proteins. [10]

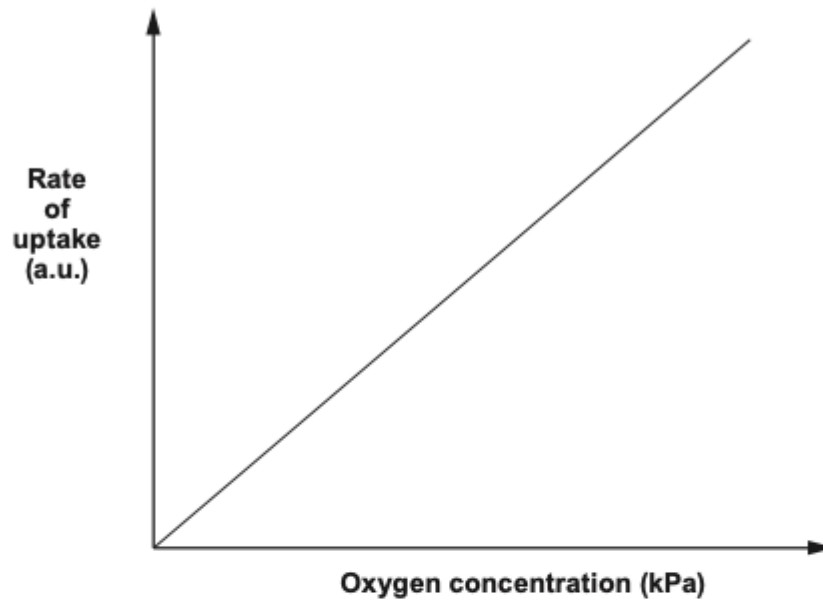
- 8 (b) A globular proteins show tertiary / quaternary structure;
- B they have a {specific/precise} 3D shape;
- C their shape is maintained by bonds between (atoms within the) R-groups;
- D disulphide bridges / ionic bonds / hydrogen bonds / Van der Waals forces / hydrophobic interactions; (any 2) NOT peptide
- E intrinsic proteins span the membrane;
- F extrinsic proteins are {embedded in one half of the membrane / on the surface of the membrane};
- G correct reference {made to the distribution of charge / polar and non-polar groups} on the {intrinsic/extrinsic} proteins;
- H channel proteins have a hydrophilic pore;
- I this allows {polar molecules/ions} to pass through the membrane;
- J by (facilitated) diffusion; NOT active transport
- K carrier proteins allow the passage of molecules {with a complementary shape/ by the protein changing shape};
- L by (facilitated diffusion and) active transport;
- M Glycoproteins contain a carbohydrate chain attached to a protein;
- N {Glycoproteins/ extrinsic proteins} act as hormone receptors / are involved in cell recognition;
- O enzymes may be located in the membrane / catalyse reactions / carry out digestion / synthesise ATP;

Question 7 Total

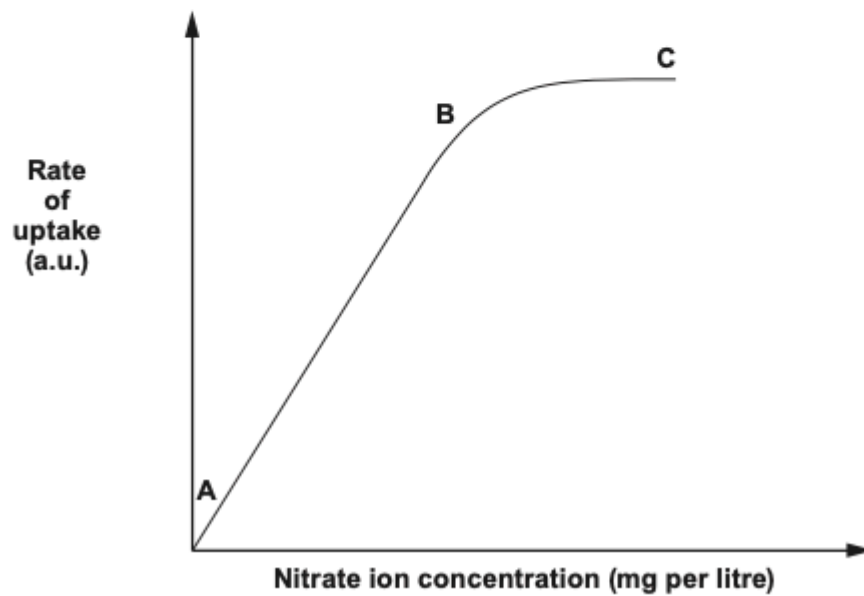
[10]

5. The graphs below show the uptake of different molecules into the roots of plants.

I. Oxygen



II. Nitrate ions



- (a) Using graph I, name the process by which oxygen is absorbed by the roots, giving a reason for your answer. [2]

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- (b) Explain why the rate of uptake of nitrate ions increases between points **A** and **B** shown on graph II. [1]

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- (c) In the presence of a respiratory inhibitor such as cyanide, the rate of nitrate uptake falls to zero. Name the process by which nitrate ions are taken up. [1]

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- (d) Water enters root hair cells by osmosis. Calculate the solute potential (Ψ_s) of the root hair cell, when there is no net movement of water, the water potential of the soil water is -100 kPa and the pressure potential (Ψ_p) inside the root hair cell is $+200\text{ kPa}$. Use the formula $\Psi = \Psi_s + \Psi_p$.

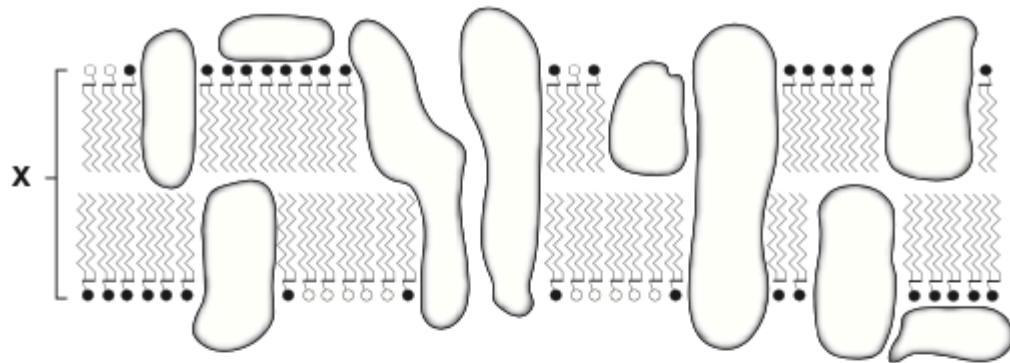
Show your working and units. [2]

Answer

5	(a)	(Method) Diffusion; (Reason) <u>Rate</u> is <u>proportional</u> to <u>concentration</u> ; NOT graph is proportional	2
	(b)	(Increasing ion concentration) increases chance that (a molecule will) {collide with/ pass through} {pump/carrier/protein};	1
	(c)	Active transport;	1
	(d)	$\Psi_s = \Psi - \Psi_p$ / i.e. -100 -200; -300 kPa; (Must have units) Correct answer + unit = 2 marks Correct answer + no unit = 1 mark	2

Question 5 Total [6]

6. The diagram below shows the fluid mosaic model proposed by Singer and Nicolson in 1972.



- (a) The width of the membrane as shown by **X** has been measured using transmission electron microscopes. Membrane width does not vary greatly between different organisms. State a value for this width. [1]

Membrane width =

- (b) Glucose is water soluble. Vitamin A is lipid soluble. Describe and explain how each molecule crosses the membrane shown above. [4]

Vitamin A

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Glucose

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(c) Beetroot vacuoles contain a red pigment called *betacyanin*. When beetroot discs are cut with a borer and immersed in a solution of 70% ethanol (an organic solvent) at 15°C, the red pigment begins to leak out of the cells into the ethanol turning it red.

(i) Using your knowledge of the structure of cell membranes, explain why this leakage of pigment occurs.

[2]

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(ii) When the experiment was repeated at 30°C, the time taken for the ethanol to turn red decreased. Explain why. [2]

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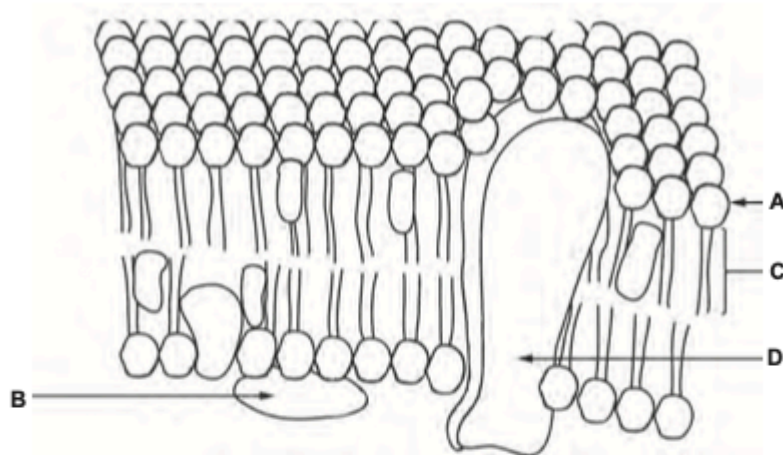
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- | | | | |
|---|-----|---|---|
| 6 | (a) | 7/8 nm (must have correct units);
Accept range of 6-10 | 1 |
| | (b) | Vitamin A -
{Dissolves in /can pass directly through} { <u>phospholipid</u> layer/
<u>hydrophobic</u> regions};
By diffusion;

Glucose –
Cannot pass through <u>phospholipid</u> layer therefore uses {protein
channels/ carriers/ transport proteins/ hydrophilic
channels/intrinsic proteins};
By facilitated diffusion; Accept active transport | 4 |
| | (c) | (i) Ethanol {dissolves/emulsifies} (phospho) <u>lipids</u> / denatures
protein; NOT cell membranes
Creates {gaps/holes/pores} in the membrane/ makes
membrane more porous;
NOT makes membrane more permeable | 2 |
| | | (ii) Increased temperature increases <u>kinetic</u> energy of
{dye/membrane} molecules;
Increases (rate of) <u>diffusion</u> (of dye across membrane)/dye
<u>diffuses</u> across the membrane more rapidly; | 2 |

Question 6 Total **[9]**

3. The diagram below shows part of the plasma membrane from an animal cell.



- (a) Complete the table below to identify the structures **A-D** shown in the diagram.

[4]

Structure	Name
A	
B	
C	
D	

- (b) Describe how structure **D** would transport a molecule against a concentration gradient.

[2]

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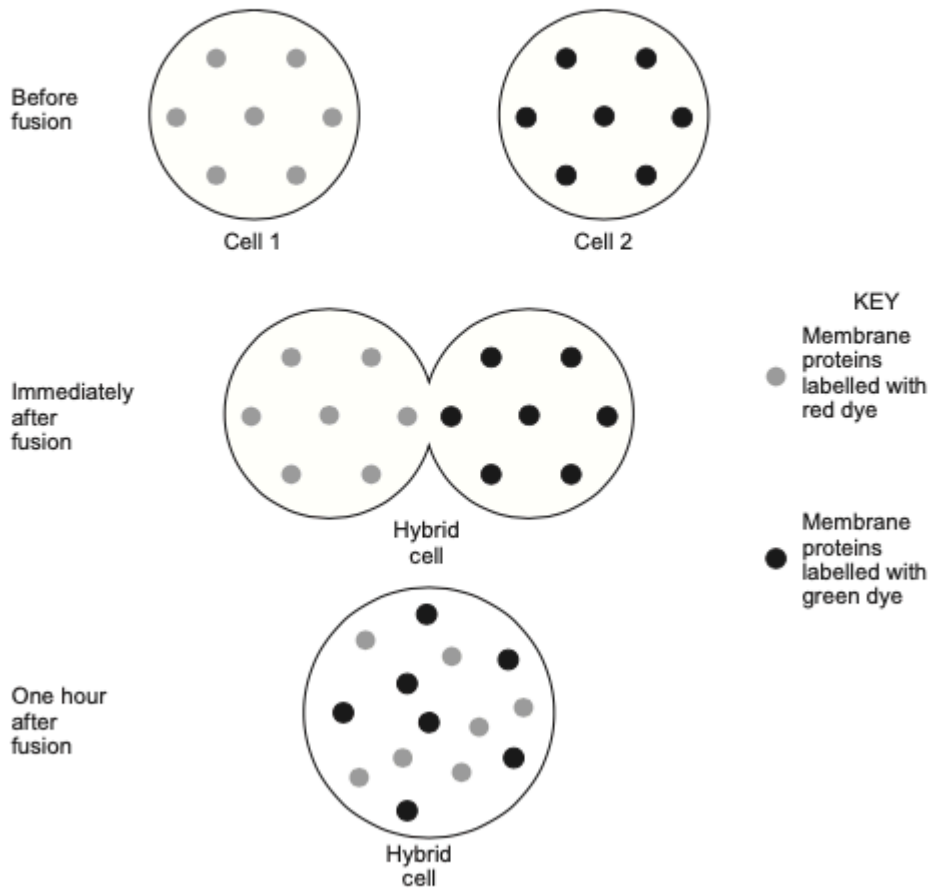
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- (c) In an experiment to determine the structure of the plasma membrane, scientists labelled the membrane proteins from two different cells using coloured dyes. One cell had its membrane proteins labelled with a red dye, whilst a second had its membrane proteins labelled with a green dye. The two cells were then fused to become a hybrid cell. This cell was viewed immediately after fusion and again after one hour. The results are shown below.



3	(a)	A {polar / hydrophilic / phosphate} {head / group}; NOT layer Glycerol = neutral B extrinsic protein; C {non-polar / hydrophobic} tails / fatty acids ; NOT lipid tail D transmembrane protein / carrier protein / channel protein / intrinsic protein;	4
	(b)	active transport / reference to {<u>carrier</u> protein / protein pump} changing shape; NOT channel protein NOT reference to more than one type of transport using ATP / energy;	2
	(c)	1. ref to fluid mosaic (model); 2. proteins free to move (within membrane / bilayer); NOT in the cell 3. {after fusion / 1 hour} there is a new arrangement of proteins / OWTTE;	3
		Question 3 total	[9]

5. (a) Define the term *water potential*.

[2]

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A student prepared a microscope slide containing a piece of red onion epidermis which had been soaked in a 0.4 M sucrose solution for 30 minutes. The slide was viewed under high-power and the following drawing was made.

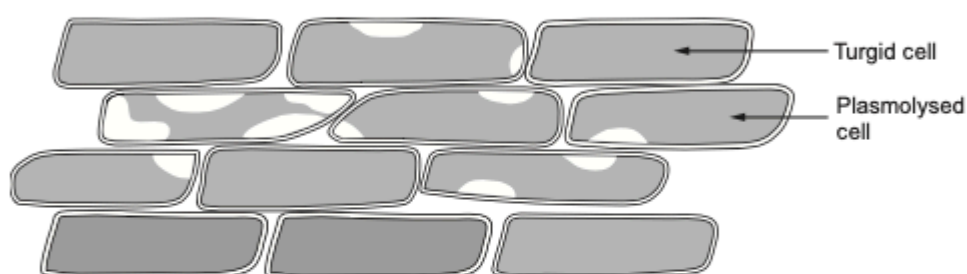


Table to convert molarity to solute potential (kPa).

Molarity of sucrose solution (M)	Solute potential kPa
0.05	-130
0.10	-260
0.15	-410
0.20	-540
0.25	-680
0.30	-860
0.35	-970
0.40	-1120
0.45	-1280
0.50	-1450
0.55	-1620
0.60	-1800

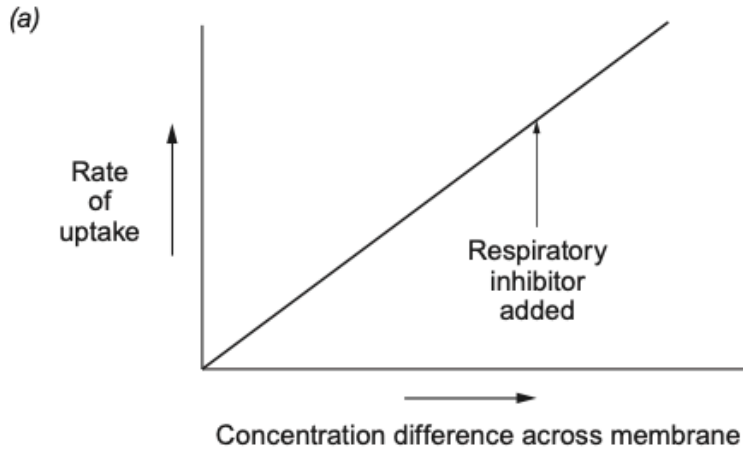
(b) The student concluded that the water potential of the epidermal tissue was -1120 kPa . Using all the information provided, explain why. [4]

(c) Explain the role of the cell wall in maintaining turgidity. [3]

5	(a)	The {potential / tendency / ability / capacity} for water to move out of* a {solution / system / cell}/ pressure exerted by water molecules in a {system / cell}; *Accept 'in to' by osmosis / correct definition of osmosis;	2
	(b)	Any 4 from 1. 6 cells turgid, 6 cells plasmolysed / {50%/half} of viewed cells were plasmolysed; 2. reference to incipient plasmolysis; 3. Pressure potential of tissue = zero; 4. No net movement of water; 5. solute potential = {water potential / -1120kPa}; Accept symbols 6. {so molarity of solution / 0.4M} = <u>molarity of tissue</u>;	Max 4
	(c)	Any 3 from 1. As water passes into cell (by osmosis); 2. As {protoplast / cell contents / cytoplasm / vacuole} {expand / swell}; 3. (cellulose) cell wall {is inelastic / won't stretch / won't expand}; rigid = neutral 4. (cell wall)pushes against (expanding) {protoplast / cytoplasm / cell contents};NOT cell wall pushes against {vacuole / cell membrane} 5. generating <u>pressure potential</u> (by resistance of cell wall);	Max 3
		Question 5 total	[9]

5. The following graphs show the effect of increasing the concentration gradient on the rate of uptake of substances across a cell membrane. The effect of adding a respiratory inhibitor on the rate of uptake is also shown.

For each graph, name the type of uptake involved and give reasons for your choice.



Type of uptake

[3]

Reasons for choice

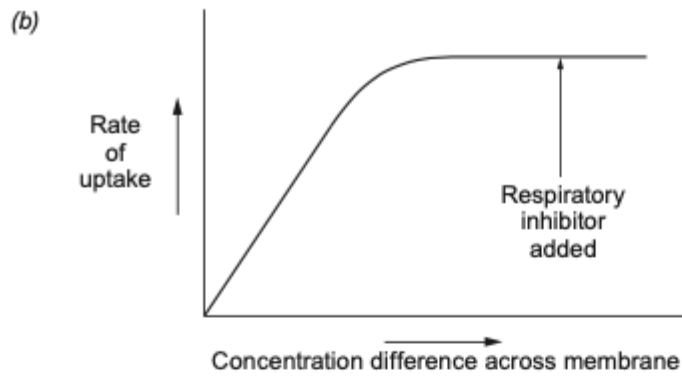
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Type of uptake

[3]

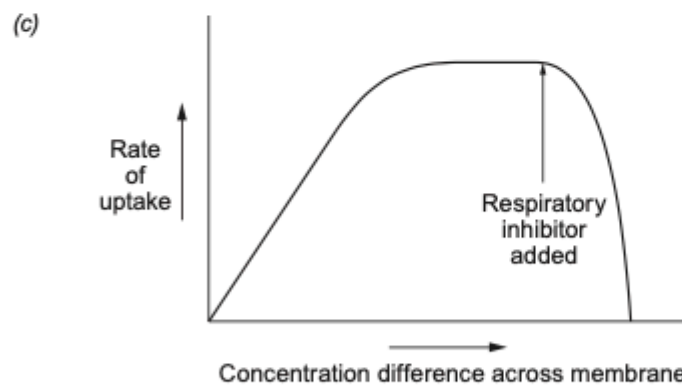
Reasons for choice

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Type of uptake

[3]

Reasons for choice

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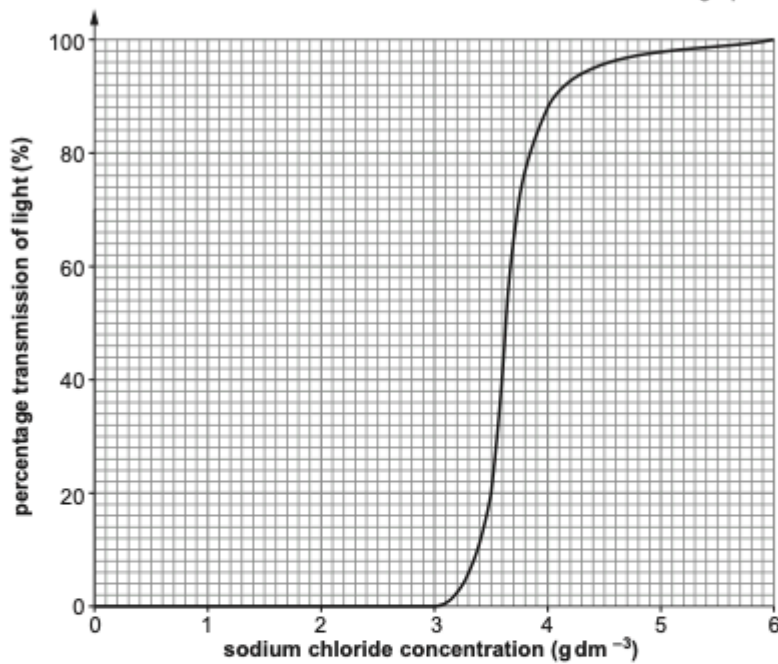
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5	(a)	1. Diffusion ; 2. As concentration (difference) increases rate of uptake increases ; 3. {Not limited by carriers/Not affected by respiratory inhibitors} because {No ATP / energy required / passive} ;	3
	(b)	1. Facilitated diffusion ; 2. At high concentration (differences), rate of uptake levels/ plateaus 3. {transport/carrier/channel} proteins / pores} {saturated / full/ are a limiting factor} ; 4. Not affected by respiratory inhibitors because {no ATP / energy required / passive} ;	Max 3
	(c)	1. Active transport ; 2. At high concentration (differences), {rate of uptake levels/ carriers are saturated} ; 3. {Affected / slowed} by respiratory inhibitors because process needs {ATP / energy};	Max 3
		Question 5 Total	[9]

6. (a) A student investigated the effect of sodium chloride concentration on the haemolysis of red blood cells. Red blood cells were mixed with a range of concentrations of sodium chloride solution for 30 minutes. The cell suspensions were then centrifuged to obtain cell free solutions. Any haemoglobin released from the cells would reduce the percentage transmission of light through the solution. Using a colorimeter, the percentage transmission was measured for each concentration. The results are shown in the graph below.



- (i) Explain, in terms of water potential, why the percentage transmission was lowest between sodium chloride concentrations of 0 and 3 g dm⁻³. [4]

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- (ii) The graph shows that as the concentration of sodium chloride increases from 3 to 6 g dm^{-3} the number of haemolysed cells decreases. Explain why there is a range of concentrations at which haemolysis occurs. [2]

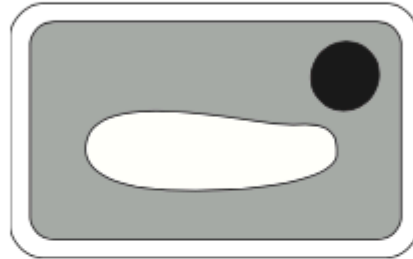
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- (b) The diagram below shows a plant cell that has been in pure water.



The cell was then placed in a concentrated salt solution for 30 minutes.

- (i) Describe the differences that you would now observe in this cell. [3]

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- (ii) State the pressure potential of the cell after it had been in the concentrated salt solution for 30 minutes. [1]

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6	(a)	(i) 1. water moves in by osmosis ; 2. {higher <u>water potential</u> outside rbc / lower inside} / down <u>water potential</u> gradient into the cell; Accept: correct ref to water potential less negative outside cell Reject: ref water concentration Neutral: ref to solute concentration / hypotonic/hypertonic 3. ref. no cell wall to prevent bursting / cell membrane unable to withstand pressure ; 4. Most haemoglobin is released therefore lowest transmission of light / less light reaches sensor; (ii) Different cells have different {(solute) concentrations / solute potential / water potential}; Reject: water concentration Each cell would require a different {external water potential / solute concentration}, before haemolysis / bursting ;	4
	(b)	(i) Any 3 from 1. cell plasmolysis ; 2. cytoplasm decreases in volume ; 3. cell membrane pulls away from cell wall ; 4. vacuole decreases in volume ; ii 0 <u>kPa</u> Question 6 Total	3 1 [10]

1.4

WJEC 2009

- (b) One species of bacterium *Streptococcus mutans* produces the enzyme glucosyl transferase that catalyses the polymerisation of glucose molecules. Polymers of glucose form part of the plaque on teeth which leads to dental disease.

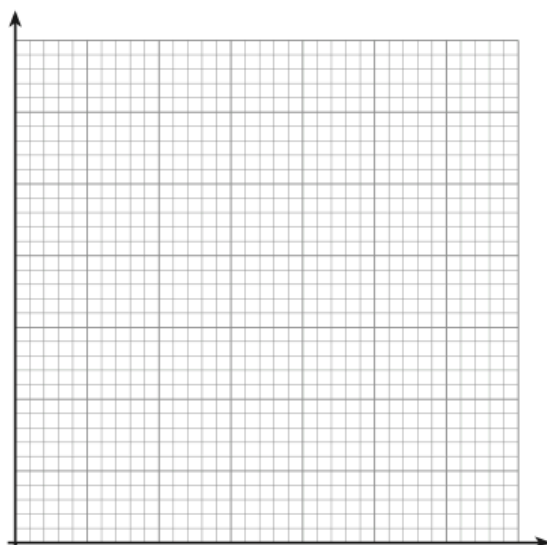
(i) What is polymerisation? [1]

- (ii) Two scientists, Devulapalle and Mooser carried out experiments using the enzyme glucosyl transferase and found that its activity was affected by the presence of iron sulphate. They published the following results.

<i>Iron II sulphate concentration (mM)</i>	<i>glucosyl transferase activity (labelled product μmoles)</i>
0.0	60.0
1.0	25.2
2.0	12.0
3.0	5.2
4.0	2.4
5.0	1.6
6.0	1.2

(*J Dent Res* 80: 466-469, 2001)

Plot the data as a suitable graph on the grid provided. [3]



- (iii) Use the graph to calculate the concentration of iron sulphate that would decrease the activity of the enzyme by 50%. [1]

.....

- (iv) Calculate the percentage by which a 3mM iron sulphate solution would decrease the activity of the enzyme. Show your workings. [2]

Answer

- (v) Devulapalle and Mooser discovered that the iron sulphate acted as a competitive inhibitor. Describe the mechanism by which competitive inhibitors decrease the activity of enzymes. [3]

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- (vi) Suggest a possible use for the discoveries made by Devulapalle and Mooser. [1]

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(Total 17 marks)

6. (a) (i) Showing, 1 O and 2 H s removed.
Elimination of water, stated.
Molecules joined by oxygen bridge. [3]
- (ii) Maltose (not: disaccharide). [1]
- (iii) Water. [1]
- (iv) Condensation. [1]
- (b) (i) Joining together sub units / monomers /repeating units/ residues
(to make a larger molecule)
(not: joining molecules into a chain/ specific example) [1]
- (ii) Correct axes – iron sulphate concentrate on horizontal, both labelled
and units given.

Suitable scale using at least half available space;

plots visible and clear line correct shape. [3]
(not: extrapolation/line of best fit)
- (iii) 0.9mM (allow: between 0.7 and 0.9mM). [1]
- (iv) $60 - 5.2 = 54.8 / 60 \times 100 = 91.3(\%)$ (allow: 91)
(2 for correct answer 1 for correct working but wrong answer.) [2]
- (v) Inhibitor competes with substrate (to bind with active site);

inhibitor binds to/fits into active site;

with inhibitor bound substrate is unable to bind/less E-S complexes;

inhibitor same/complementary shape as substrate;

the greater the concentration of substrate the less inhibition / ra / owtte

(Any 3) [3]
- (vi) (Add iron sulphate to toothpaste / mouthwash / sugary drinks.)
to prevent formation of plaque / tooth decay. [1]

[Total 17 marks]

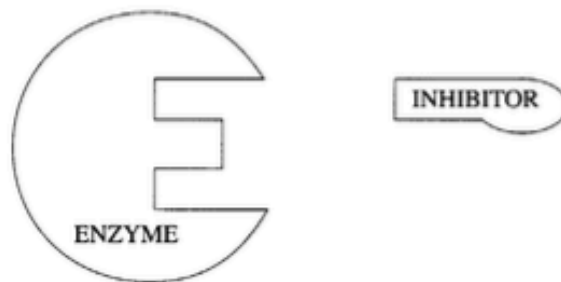
- Or** (b) What are immobilised enzymes? Describe the advantages of their use and how they are used in biosensors. [10]

- (b)
- A. Enzyme molecules that are fixed / bound / Trapped (not: immobilised/do not move) [1]
 - B. to an inert Matrix/alginate bead. [1]
 - C. They are more stable at higher temperatures (therefore reaction rates may be faster by using higher temps.) [1]
 - D. They can tolerate wider range of pH. [1]
 - E. They are more easily recovered for re-use/separated from product. [1]
 - F. Several enzymes with different pH or temp. optima may be used at one time. [1]
 - G. Reaction can be more easily controlled by adding or removing enzymes. [1]
 - H. They are specific so can select one type of molecule in a mixture. [1]
 - I. So can be used for rapid detection of biologically important molecules. [1]
 - J. They can also accurately measure the quantities present / are sensitive. [1]
 - K. Used in medical diagnosis / named condition eg diabetes. [1]
 - L. And environmental monitoring, [1]
 - M. Description of mechanism, some use a transducer to generate an electrical impulse that can be measured with a meter. [1]
 - N. eg. Blood sugar meter as used by diabetics / AVP. [1]
 - O. AVP/ easier to make pure product (not contaminated by enzyme.)

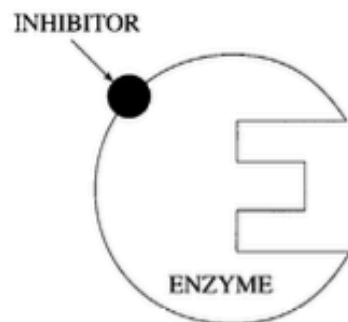
[Total 10 marks]

6. (a) Diagrams A and B represent two different types of enzyme inhibition.

A.



B.



- (i) State the type of inhibition shown in A and B.

[2]

A

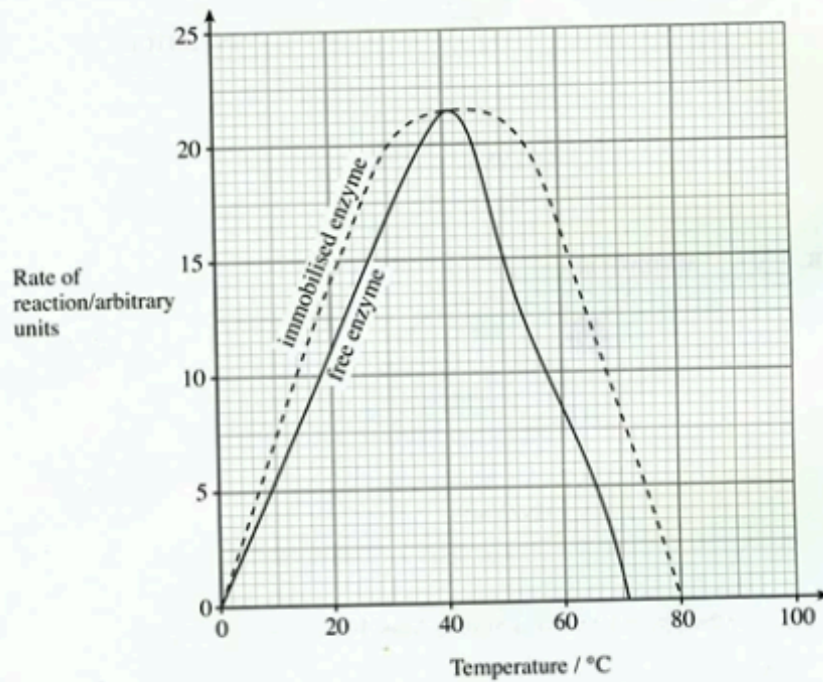
B

- (ii) What type of inhibition, A or B, would be **decreased** by increasing the concentration of substrate?

[1]

.....

- (b) Immobilised enzymes are enzyme molecules that are trapped on an inert matrix such as a gel capsule.
The graph shows the effect of temperature on the maximum rate of reaction of the **same** enzyme in its free and in its immobilised state.



- (i) Explain the rate of reaction at 5°C and 70°C for the **free** enzyme.

[2]

5°C

.....

70°C

.....

(ii) Describe **three** differences between the effects of temperature on the immobilised and the 'free' enzyme. [3]

1.

.....

2.

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3.

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(iii) Suggest how trapping the enzyme to an inert matrix can explain the differences you have described in part (b) (ii). [1]

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(iv) Describe **one** use of immobilised enzymes in medicine. [1]

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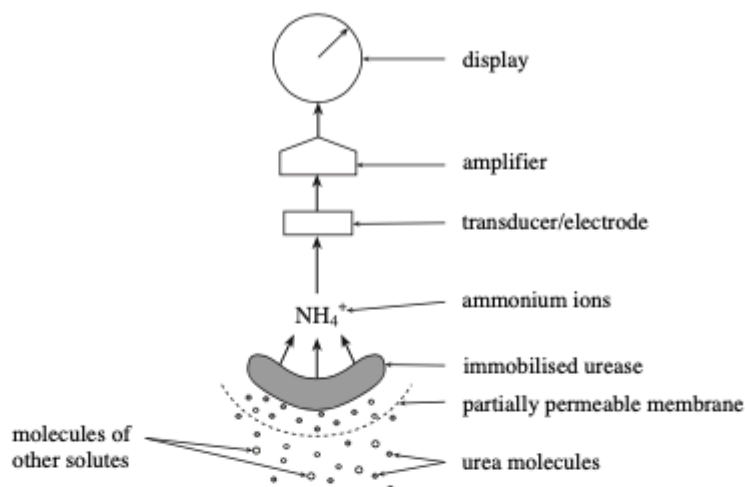
(Total 10 marks)

6. (a) (i) A competitive [1]
 B non competitive [1]
 (ii) A [1]
- (b) (i) 5°C kinetic energy is low / few collisions between the (active site) of the enzyme and the substrate; [1]
 Allow: ref. to increasing temp and kinetic energy i.e. assume 0 to 5°
 70°C the hydrogen bonds are broken (as vibrations are strong) /
active site of the enzyme is denatured/ fewer ES complexes formed due to denaturation. [1]
- (ii) A. activity of immobilised enzyme is greater between 0°C and 40°C or at lower temperatures/ rate of reaction greater
 B. optimum temperature of IE covers a wider range / 40°C - 50°C
 C. above 40°C the free enzyme begins to denature whereas the IE starts to denature at 50°C
 D. IE is more active at all temperatures except 40°C
 E. free enzyme is (completely) denatured at 70°C IE is completely denatured at 80°C
 (any three) [3]
- (iii) The shape of the enzyme / 3-D structure is maintained or it is stabilised – molecular movement is 'reduced' [1]
 (not: shielded/protected enzyme)
- (iv) Detection of blood sugar / testing blood sugar (in diabetics) [1]
 Allow : lactose free milk/Clinistix qual
 (not: diabetics/ biosensor)

[Total 10 marks]

WJEC 2010

3. The diagram below shows a biosensor that uses the enzyme urease to measure urea in either the blood or urine.



- (a) (i) Name **one** method that could be used to immobilise the urease. [1]

- (ii) State **three** advantages of using immobilised enzymes. [3]

- (b) (i) Describe the function of the partially permeable membrane in this biosensor. [2]

- (ii) With reference to the diagram, describe the role of this transducer. [2]

- (c) If the biosensor was used to test two blood samples, explain why the temperature of the two samples should be the same. [2]

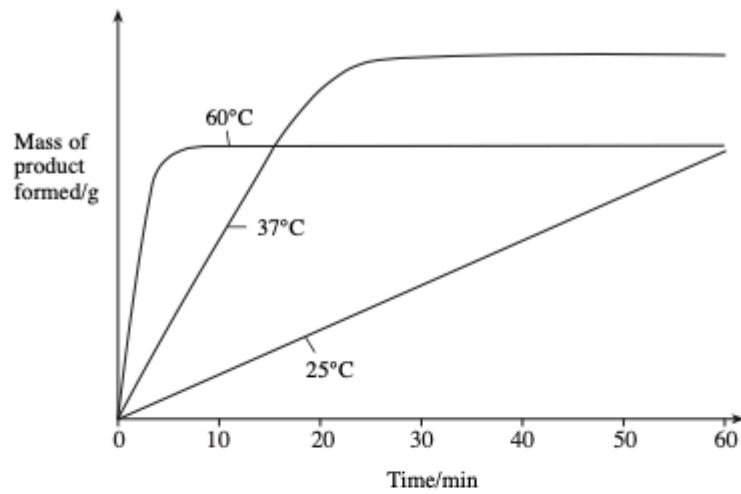
- (d) Name a medical condition which a biosensor can detect. [1]

(Total 11 marks)

3. (a) (i) alginate beads / gel membrane / meshwork of inert material /
cellulose (not: entrapment unqualified) [1]
- (ii) product easily recovered/not contaminated by enzyme;
so cheaper to use;
greater stability;
despite variations in/higher temperature / pH;
enzyme easily removed / added;
can control rate.
more than one enzyme can be used [3]
- (b) (i) allows urea to pass through;
prevents passage of blood cells / other molecules/solutes;
so they can't affect results / enzyme / reduce enzyme activity. [2 max]
- (ii) absorb/ref. ammonium ions;
converts into an electrical signal / changes chemical to electrical signal;
to record levels of urea. [2]
- (c) increased temperature increases enzyme activity/rate of reaction;
more ammonium ions formed;
greater electrical current generated;
reference fair testing. [2 max]
- (d) diabetes. [1]

[Total 11 marks]

3. The graph shows the mass of product formed when a fixed concentration of enzyme is added to a fixed concentration of substrate and then placed in water baths at three different temperatures.



- (a) (i) Explain why the mass of product formed at 60°C is greater during the first five minutes than the masses formed at 25°C and 37°C. [3]

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- (ii) Explain why there is less overall product formed at 60°C than at 37°C. [3]

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- (b) Explain why the mass of product formed at 37°C levels off after approximately 20 minutes. [1]

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(c) Explain why the curve at 25°C has not levelled off after 60 minutes. [3]

(Total 10 marks)

- | | | | | |
|---|-----|------|--|---------|
| 3 | (a) | (i) | At higher temperature/60° enzyme/substrate has more kinetic energy/vibrates more; (not: ref. movement)

More Enzyme substrate complexes formed/ more <u>successful</u> collisions;

More product formed/greater rate of reaction. | 3 |
| | | (ii) | At 60°C enzyme reacts rapidly;

(Gradual) denaturation of enzyme occurs or description;

All substrate not reacted; | 3 |
| | (b) | | All substrate converted to product.

(not: active sites full) | 1 |
| | (c) | | Lower temperature, less kinetic energy/fewer vibrations;

Fewer enzyme substrate complexes formed/fewer <u>successful</u> collisions;

Some substrate remains after 60 minutes; (not: reaction has not ended)

Maximum product formation not yet achieved. | (3 max) |

10 MARKS

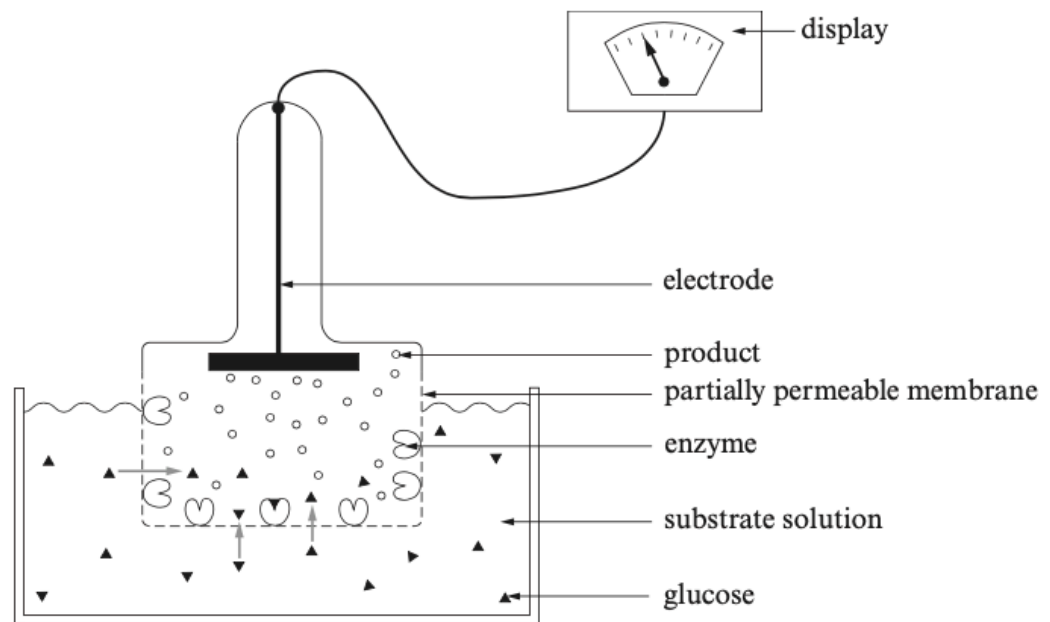
WJEC 2011

- (b) (i) Describe how inhibitors affect the rate of an enzyme catalysed reaction. [7]
- (ii) Describe the advantages of using immobilised enzymes in industrial processes. [3]

- (b) (i) A. two types, competitive and non-competitive;
- B. both types of inhibitors reduce rate of reaction;
- C. competitive inhibitor complementary to active site / structurally similar to substrate;
- D. competes with substrate for active site of enzyme;
- E. blocks active site/prevents substrate from binding to active site
- F. fewer/ no enzyme substrate complexes formed;
- G. increase substrate concentration reduces effect of inhibitor;
- H. non-competitive binds away from active site/ binds at allosteric site
- I. changes shape/conformation of enzyme molecule;
- J. shape/conformation of active site changed;
- K. increasing substrate concentration has no effect on rate of reaction [7]
- (ii) L. enzymes tolerate wider range of conditions/temp/pH/thermostable/ Owtte (not: stable unqual)
- M. enzyme easily reused;
- N. several enzymes can be used together;
- O. product not contaminated / easier purification of product
- P. greater central of reaction achieved/ enzymes easily added or Removed qual. [3]

(Total 10 Marks)

2. The diagram below shows a possible structure for a biosensor that uses enzymes to detect glucose.



- (a) Immobilised enzymes that are used in biosensors must have certain properties. Suggest **two** of these properties.

[2]

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- (b) Explain the function of the partially permeable membrane.

[2]

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- (c) Describe how the concentration of glucose is transmitted to the display.

[2]

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- (d) Explain why the temperature of the biosensor should be kept constant when using it to measure the concentration of glucose in different solutions.

[2]

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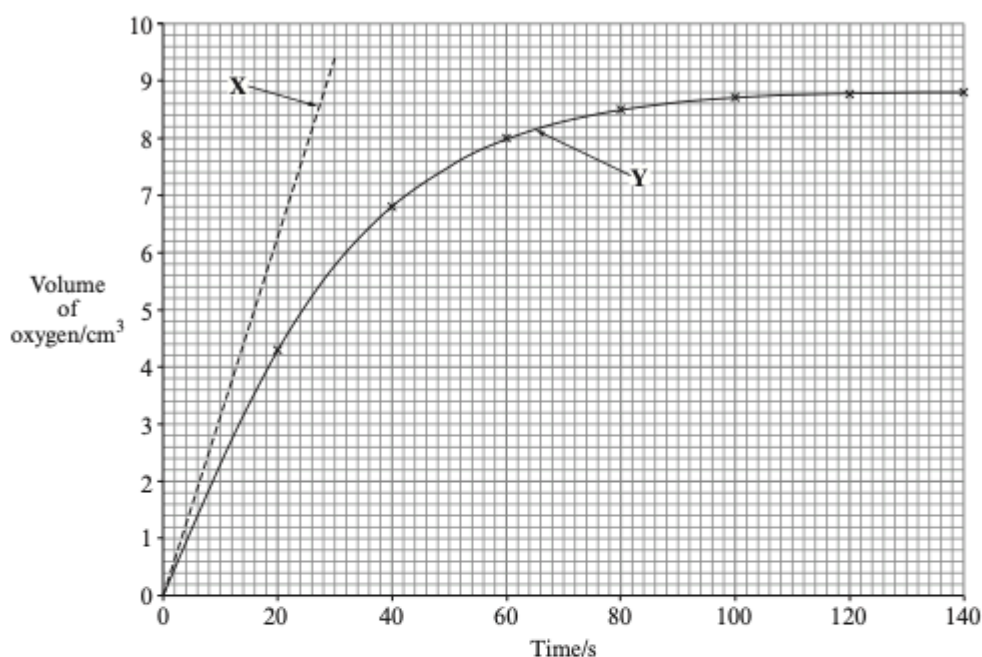
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(Total 8 Marks)

2. (a) capable of immobilisation/fixed to inert matrix or named;
Stable/able to withstand changes in temperature or pH;
specific to test or substrate;
(not: ref. turn over number) **2 max**
- (b) allows glucose through; (not: ref. small molecules)
prevents passage of other molecules/solutes
(not: ref. substances) **2**
- (c) glucose broken down by enzyme;
Products/oxygen affect/detected by electrode;
(not: measured by)
electric signal generated/chemical to electrical;
greater conc. glucose the greater the signal; **2 max**
- (d) enzyme activity/ rate of diffusion of glucose affected;
change rate of reaction;
unreliable result;
(not: ref. enzyme denaturation/fair experiment/control/false
reading/confidence) **2 max**

(Total 8 marks)

4. A student investigated the action of the enzyme catalase. This enzyme catalyses the breakdown of hydrogen peroxide into oxygen and water. The student collected the oxygen given off in a measuring cylinder. The volume of gas was recorded every 20 seconds as shown on the graph labelled Y below.



- (a) The rate of reaction can be calculated using the formula:

$$\frac{\text{Volume of oxygen collected}}{\text{Time taken to collect}}$$

Use the formula to calculate the rate in $\text{cm}^3 \text{min}^{-1}$ for the first 30 seconds.

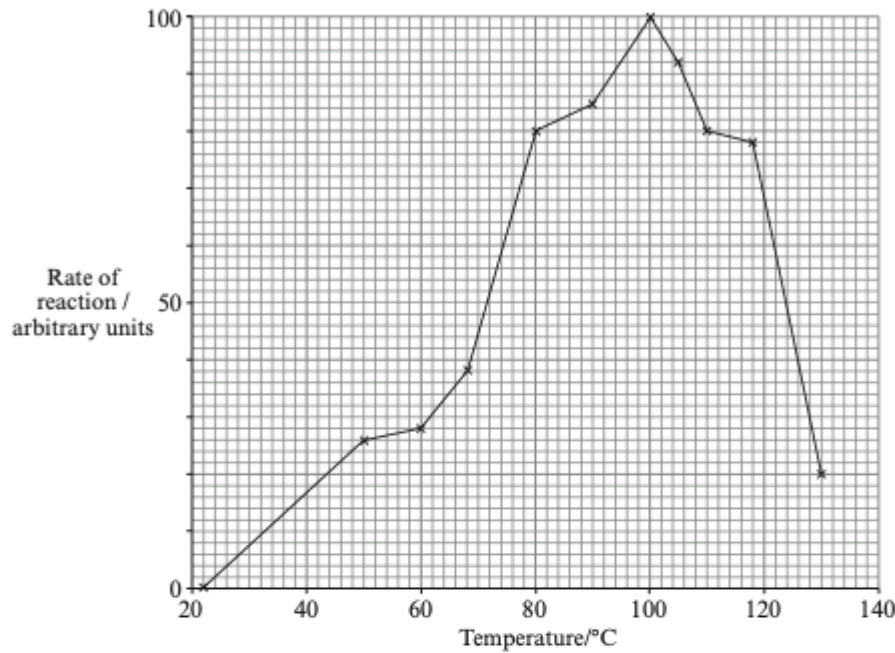
[2]

- (b) The initial rate is the rate of reaction at the beginning and is the maximum rate. It is shown by line X. The initial rate is $19 \text{ cm}^3 \text{min}^{-1}$.

Explain why the initial rate is greater than the rate calculated in (a).

[2]

- (c) The graph below shows the effect of temperature on the activity of an amylase enzyme found in bacteria that live in hot water in volcanic regions.



- (i) Using the graph, describe and explain the effect of temperature on the rate of activity of the amylase. [6]

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- (ii) State the difference between bacterial amylase and an amylase found in humans. [2]

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(Total 12 Marks)

4. (a)

11.6 cm³ min⁻¹;
(allow: 5.8/30 x 60) correct answer + units =2;
correct answer - units =1; incorrect answer, correct working
= 1

2

(b) Maximum/higher concentration of substrate;
all active sites occupied;
(not: ref. unoccupied at start)

2

(c) (i) increase in rate from 20 - 100°C/up to 100°C;
fall from 100 - 130°C;
increase in kinetic energy;
molecules move faster; (not: more)
More successful collisions/more enzyme-substrate
complexes formed;
up to optimum; (not: 100°C unqualified)
above optimum increased vibrations;
hydrogen bonds break;
Loss/change of shape of active site; (not: ref. enzyme)
denature;

6 max

(ii) enzymes have different optimum temperatures/
human amylase has optimum of 37°C, bacterial 100°C;
human amylase denatures at a lower temperature;

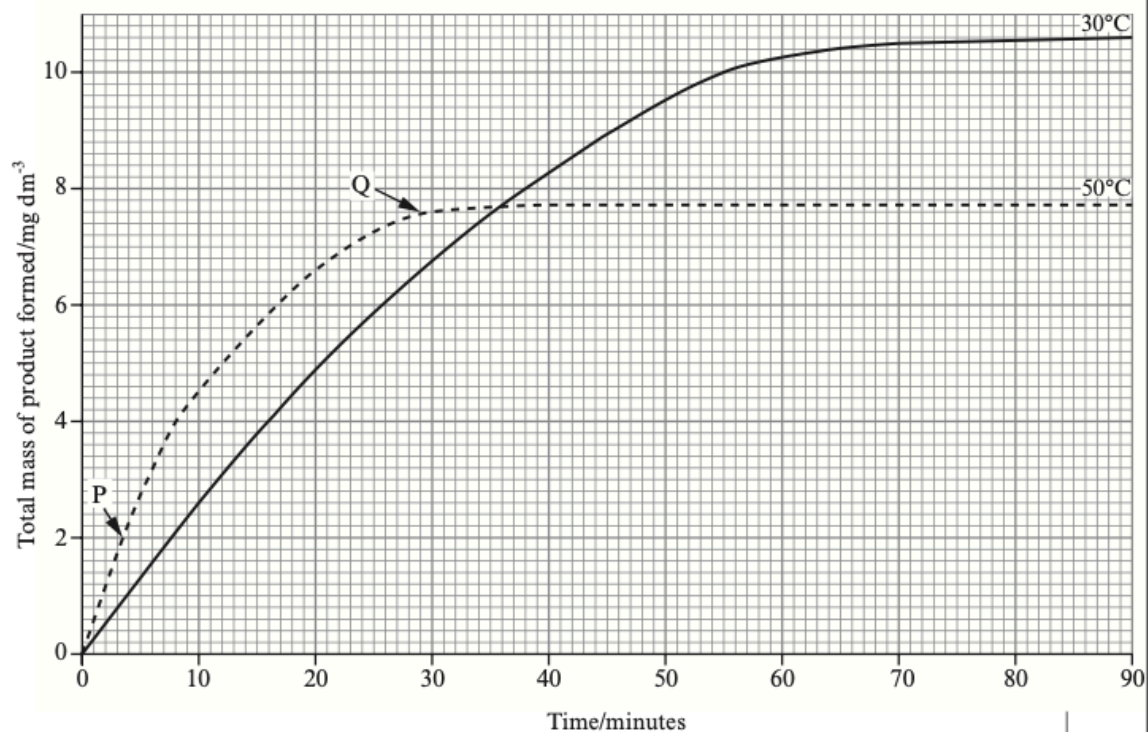
1

1

(Total 12
marks)

WJEC 2012

4. An investigation was carried out to determine the mass of product formed in an enzyme-controlled reaction at two different temperatures, with an excess concentration of substrate. The results are shown in the graph.



- (a) (i) Calculate the rate of reaction in the first 10 minutes at 30°C. [1]

Rate = $\text{mg dm}^{-3} \text{ min}^{-1}$

- (ii) State the factor which determines the rate of reaction between points P and Q on the graph. [1]

- (b) (i) Explain why the initial rate of reaction was slower at 30°C than at 50°C. [2]

- (ii) Explain the shape of the curve between 30 minutes and 60 minutes at 50°C. [3]

- (c) The investigation was repeated at 30°C with the addition of a competitive inhibitor.

- (i) Draw the expected curve on the graph. [1]

- (ii) Explain how a competitive inhibitor would bring about this effect. [3]

4. (a) (i)	0.26	1
(ii)	concentration of substrate NOT amount/ availability of Active sites	1
(b) (i)	Less/low <u>kinetic</u> energy ;	1
	fewer successful collisions/ {enzyme substrate/ES} complexes formed/ ORA	1
(ii)	enzymes denatured/ alteration in tertiary structure/ 3D structure; breaking of H/ hydrogen bonds; NOT disulphide active site altered/ active site denatured; substrate cannot bind/ less enzyme substrate complexes formed (any three)	3
(c) (i)	Must be a curve starting at origin and may meet 30°C line but not levelling off	1
(ii)	{shape/structure} of inhibitor similar to substrate/complementary to active site; {Fits/ fills/ bonds/ attaches} to active site/ competes for active site; (As it has a similar shape to the substrate it competes for the active site = 2 marks.) <u>At higher substrate concentration</u> there is a greater chance of Enzyme substrate complexes forming / effect of inhibitor is diminished/ the substrate outcompetes the inhibitor/ ORA	3

(Total 11 marks)

- (a) Describe the biological principles involved in the use of immobilised enzymes including the detection of blood sugar using biosensors. [10]

8. (a)
- A. enzyme (molecules) {fixed/ bound/ trapped} in an {inert support/ matrix}
 - B alginate beads/ gel membrane, /adsorbed (NOT absorbed) onto nylon/ gel capsule/ cellulose
 - C Product not contaminated
 - D reuse of enzymes/recovery/ easily separated.
 - E stable/ tolerate wider range of conditions
 - F for example pH, temperature/ higher temperatures than normal/ denatured at higher temperatures
 - G several enzymes can be used together/ with differing pH or temperature optima.
 - H rapid/ greater productivity

Biosensors

- I accurate/ specific
- J detect/sensitive to low concentrations/ clinistix
- K used in diagnosis of diabetics/ diabetes
- L {Biosensor/electrode probe} has a specific enzyme immobilised in a membrane/ glucose oxidase in context
- M glucose diffuses into the immobilised enzyme layer/ through selectively permeable membrane
- N (enzyme together with transducer) produces an electrical signal in response to substrate transformation/ chemical to electrical signal
- O size of signal proportional to concentration of product/ substrate

(Any 10 out of 15 points)

- (a) Some cells produce and secrete digestive enzymes. Describe the part played by each of the organelles involved in the production and secretion of the enzymes. [10]

7. (a)	{ A	Nucleus;	1
	{ B	Contains DNA code for amino acid sequence;	1
	{	NOT genetic information alone;	
	{ C	Carries out transcription / makes RNA copy;	1
	{ D	Nucleolus;	1
	{ E	Makes ribosomes / organises transcription / makes rRNA;	1
	F	{ <u>Rough</u> ER / Ribosomes} { translate mRNA / put amino acids together / protein synthesis};	1
	{ G	Endoplasmic reticulum;	1
	{ H	Transports protein;	1
	{ I	(To) Golgi;	1
	{ J	Packages protein into vesicle;	1
	{ K	Modifies protein or description;	1
	{ L	<u>Secretory</u> vesicle;	1
	{ M	Vesicle migrates towards plasma membrane; (can award M and N if use vesicle instead of secretory vesicle)	1
	N	Vesicle fuses / merges with plasma membrane;	1
	O	Contents of vesicle emptied by <u>exocytosis</u> ;	1

Question total 10

- (b) Describe and explain how different factors affect the rate of enzyme reactions. (Effects of enzyme inhibitors are not required.)

7. (b)	A	Temperature;	1
	B	description of (exponential) increase to optimum / maximum / certain temperature then (sudden) decline / sketch graph showing;	1
	C	Increasing temperature increases rate because of increased energy / moving molecules faster / kinetic energy / ORA;	1
	D	{Increasing frequency of / more / more likely} <u>successful</u> collisions / Enzyme Substrate Complexes forming / ORA;	1
	E	pH;	1
	F	description of optimum pH and declining activity further from optimum in both directions / sketch graph / optimum pH and narrow range;	1

(Award G, H, I, J in context for Temp and/or pH)

{	G	(3D) <u>shape</u> of <u>active site</u> changes;	1
	H	Changing away from optimum affects bonds holding <u>tertiary</u> structure / structure of enzyme molecules;	1
	I	Correct reference to hydrogen / covalent / ionic bonds; NOT disulphide / peptide	1
	J	Substrates do not fit into active site / is not complementary (so rate reduced);	1
	K	Substrate concentration; NOT amount;	1
	L	Enzyme concentration; NOT amount;	1
		(Award M,N, O in context for Enzyme conc and/or Substrate conc)	
{	M	Activity increases up to maximum when it levels off / sketch graph showing / ORA;	1
	N	Increasing substrate / enzyme conc. increases number of active sites occupied / Enzyme Substrate complexes / successful collisions / ORA;	1
	O	Maximum rate when <u>all</u> active sites <u>occupied</u> / <u>saturated</u> correct reference to limiting factors;	1

WJEC 2013

Either, (a) Describe and explain the effect of inhibitors on enzyme action.

[10] |

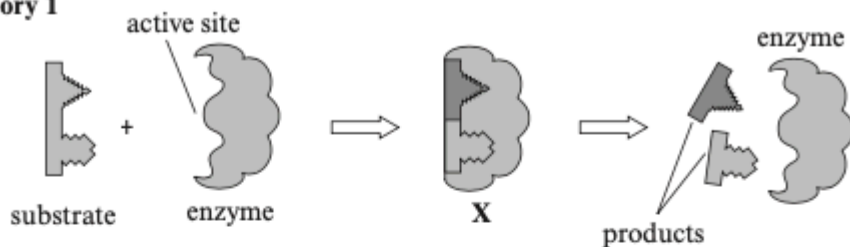
7. (a) **Describe and explain the effect of inhibitors on enzyme action.**

[10]

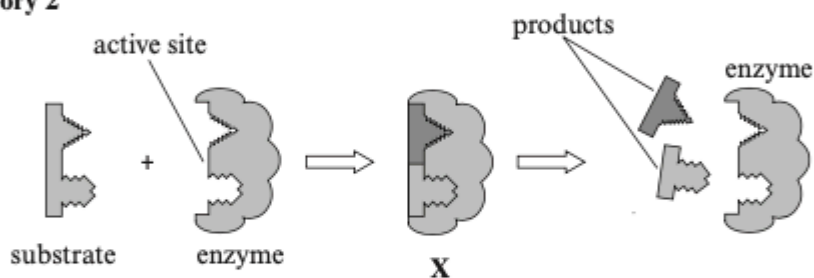
- | | | | |
|-----------------|---|---|---|
| enzymes | { | A | Enzymes are globular proteins/ <u>biological</u> catalysts; |
| | | B | Active site (of the enzyme) has a specific 3D/ tertiary shape; |
| | | C | lower activation energy of a reaction; |
| | | D | Inhibitors reduce the rate of (an enzyme catalysed) reaction; |
| competitive | { | E | Competitive inhibitors; |
| | | F | Have a shape similar to the substrate/complementary to the active site; NOT same shape |
| | | G | Fit/ bind into the active site; |
| | | H | Prevent the substrate molecule entering the active site/block the active site; |
| | | I | Max. rate of reaction can be achieved at higher substrate concentrations/ Increasing the concentration of the substrate reduces the effect of the inhibitor; allow correctly labelled graph |
| non-competitive | { | J | Non-competitive inhibitors; |
| | | K | Bind to the allosteric site/site other than the active site; |
| | | L | Causes a change in the shape of the <u>active site</u> ; |
| | | M | Substrate can no longer fit into the <u>active site/ active site is no longer complementary</u> ; |
| | | N | Fewer/ no enzyme-substrate complexes form/ fewer successful collisions; |
| | | O | Max. rate of reaction cannot be achieved/increasing the concentration of the substrate has no effect on inhibition; allow correctly labelled graph |

2. The diagram below shows two theories used to explain enzyme activity.

Theory 1



Theory 2



- (a) (i) **Theory 1** shows the induced fit hypothesis. What name is given to **Theory 2**? [1]

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- (ii) Which theory represents the activity of lysozyme? [1]

.....

- (b) Name **X** as shown in both theories. [1]

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- (c) Enzymes are biological catalysts. How do they bring about their effect of speeding up a reaction? [1]

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(d) What characteristic of an enzyme at the **end of a reaction** is visible in both diagrams? [1]

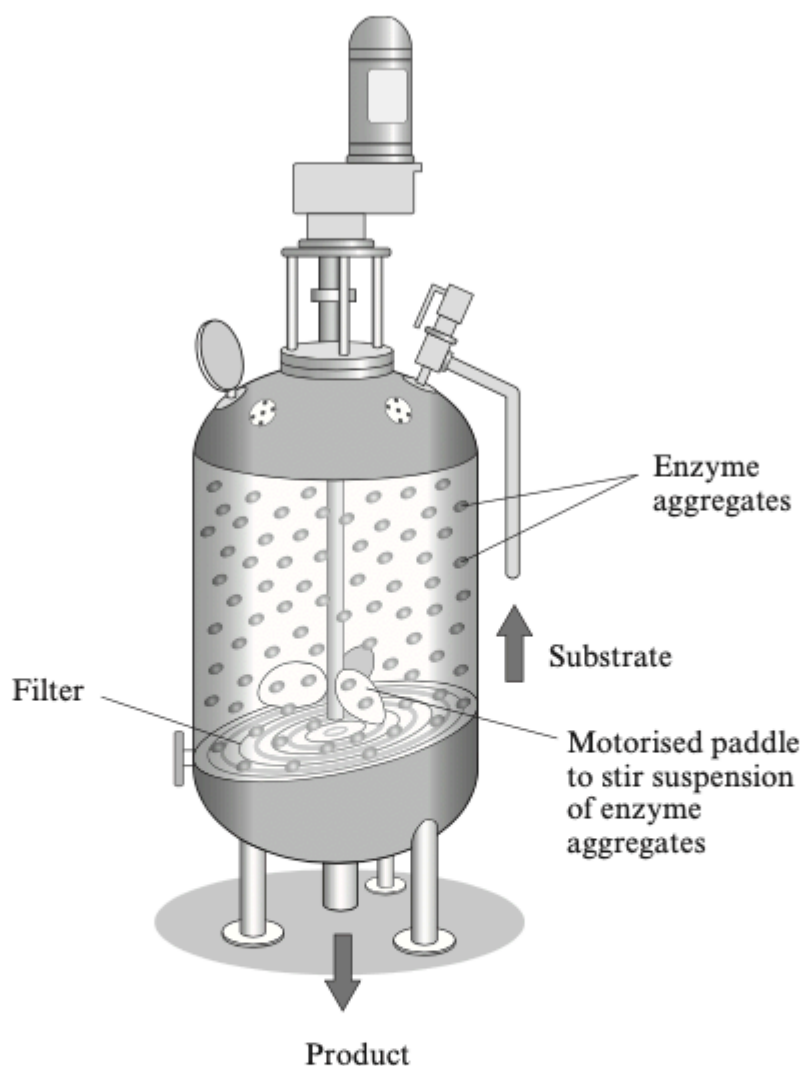
(e) State **three** factors which affect enzyme activity, excluding the presence of inhibitors. [3]

(f) Distinguish between intracellular and extracellular enzymes. [1]

(Total 9 marks)

2.	(a)	(i)	Lock and key;	1
		(ii)	Theory 1/ induced fit;	1
	(b)		Enzyme substrate complex; NOT ESC/ ES complex	1
	(c)		Lower the <u>activation</u> energy/eq;	1
	(d)		Enzyme/ active site is unchanged/can be re-used; NOT active sites are a specific shape unqualified	1
	(e)		Temperature (not heat); pH; NOT acidity Enzyme concentration; Substrate concentration; NOT amount	3
	(f)		Intracellular: inside the <u>cell</u> + Extracellular:outside the <u>cell</u> ; NOT inside body	1
			Question 2 total	[9]

6. Immobilised enzymes are prepared for industrial use in a number of ways. In the vessel shown below, the enzymes have been formed into clumps called enzyme aggregates. These are held together by cross-linking without altering their tertiary structure. They are permanently insoluble but maintain their catalytic activity.



- (a) Why is it important that the tertiary structure of these enzymes is not altered by the cross-linking? [2]

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(b) Using your own knowledge and the diagram opposite, explain why it is necessary for these enzyme aggregates to be insoluble. [2]

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(c) State **three** advantages in using immobilised enzymes in industry. [3]

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(d) Name another method of immobilising enzymes, other than cross-linking. [1]

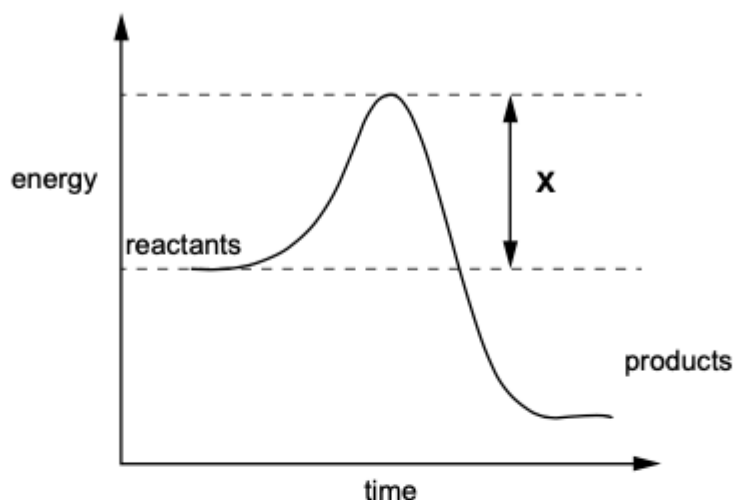
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(Total 8 marks)

6.	(a)	Causes change in <u>shape</u> of enzyme/active site; So substrate no longer fits into active site; {No/ fewer} enzyme substrate complexes;	2 max
	(b)	{{(Insoluble) enzymes/ (enzyme) aggregates} cannot pass through the filter/ ORA; So the product is uncontaminated with enzymes/ ORA;	2
	(c)	Can tolerate { <u>higher</u> temperatures/greater <u>range</u> of pHs}; NOT range of temperatures Easily <u>recovered</u> for reuse/ enzymes stay in aggregates/ reused qualified/ uncontaminated product/ separated from product; NOT reused unqualified/ enzymes reused Several enzymes can be used together; Easy addition/removal of enzymes;	3 max
	(d)	Any one from : Gel capsule/alginate beads/ gel beads; cellulose fibres; gel membrane; porous glass beads; NOT inert matrix unqualified/ encapsulation unqualified	1 max
Question 6 Total			[8]

WJEC 2014

4. The graph below shows the energy changes that take place during a chemical reaction.



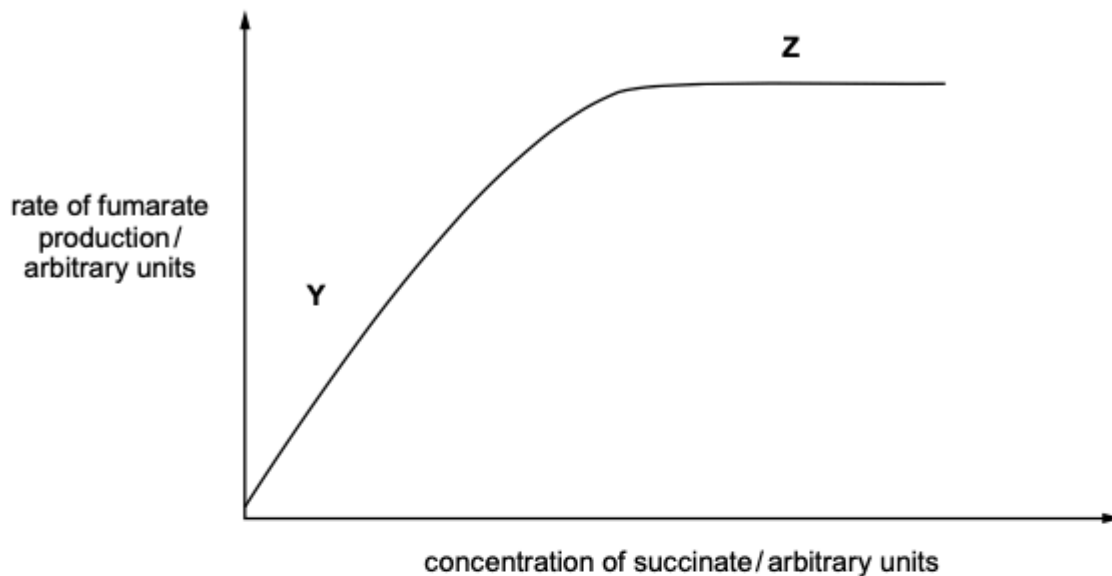
- (a) (i) What is represented by X on the graph above? [1]

- (ii) Enzymes are biological catalysts.
Draw a line on the graph above to show the energy changes that would take place if an enzyme was present during the reaction.

[1]

- (b) Succinate dehydrogenase is an enzyme found in mitochondria and is involved in respiration. The enzyme catalyses the conversion of succinate into fumarate. Using your knowledge of enzyme structure, explain why this is the **only** reaction succinate dehydrogenase can catalyse. [2]

- (c) The graph below shows the rate of fumarate production at varying concentrations of succinate, at optimum temperature and pH with no inhibitors present.

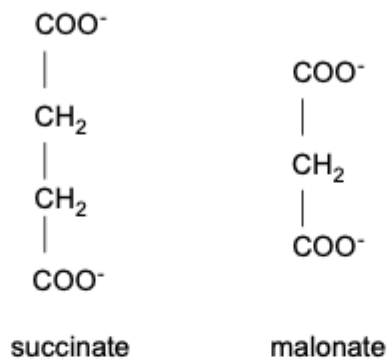


- (i) I State what factor is limiting the rate of reaction in the region marked **Y** on the graph. [1]

- II Use evidence from the graph to support your answer. [1]

- (ii) Explain what is limiting the rate of reaction in the region marked **Z** on the graph. [2]

- (d) Malonate is a competitive inhibitor of succinate dehydrogenase. The diagrams below show the structural formulae of succinate and malonate.



- (i) Using the information in the diagram above and your own knowledge, explain how malonate inhibits succinate dehydrogenase. [3]

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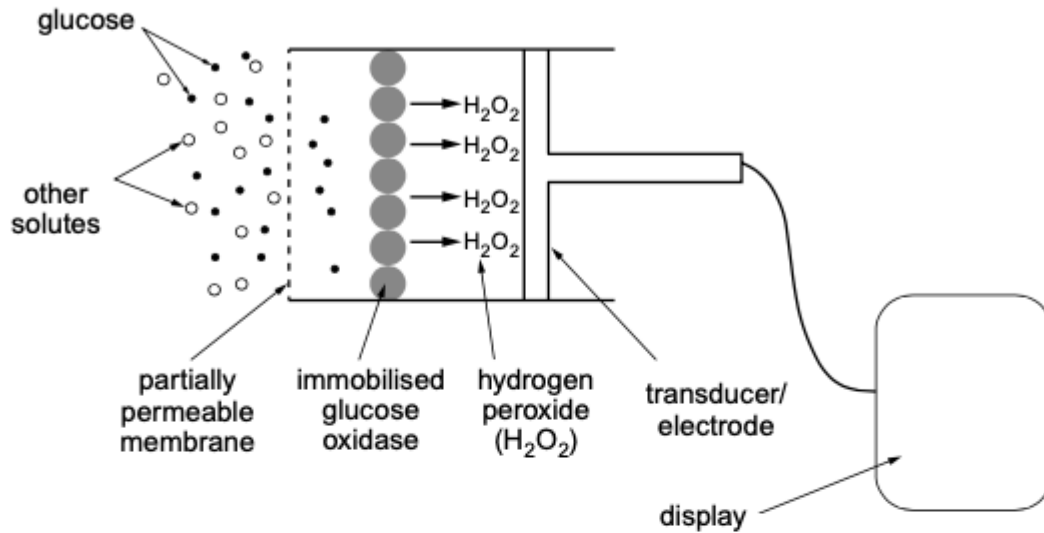
- (ii) On the graph in part (c) opposite draw a curve to show the rate of reaction when malonate is present. [1]

4	(a)	(i) Activation energy;	1
		(ii) Line starting and finishing at the same point but with a lower activation energy;	1
	(b)	The <u>active site</u> (of succinate dehydrogenase) has a <u>specific shape</u> ; Succinate has a <u>complementary</u> shape; (and therefore) {fits/ binds/ bonds to} into the active site; NOT attaches	Max 2
	(c)	(i) I The concentration of succinate/ substrate;	1
		II As the concentration of the {succinate/substrate} increases {the rate of reaction/production of fumarate increases};	1
		(ii) The concentration of succinate dehydrogenase/ enzyme; all of its active sites are occupied (at any given moment);	2
	(d)	(i) Malonate has a similar {shape/structure} to {succinate/ substrate} / malonate has a complementary {shape/structure} the active site; NOT same shape Malonate {binds/ competes} to the active site; Prevents succinate binding / fewer enzyme-substrate complexes are formed; (MP3 must be in context of competitive inhibition)	3
		(ii) Curve rising at a lower rate and plateaus at the max rate at a higher concentration; Accept max rate may not be reached	1

Question 4 Total

[12]

6. Biosensors make use of immobilised enzymes to detect specific molecules in a mixture. The diagram below shows a possible structure of a biosensor used to monitor blood glucose concentration.



- (a) (i) Describe the function of the partially permeable membrane in this biosensor. [2]

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- (ii) With reference to the diagram above, describe how the concentration of glucose is transmitted to the display. [3]

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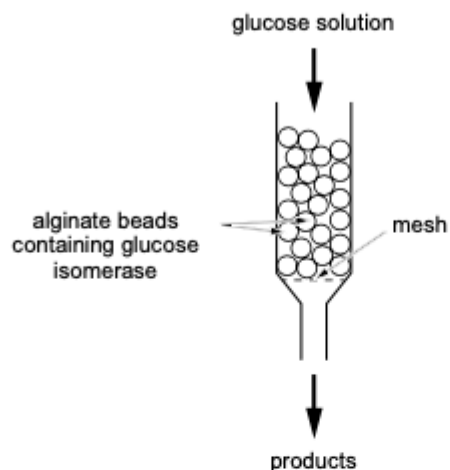
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Immobilised enzymes are also used in the food industry to produce many useful substances, for example fructose syrup. The diagram below shows a simplified version of this process. A glucose solution is passed through a column of the immobilised enzyme glucose isomerase and fructose is released as a product.



- (b) (i) Suggest why the enzyme involved is called glucose isomerase. [1]

- (ii) One of the advantages of using immobilised enzymes is that the product does not contain the enzyme and therefore does not need to be purified. Describe a biochemical test that could be used to show that the product has not been contaminated by the **enzyme**. [2]

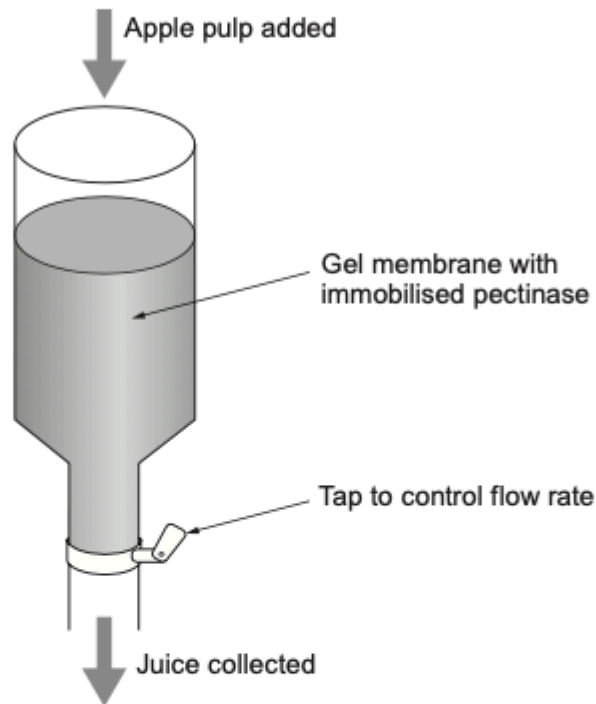
- (iii) Describe **two other** advantages of using immobilised enzymes. [2]

- | | | | | |
|---|-----|-------|---|-------|
| 6 | (a) | (i) | Allows the <u>glucose</u> molecules to pass through (to the enzyme layer);
Prevents the passage of other solutes ;
so they can't {affect results / affect enzyme / reduce enzyme activity}; | 2 max |
| | | (ii) | glucose broken down by <u>enzyme</u> ;
the {hydrogen peroxide/oxygen} is {detected/absorbed} by electrode;
an electric signal is generated/ changes chemical to electrical signal;
the greater the concentration of {glucose/hydrogen peroxide/oxygen}
the greater the signal; | 3 max |
| | (b) | (i) | The enzyme converts glucose into it's <u>isomer fructose</u> / glucose and
<u>fructose are isomers</u> ; | 1 |
| | | (ii) | Add Biuret solution / sodium hydroxide solution & copper sulphate;
(reject if reference to heat)
The solution would remain blue / no colour change would occur; | 2 |
| | | (iii) | can be re-used;
has greater stability/denature at higher temperatures;
can catalyse reactions/greater stability over a wider range of pH;
More than one enzyme can be used/enzymes added or removed
easily/ greater control over process/ can be used in a continuous
process;
(Reference to cost is neutral) | 2 max |

Question 6 Total

[10]

7. Pectin is a structural polysaccharide found in the cell walls of plant cells and in the middle lamella between cells, where it helps to bind cells together. Pectinases are enzymes that are routinely used in industry to increase the volume and clarity of fruit juice extracted from apples. The enzyme is immobilised onto the surface of a gel membrane which is then placed inside a column. Apple pulp is added at the top, and juice is collected at the bottom. The process is shown in the diagram below.



- (a) Immobilising enzymes can increase the temperature range over which they can be used. Describe **two other** advantages of immobilising pectinases. [2]

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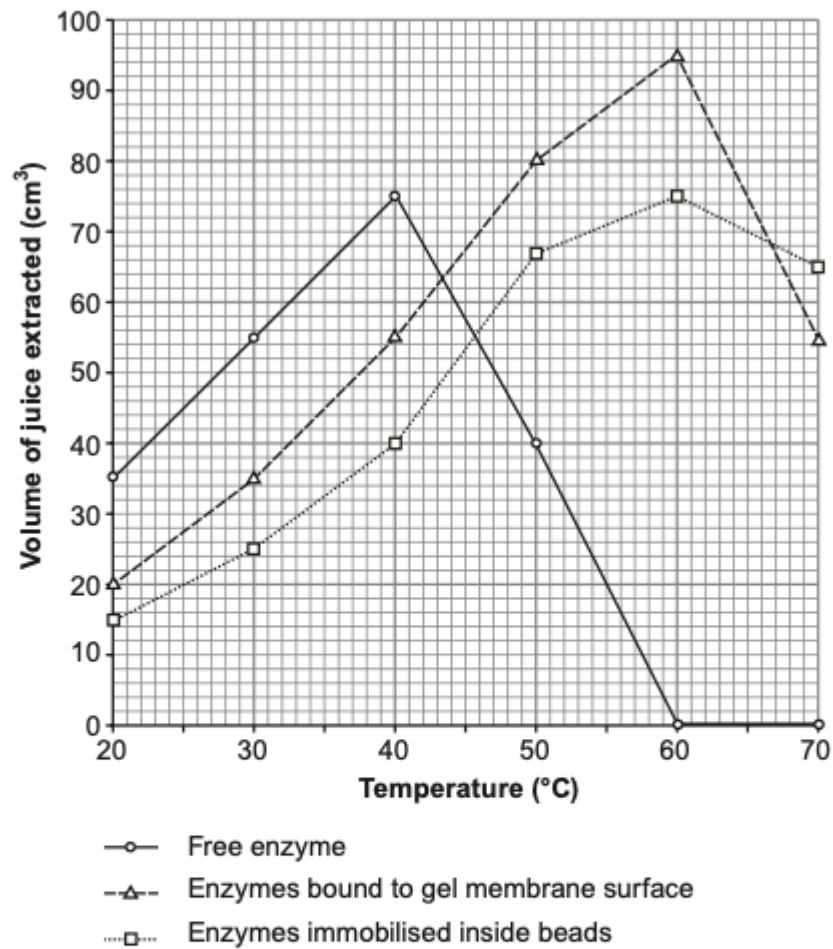
- (b) Suggest why reducing the flow rate of material through the column would result in an increased volume of juice being collected. [2]

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- (c) The extraction of juice using pectinase was compared using equal volumes and concentrations of free enzyme, enzymes bound to the surface of a gel membrane and enzymes immobilised inside alginate beads. The results are shown in the graph below.



Using the graph and your own knowledge of enzymes, answer the following questions.

- (i) Describe and explain the results for the free enzyme at temperatures above 40°C. [4]

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- (ii) Explain why a higher yield of juice was obtained when using free enzyme between temperatures of 20°C and 40°C than when using immobilised enzyme. [2]

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- (iii) Suggest a reason for the differences seen in the results for the enzymes bound to the gel membrane surface with those immobilised inside the beads, between temperatures of 20°C and 60°C. [2]

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7	(a)	Any two from <u>Product</u> not contaminated with enzyme; Enzyme can be re-used/ small quantity of enzyme required; Can {withstand/tolerate} a <u>wider</u> range of pH; Can be used in a continuous process;	Max 2
	(b)	Increases (contact) time between enzymes and substrate/ more time for pectinase to digest {apple pulp/pectin}; More <u>successful</u> collisions/more enzyme substrate complexes formed; NOT ESC	2
	(c) (i)	<u>40°C to 60°C</u> {decrease in/less} (volume of) juice extracted; NOT less juice extracted above 40 °C Above 60 °C no juice extracted; Between 40 °C and 60 °C enzymes are denaturing/ above 60°C they are denatured; <u>Hydrogen</u> bonds break; {Tertiary structure deformed / active site changes shape} {Substrate can no longer fit/ fewer enzyme substrate complexes formed};	Max 4
	(ii)	(Free enzymes) can move; Increased chance of <u>successful</u> collision / more enzyme substrate complexes formed;	2
	(iii)	(Increased juice extracted with membrane bound enzymes) because membrane bound enzymes are {more accessible/OWTTE} to substrate; (Enzymes immobilised inside bead) substrate has to {diffuse/pass} into bead;	2

WJEC 2015

6. A new biological washing powder has been developed containing three protease enzymes (**A**, **B** and **C**), each removing different protein based stains. The manufacturer claims their revolutionary new washing powder works "best at 35°C, but offers superior removal of tough protein stains at higher temperatures".

(a) Enzymes are biological catalysts. Describe what is meant by the term *biological catalyst*. [2]

Graph showing protease activity

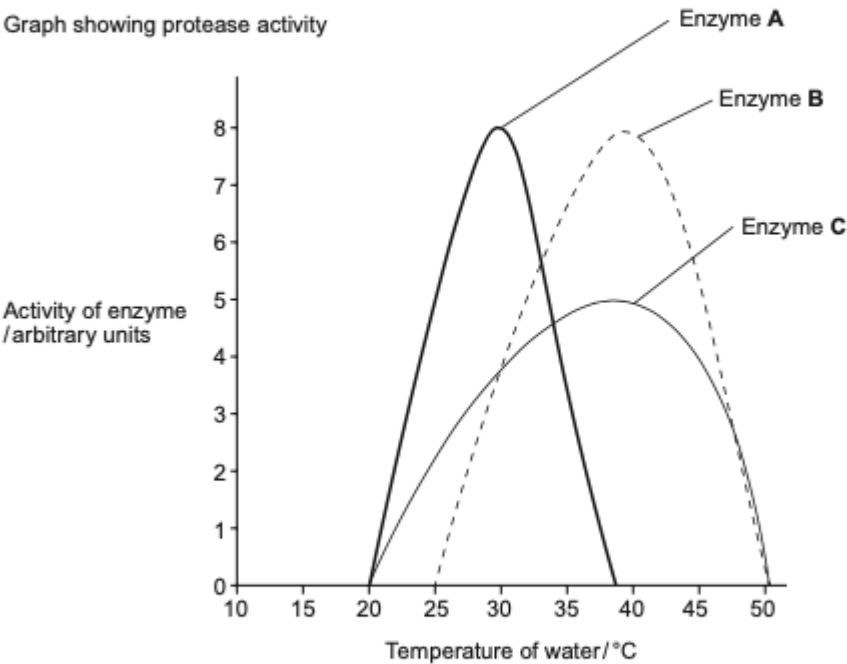


Table showing type of food stain removed

Protease enzyme	Food stain removed		
	Blood	Meat	Egg
A	Yes	No	No
B	No	Yes	No
C	No	No	Yes

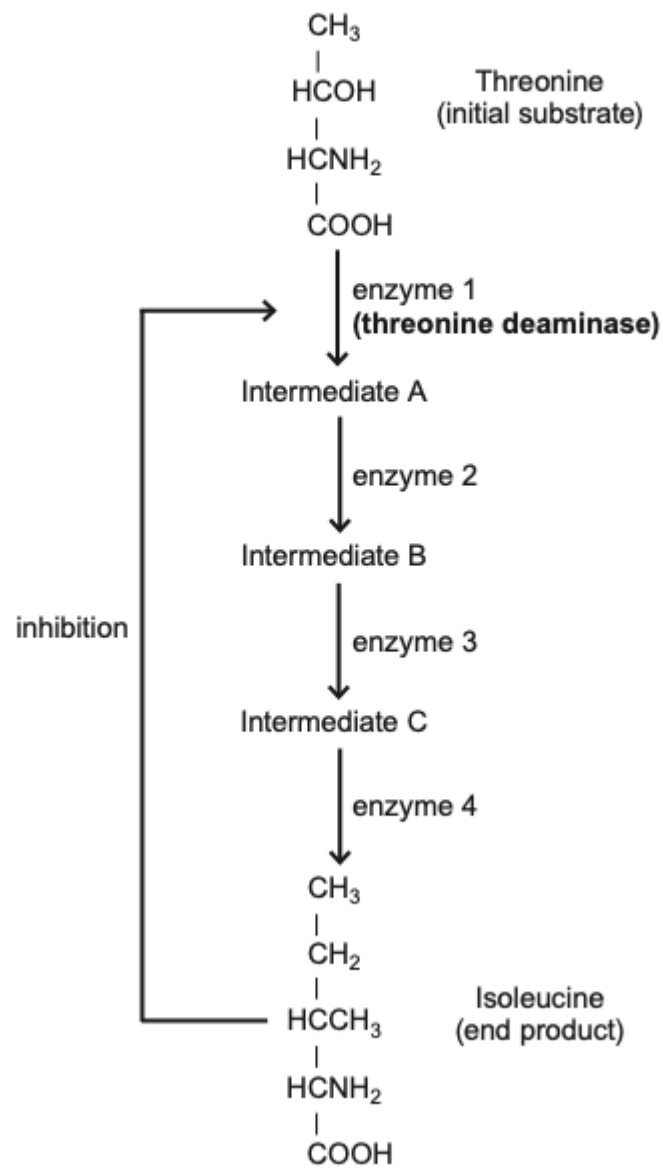
(i) Explain why the washing powder works best at 35°C. [1]

(ii) Which stain would not be removed when washing at 40°C? [1]

stains. [2]

stains when using this washing powder. [4]

- (c) Many metabolic pathways are switched off by their end products which act as competitive inhibitors of the first enzyme in the pathway. In the example below, the end product isoleucine inhibits threonine deaminase.



6	(a)	Produced by cells / is a protein; speed up (the rate of a) reaction (without being used or changed themselves) ; NOT activation energy	2
	(b)	(i) <u>all</u> (three) enzymes {have high activity / are working well};	1
		(ii) blood;	1
		(iii) 1. enzymes are <u>specific</u>; 2. each {stain / protein} has a different <u>shape</u> / different <u>shaped</u> substrates; NOT structure alone 3. would not fit one <u>active site</u> / three <u>different active sites</u> are needed;	Max 2
		(iv) 1. (at this temperature) <u>all</u> (three) enzymes {non-functional / denatured / no activity}; 2. due to breaking of <u>hydrogen</u> bonds; Allow H bonds REJECT listing of all bonds 3. active site deforms; 4. {prevents / no} { enzyme-substrate complexes forming / successful collisions}; REJECT: less enzyme-substrate complexes	4

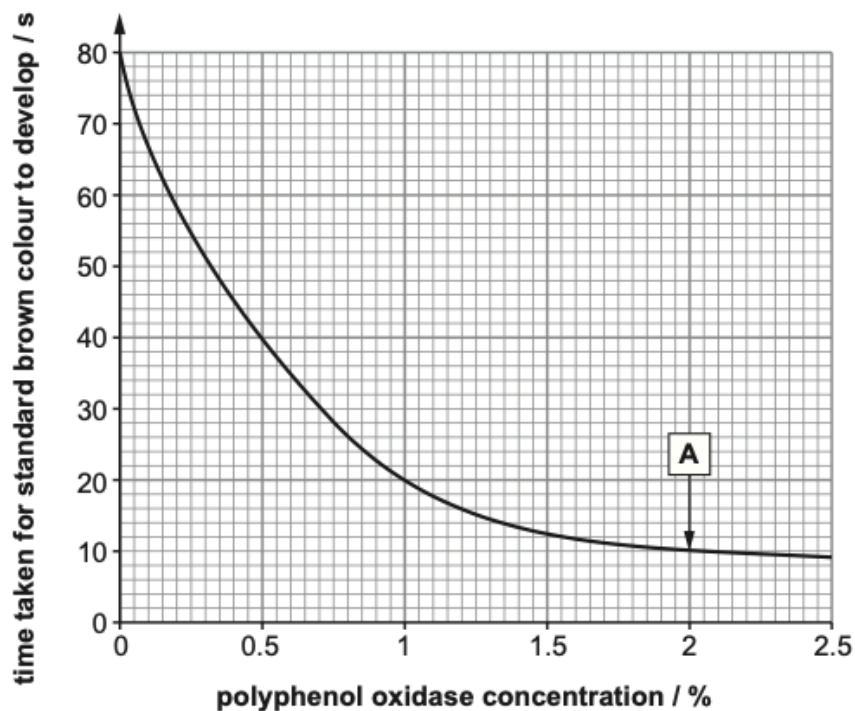
- (a) Explain how a biosensor may be used to measure blood glucose concentration. Outline the advantages of using a biosensor. [10]

7	(a)	Biosensor essay A. A biosensor can detect a <u>specific</u> molecule; B. in a mixture (of blood); C. ref to glucose oxidase; D. enzyme immobilised; E. on inert support / matrix; F. e.g. alginate beads or gel membrane; G. ref to selectively permeable membrane separating (blood) from sensor; H. When {a mixture / glucose} is passed over the enzyme {an enzyme substrate complex is formed/ correct reference to glucose oxidase reaction (glucose + oxygen = gluconic acid + hydrogen peroxide)}; I. {product / fall in oxygen} is proportional to the concentration of the {substrate / glucose}; J. detected by electrode / converted into electrical impulses by transducer; K. gives rapid result; L. gives {quantitative data / numerical value}; M. can detect at very low concentrations; N. accurate {concentration / measurement} is produced; O. for {diagnosis / monitoring} of diabetes; Question 7 total	[10]
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4. The enzyme, polyphenol oxidase, catalyses the production of dark brown coloured pigments from naturally occurring phenolic compounds (e.g. catechol). This causes fruit to turn brown when exposed to the air.

An experiment was carried out in which different concentrations of polyphenol oxidase were added to tubes containing catechol solution. The tubes were kept at a constant temperature and shaken periodically during the experiment.

The time taken for a standard brown colour to develop was recorded. The results are shown in the graph below.



- (a) State the relationship between the variables as shown in these results.

[2]

- (b) (i) Calculate the rate of the reaction at point **A** on the graph.

Rate is calculated as $\frac{1}{\text{time taken for brown colour to develop}}$ [1]

Rate = s⁻¹

- (ii) State what factor would be limiting the reaction at point **A**. [1]

.....

- (c) Banana puree is often given to young babies to eat, but it turns brown after preparation. Several methods can be used to prevent the puree from going brown. Explain how each of the methods given below prevent this from happening.

- I. Addition of lemon juice (citric acid) [2]

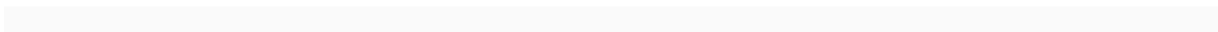
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- II. Freezing immediately after preparation [2]

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.....
.....

- III. Vacuum packing (air-free) [2]

.....
.....
.....



4	(a)		1) (Between 0 and 2% as polyphenol oxidase) concentration doubles the time taken (for the standard brown colour to develop) halves ; Accept: correct ref to data e.g.: 0.5% takes 40s, 1% takes 20s. Reject: rate 2) Above 1% time taken plateaus/ flattens out ;	2
	(b)	i	0.1 ;	1
		ii	Substrate/ catechol concentration ;	1
	(c)	I	Any 2 from: 1) enzyme {not at / below} optimum pH ; 2) enzymes {inactivated / denatured} / change to {shape of active site / charge} ; 3) {less/no} Enzyme-Substrate complexes formed/ or description ; NOT: ESC (abbreviation)	Max 2
		II	1) Low <u>kinetic</u> energy (or correct description) ; 2) {less/no/low} frequency of Enzyme - Substrate complex formation / fewer successful collisions;	2
		III	1) No oxygen present ; 2) Oxygen needed for {oxidase/ enzyme} activity / Oxidation (of phenolic compounds / banana puree) cannot take place ;	2
			Question 4 Total	[10]

- (b) Explain what is meant by an immobilised enzyme and with reference to suitable examples, discuss the advantages of their application in fields such as medicine and industry. [10]

(b)	(i)	Explain what is meant by an immobilised enzyme and with reference to suitable examples, discuss the advantages of their application in fields such as medicine and industry. A Enzyme molecules that are fixed / bound /trapped ; (not: immobilised/do not move) B alginate beads / gel membrane / meshwork of inert material / cellulose ; <i>The following are general; award once at any point throughout essay</i> C They are more stable at higher <u>temperatures</u> ; D They can tolerate <u>wider range</u> of pH ; E enzyme easily recovered for reuse ; F Product not contaminated by enzyme/no need for product to be separated from enzyme ; G {More than one enzyme type / differing optimal pH} can be used at a time ; H Reaction can be more easily controlled by adding or removing enzymes ; <i>Medical</i> I Biosensor (in medical diagnosis) / named condition e.g. diabetes ; J They are specific so can select one type of molecule in a mixture ; K rapid detection ; L quantitative result/ detected in low concentrations ; <i>Industrial/commercial</i> M Named industrial application e.g.: environmental monitoring, lactose free products, pectinase digestion etc. ; N reaction rates may be faster by using higher temps ; O {Cost effective / cheaper} because {more rapid production / continuous process}; Question 7 b Total	[10]
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1.5

WJEC 2009

3. (a) (i) Draw a simple **labelled** diagram to show how the three parts of a nucleotide are arranged. [3]

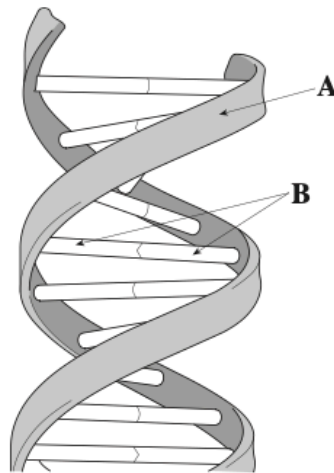
- (ii) How do nucleotides of RNA differ from nucleotides of DNA? [2]

.....

.....

.....

- (b) The DNA molecule is a double helix. It may be described as a coiled ladder.



- (i) What are the 'uprights' of the ladder, labelled **A**, made of? [1]

.....

- (ii) The 'rungs' are made by the pairing of components labelled **B**. Name the components in their complementary pairs. [2]

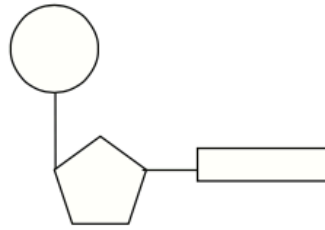
.....

.....

- (iii) Name the type of bonds that hold the pairs together. [1]

.....

3. (a) (i)



Pentose shown as pentagon and labelled, sugar/ribose/deoxyribose [1]
Phosphate on C5 and labelled, phosphate/phosphoric acid [1]
Base on C1 and labelled, (nitrogenous) base/named base [1]

(ii) The pentose is ribose in RNA deoxyribose in DNA; [1]
(allow: clear description of extra oxygen e.g. sugar in DNA contains
one less oxygen atom than sugar in RNA))
the base thymine is only found in DNA / uracil in RNA. [1]
(not: ref. helix/strands/uracil and thymine) Comparison needed

(b) (i) (Alternating) sugar / pentose or deoxyribose and phosphate. [1]

(ii) Adenine with thymine.
Cytosine with guanine. [2]
(not: abbreviations) Correct spelling thymine/cytosine.

(iii) Hydrogen. (not: H) [1]

[Total 9 marks]

Either, (a) (i) Describe the structure of DNA.
(ii) What are the differences between DNA and RNA?

8. (a) (i) A DNA is a polymer of many nucleotides / nucleotide chains / polynucleotide; [1]
- B The nucleotide contain the (5-carbon sugar / pentose sugar) deoxyribose; [1]
- C Attached to which is a base, either thymine, cytosine, adenine or guanine; [1]
- D The base is either a purine or a pyrimidine; [1]
- E T and C are pyrimidines, A and G are purines; [1]
- F The nucleotides are linked in a chain by alternate phosphate / sugar links/sugar phosphate backbone; [1]
- G One nucleotide can join to another by a condensation reaction; [1]
- H DNA consists of two of these chains twisted helically / double helix [1]
- I (Diagram showing) strands linked through correct A-T, G-C base pairings; [1]
- J Complimentary bases linked by hydrogen bonds. [1]

(Any 7 from A-J)

- (ii) K DNA contains deoxyribose sugar and RNA contains ribose; [1]
- L DNA contains thymine (base) and RNA contains uracil; [1]
- M DNA the bases are paired, in RNA they are unpaired; [1]
- N DNA is a double helix/stranded and RNA is single stranded; [1]
- O DNA is longer than RNA; [1]

(Any 3 comparative points from K-O)

[Total 10 marks]

WJEC 2010

1. Two types of nucleic acid, DNA and RNA are found in cells. Statements in the table, may apply to DNA, RNA or both. Complete the table by putting a tick (✓) if the statement is true or a cross (×) if the statement is not true. [6]

	<i>DNA</i>	<i>RNA</i>
Contains a pentose sugar		
Found in the nucleus		
Thymine is never present		
Consists of a double helix		
Molecules short lived		
Associated with ribosomes		

(Total 6 marks)

1.

	DNA	RNA
Contains a pentose sugar	✓	✓
Found in the nucleus	✓	✓
Thymine is never present	×	✓
Consists of a double helix	✓	×
Molecules short lived	×	✓
Associated with ribosomes	×	✓

[Total 6 marks]

(b) Describe the similarities and differences in the structure of amino acids and nucleotides. [10]

- (b)
- A Both contain the elements CHON;
 - B Both can link to form larger molecules/polymers/ref. monomers;
 - C Nucleotides consist of nitrogenous base;
 - D plus pentose and phosphate; (not: 5C sugar)
 - E bases are pyrimidines and purines;
 - F Amino acids possess an amine/ NH_2 group/carboxylic group;
 - G Variable R group;
 - H More/20 types of amino acid;
 - I Amino acids link together by peptide bond formation/sugar phosphate backbone;
 - J Five different bases in nucleotides/5 named; (not: letters only)
 - K Bases can undergo complementary base pairing;
 - L Adenine with thymine or uracil and guanine with cytosine;
 - M By hydrogen bonds;
 - N Nucleotides carry genetic information;
 - O Sulphur containing vs. phosphate containing.

WJEC 2011

7. (a) Complete the table below which compares DNA with messenger RNA (mRNA). [4]

<i>Feature</i>	<i>DNA</i>	<i>mRNA</i>
Name of sugar		
Number of carbon atoms in sugar		
Number of polynucleotide chains in molecule		
Location in cell		

- (b) The table below shows the relative amounts of the four bases in DNA taken from three sources.

<i>Cellular source of DNA</i>	<i>Nitrogenous base (relative amounts)</i>			
	<i>Adenine</i>	<i>Guanine</i>	<i>Cytosine</i>	<i>Thymine</i>
rat muscle	28.6	21.4	21.5	28.4
wheat seed	27.3	22.7	22.9	27.1
yeast	31.3	18.7	17.1	32.9

- (i) Explain why the relative amount of adenine is almost the same as the relative amount of thymine in **each** source. [3]

.....

.....

.....

- (ii) Explain why the base sequence of the DNA samples taken from a rat's bone marrow would be the same as those taken from the muscle of the **same** rat. [3]

.....

.....

.....

- (iii) Explain how a sample of DNA from a rat sperm cell differs from that of a muscle cell from the **same** rat. [3]

.....

.....

.....

(Total 13 marks)

Q.7 (a)

	DNA	m-RNA
Name of sugar	deoxyribose	ribose;
Number of carbon atoms in sugar	five	five;
Number of polynucleotide chains in molecule	two	one;
Location in cell	nucleus	nucleus + cytoplasm; (allow: RER/ ribosomes)

[4]

- (b) (i) base pairing;
Complementary/ adenine with thymine;
not identical because of experimental error; **[3]**
- (ii) passed on from parents/during fertilization/inherited/zygote formation;
from same cell/mitosis;
DNA replication;
genetically identical / same base sequence/ all body cells have same DNA **[3]**
- (iii) half as much; DNA (not: ref chromosomes)
variation/ genetically different
produced by meiosis; **[3]**

(Total 13 Marks)

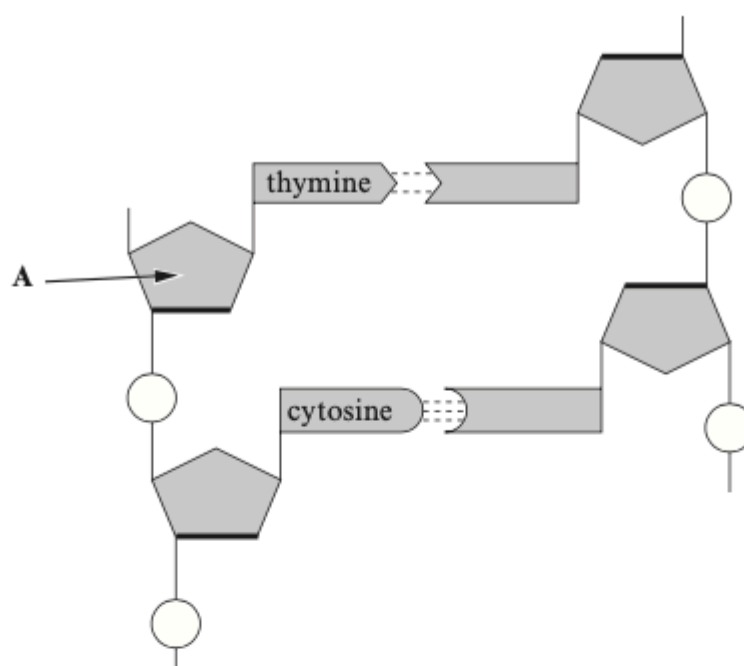
(b) Describe the structure of DNA.

- (b)
- A polynucleotide/chain of nucleotides;
 - B nucleotide consists of phosphate, sugar plus base;
 - C sugar is deoxyribose;
 - D base contains nitrogen; (allow: ref. nitrogenous)
 - E four bases are adenine, guanine, cytosine and thymine; (not: letters/ref. uracil)
 - F sugar phosphate backbone;
 - G two polynucleotide chains linked;
 - H antiparallel (stated, in context);
 - I ref. polynucleotide chains/base pairs held together by hydrogen bonds;
 - J complementary base pairing;
 - K A-T, G-C;
 - L pairing of purines and pyrimidines;
 - M double helix;
 - N purines double ring and pyrimidines are single ring;
 - O ref. sequence of bases is genetic code;

10

WJEC 2012

5. The diagram represents the molecular structure of part of a DNA molecule.



- (a) Name part A. [1]

.....

- (b) Part of a DNA molecule has the following sequence of bases.

T-A-T-C-G

- (i) In the table below write the letters for the sequence of bases of the complementary portion of DNA. [1]

DNA molecule	T	A	T	C	G
complementary DNA					

- (ii) Biochemical analysis of a sample of DNA showed that 30% of the bases were guanine.
Calculate the percentage of the bases in the sample which would be adenine.
Show your working. [2]

Answer

(Total 4 marks)

5. (a) deoxyribose/ pentose/ 5 C sugar 1

(b) (i) A-T-A-G-C 1

(ii) Guanine pairs with cytosine/ G pairs with C = 60%/ G+ C = 2

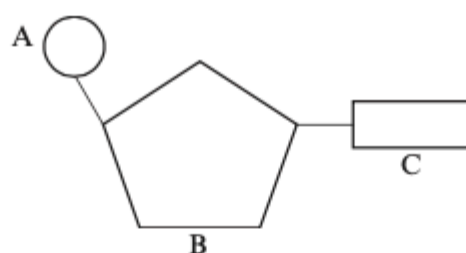
60%;

A+T= 40%

A= 20% (any two)

Correct answer = 2 marks

3. (a) The diagram below shows a unit which makes up nucleic acids.



- (i) Name the structural unit shown. [1]

.....

- (ii) Name component A. [1]

.....

- (iii) Name component B in DNA and RNA. [1]

DNA

RNA

- (iv) Name the **four** components found in DNA, represented by C. [2]

.....
.....

- (b) Describe how the structural units, drawn in part (a), are arranged in DNA molecules. [4]

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- (c) Describe the function of DNA molecules in cells. [1]

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.....

(Total 10 marks)

3. (a) (i) Nucleotide; 1
- (ii) Phosphate / phosphoric acid / PO_4 / PO_3^- ; 1
NOT phosphorus / P
- (iii) Deoxyribose in DNA **and** ribose in RNA (both); 1
- (iv) Adenine, Thymine, Cytosine, Guanine (1 if 1 error). 2
- (b) **Any 4** 4
- Pairing described A-T and C-G (both needed);
- Backbone / Chains / polynucleotide formed by alternating sugar phosphate groups;
- two chains connected / joined by base pairs;
- hydrogen bonding;
- two chains (twisted) to form a helix / double helix;
- NOT alpha helix.
- Accept labelled diagram.
- (c) {forming template / code / instructions} for {protein synthesis / mRNA 1
/ amino acid sequence / primary structure of protein / transcription}
(accept Replication in dividing cells) /
NOT genetic material alone.

Question total 10

WJEC 2013

2. There are 2 types of nucleic acid: DNA and RNA.

- (a) Complete the table below to describe **three** differences between the structure of DNA and RNA. [3]

DNA	RNA

- (b) A sample of DNA was analysed, 23% of the nucleotides contained guanine. Calculate the percentage of nucleotides which contained adenine. Show your working. [2]

.....

.....

.....

(Total 5 marks)

2. (a)

DNA	RNA
Double <u>stranded</u>	Single <u>stranded</u>
helical	Not helical
Deoxyribose/ $C_5H_{10}O_4$ / one less oxygen atom in pentose NOT deoxyribonucleic acid	Ribose/ $C_5H_{10}O_5$ / one more oxygen atom in pentose NOT ribonucleic acid
Contains thymine Not letters Can list all bases present	Contains uracil Not letters Can list all bases present
Only one type	3 types (mRNA, tRNA & rRNA)
(Relatively) long/ larger molecule	(relatively) short/ smaller molecule

Max 3

(b)

23% guanine therefore 23% cytosine;
(54% made up of adenine and thymine)
Adenine = 27(%)
Correct answer = 2 marks

2

Question 2 total

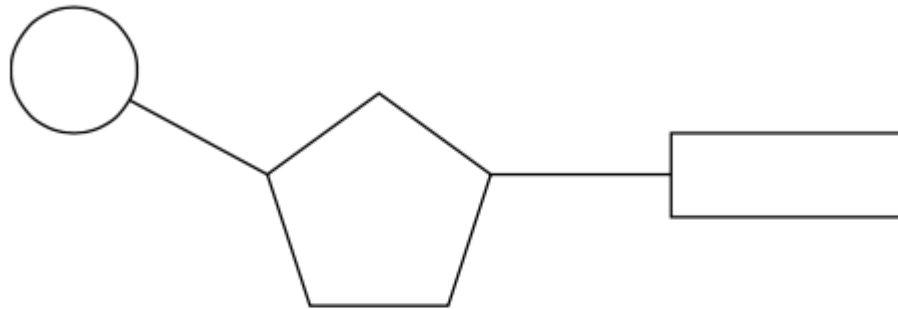
[5]

(b) Give an account of the structure and function of nucleic acids.

[10]

(b)	A. Ref to DNA and RNA; B. Diagram/description of a nucleotide with correct labels/terms (phosphate & pentose sugar & nitrogenous/eq base); C. DNA named sugar Deoxyribose; must link to DNA D. Ref to purines and pyrimidines; E. Correct identification of purines and pyrimidines (Full names only); F. Ref to Uracil replacing thymine in RNA; G. Correct base pairing A-T, C-G (<i>Allow letters;allow from diagram</i>) H. Description/labelled diagram of <u>double</u> helix in DNA; I. Held together by H – bonding; J. Functions of DNA (i) replication in dividing cells; K. (ii) code/ template for protein synthesis; L. Description of RNA as a single chain/ strand (of nucleotides); NOT single helix M. Ref correct sugar Ribose in RNA; correctly linked N. mRNA carries genetic code from the nucleus to the ribosome; O. correct reference to tRNA/ribosomal RNA; Question 8 Total	[10]
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1. The diagram below shows a simple nucleotide.



(a) On the diagram above, draw a circle around the component that contains nitrogen. [1]

(b) Describe **two** differences between a DNA nucleotide and an RNA nucleotide. [2]

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.....

.....

An experiment was carried out to determine the relative percentages of the bases in DNA from various organisms. The results are shown in the table below.

Source of DNA	Relative percentage of base in sample			
	Adenine	Guanine	Thymine	Cytosine
human	30.9	19.9	29.4	19.8
sea urchin	32.8	17.7	32.1	17.3
wheat	27.3	22.7	27.1	22.8

- (c) DNA is a double stranded molecule. Explain how the data in the table supports the concept of complementary base pairing. [2]

.....

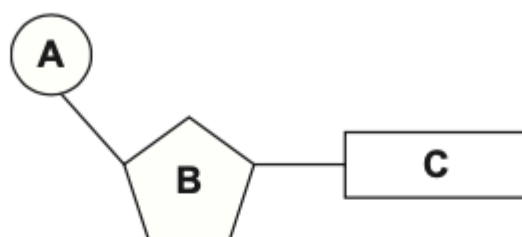
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- | | | |
|-------------------------|--|----------------------------|
| 1 | <p>(a) Base clearly circled;</p> <p>(b) (The pentose in) RNA is ribose <u>and</u> in DNA is deoxyribose;
the base thymine is only found in DNA <u>and</u> the base uracil is found in RNA;
NOT: ref. helix/strands/uracil and thymine unqualified</p> <p>(c) Adenine with thymine <u>and</u> cytosine with guanine;
Appropriate use of {data/ratios} for {human/sea urchin/wheat};
Need data on both A T and C G
NOT 'they are the same' or reference to ratio the same in all organisms</p> | <p>1</p> <p>2</p> <p>2</p> |
| Question 1 total | | [5] |

4. The diagram below shows a component of **DNA**.



(a) Name the parts **A**, **B** and **C**. [3]

A

B

C

(b) Describe how a polymer of DNA would be different from a polymer of RNA. [2]

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(c) (i) Name the stage in the cell cycle where DNA replication occurs. [1]

.....

(ii) Vincristine is a drug which prevents the spindle fibres from shortening. Name the stage in the cell cycle which would be affected. [1]

.....

(iii) State **three** differences between daughter cells produced by the process of mitosis and those produced by meiosis. [3]

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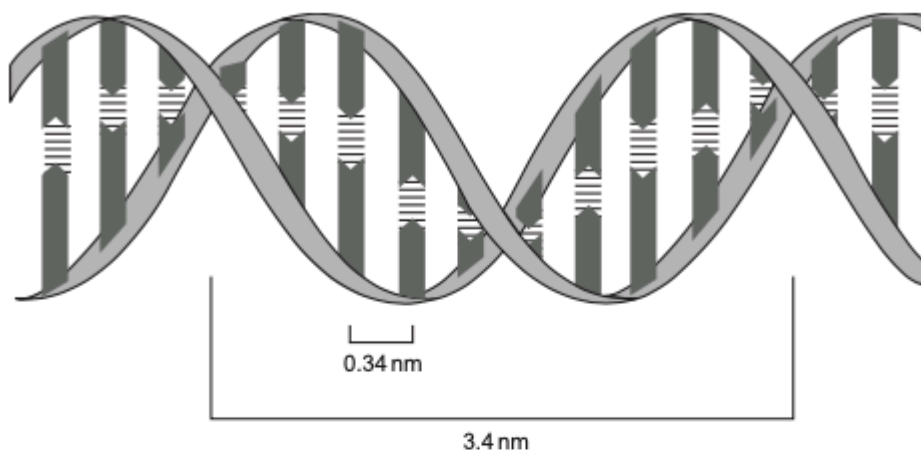
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- 4 (a) A – Phosphate; 3
Accept phosphoric acid
B – Deoxyribose;
NOT pentose
C – {Organic/nitrogenous} base;
NOT named base (can be neutral)
- (b) Uracil in RNA thymine in DNA; NOT U in RNA and T in DNA Max 2
RNA is (usually) single stranded, DNA is double stranded;
DNA is longer molecule than RNA ;
Sugar is ribose in RNA, deoxyribose in DNA;
- (c) (i) Interphase; 1
- (ii) Anaphase; 1
- (iii)
- | Meiosis | Mitosis |
|--|--|
| <u>4</u> cells | <u>2</u> cells produced; |
| Haploid/ half the number of chromosomes of the parent cell | Diploid/ same number of chromosomes as the parent cell |
| <u>Genetically</u> different; | <u>genetically</u> identical;
Accept clone |
- 3

Question 4 Total [10]

WJEC 2015

2. The diagram below shows the DNA structure proposed by Watson and Crick in 1953.



- (a) If the double helix takes 3.4 nm to make one complete turn and base pairs are 0.34 nm apart, how many base pairs would you expect to find in five complete turns? Show your working. [2]

Answer =

- (b) Give a reason why the diagram **shown** above must be DNA rather than RNA. [1]

.....

.....

.....

.....

- (c) The table below shows the bases guanine and cytosine as percentages of the total nucleotides present in three different micro-organisms.

Micro-organism	Base composition (%)	
	G	C
yeast	18.7	17.1
bacteria	36.0	35.7
virus	42.0	13.9

- (i) Suggest which type of nucleic acid is present in the virus shown in the table. Explain your answer. [2]

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- (ii) State the type of base to which cytosine belongs. [1]

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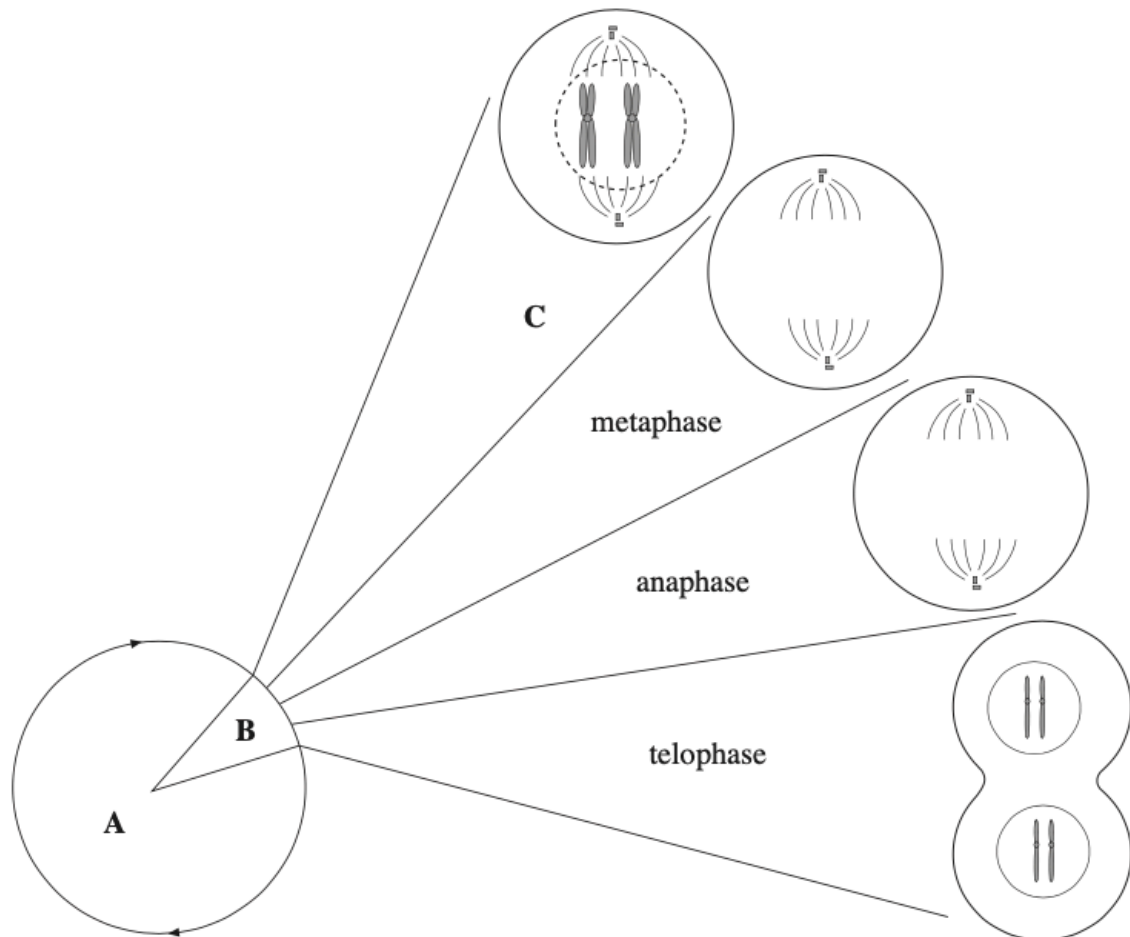
2	(a)	$(3.4/0.34) \times 5$ or $3.4 \times 5/0.34$ $= 50$ Correct Answer = 2 marks correct calculation, incorrect answer = 1 mark Wrong unit = 1 mark only	2
	(b)	Any one from: double {stranded / polynucleotide chains} / not single stranded; helical; NOT alpha helix antiparallel strands;	max1
	(c) (i)	<u>RNA</u> ; NOT mRNA, tRNA, rRNA The percentages of G and C are not equal and so there is <u>no</u> (complementary) <u>base pairing</u> ;	2
	(ii)	Pyrimidine;	1
		Question 2 total	[6]

WJEC 2016

1.6

WJEC 2009

2. The diagram shows the regular pattern of events that occur in dividing cells.



- (a) Name the stages in the cycle labelled **A**, **B** and **C**. [3]

A

B

C

- (b) Complete the drawings of the cells in metaphase and anaphase to show how the chromosomes are arranged. [2]

- (c) Describe **two** processes that take place during stage **A**. [2]

.....

.....

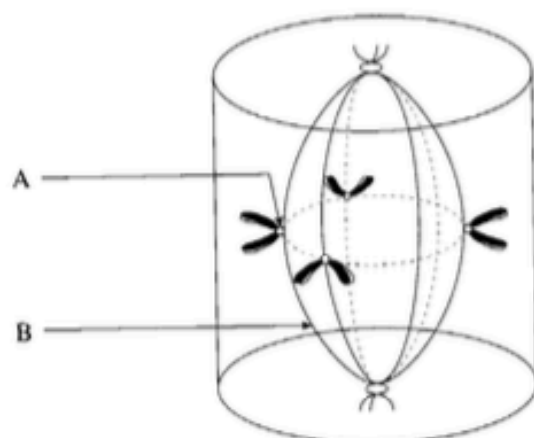
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(Total 7 marks)

2. (a) A – interphase (not: resting phase).
B – mitosis (not: cell division).
C – prophase. [3]
- (b) (i) Metaphase - both chromosomes on equator, chromatids either side. [1]
Anaphase – 4 V or U shaped, centromere pointing towards centrioles. [1]
- (ii) Nucleotide synthesis, replication of DNA, replication of organelles, protein synthesis, growth, synthesis ATP.AVP. (Any 2) [2]

[Total 7 marks]

7. The diagram shows a three-dimensional view of one of the stages of mitosis in a typical animal cell.



- (a) (i) Name the stage shown. [1]

- (ii) Name the structure labelled A. [1]

- (iii) State the function of structure B. [1]

- (b) Dividing cells undergo a regular pattern of events, known as the cell cycle. The following statements describe some of the main events taking place in animal cells. Insert the name of the appropriate stage in the box opposite each statement. [3]

Statement	Stage in cell cycle
Chromosomes shorten and thicken and spindle forms	
A period of intense activity which includes the replication of DNA	
Formation of two nuclei	

- (c) Complete the table to show **three** differences between meiosis and mitosis. [3]

Meiosis	Mitosis

(Total 9 marks)

7. (a) (i) metaphase [1]
- (ii) centromere [1]
- (iii) pulls chromatids/chromosomes to opposite poles [1]

(b)

Statement	Stage in cell cycle
Chromosomes shorten and thicken and spindle forms	Prophase
A period of intense activity which includes the replication of DNA	interphase
Formation of two nuclei	Telophase (not: cytokinesis)

[3]

(c)

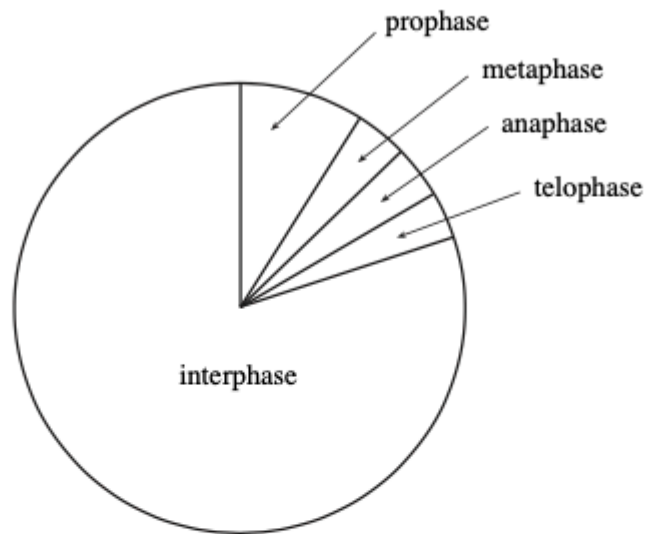
Meiosis	Mitosis
1 two divisions	One division
2 four daughter cells	Two daughter cells
3 number of chromosomes is halved	Number of chromosomes remains the same
Daughter cells <u>genetically</u> different/variation	Daughter cells genetically identical/no variation
Crossing over	No crossing over
(Homologous chromosomes pair)	(Homologous chromosomes do not associate in pairs)

(any three, last 2 points not expected on new spec.) Matched statements required

[3]

WJEC 2010

4. The drawing shows a simplified diagram of the cell cycle. The size of each segment indicates the relative length of each phase.



- (a) (i) Draw an arrow on the diagram to show the correct sequence of events in the cycle. [1]
- (ii) On the diagram add a segment to show the point at which cell division (cytokinesis) occurs. [1]
- (b) List **four** events that occur during interphase. [4]

.....

.....

.....

.....

- (c) Name the stage of mitosis where each of the following occurs.
- | | |
|---|-----|
| (i) Chromatids line up at the equator. | [1] |
| | |
| (ii) Centromeres split. | [1] |
| | |
| (iii) Spindle fibres contract and shorten. | [1] |
| | |
| (iv) Chromosomes are first visible as a pair of chromatids. | [1] |
| | |
| (v) Nuclear membrane reforms. | [1] |
| | |
- (Total 11 marks)**

4. (a) (i) arrow drawn pointing clockwise; [1]
- (ii) segment drawn after telophase of roughly similar dimensions [1]
- (b) replication of DNA;
- increase in cell size;
- chromosomes exist as chromatids;
- replication of organelles/centrioles;
- synthesis of ribosomal material;
- synthesis of ATP;
- synthesis of protein [4 max]
- (c) (i) metaphase; [1]
- (ii) anaphase [1]
- (iii) anaphase; [1]
- (iv) prophase; [1]
- (v) telophase. [1]

[Total 11 marks]

(a) Describe the events that occur during one complete cell cycle including mitosis, in an animal cell. [10]

- (a)
- A Interphase, replication of DNA; (not: DNA doubles)
 - B Also replication of organelles;
 - C Synthesis of rRNA/proteins/ATP; (not: metabolic activity)
 - D Prophase chromosomes appear as two chromatids/ ref, condensation;
 - E Joined at centromere;
 - F Nuclear membrane disappears;
 - G Chromosomes line up at equator during metaphase;
 - H Spindle formation;
 - I Centromere divides at anaphase;
 - J Chromatids/chromosomes to opposite poles at anaphase;
 - K Contraction/shortening of spindle fibres;
 - L Nuclear membrane reforms during telophase;
 - M Cytokinesis/cell division occurs by furrowing of membrane/cleavage;
 - N Cytoplasm splits/divides;
 - O Centrioles replicate / move to poles.

Note: ref. to each event must take place in correct stage

10 MARKS

WJEC 2011

4. (a) Complete the table to show whether each role applies to mitosis, meiosis or both. If the role applies put a tick (✓) and a cross (✗) where it does not apply.

<i>Role</i>	<i>Mitosis</i>	<i>Meiosis</i>
Involved in growth		
Produces variation		
Produces haploid cells		
Occurs in plants		

[4]

- (b) In the space below draw a labelled diagram to show a single chromosome as it appears during prophase of mitosis. [2]

- (c) Describe and explain what happens to chromosomes during anaphase of mitosis. [3]

.....

.....

.....

.....

- (d) Name the cell organelle responsible for the production of the spindle fibres. [1]

.....

(Total 10 marks)

Q.4 (a)

[4]

Role	Mitosis	Meiosis
	✓	X
	X	✓
	X	✓
	✓	✓

(not: hybrid ticks)

(b) joined pair of chromatids;
chromatid labelled and centromere labelled;

[2]

(c) centromere splits;
chromatids pulled to (opposite) poles;
by shortening/ contraction of spindle fibres;

[3]

(d) centrioles;

[1]

(Total 10 Marks)

(a) (i) Describe the functions and importance of mitosis to living organisms using examples where appropriate. [6]

(ii) Describe how meiosis differs from mitosis and explain the significance of these differences. [4]

8. (a) (i) A daughter cells exact copies/genetically identical cells;
B same number of chromosomes as parents;
C genetic stability;
D important for growth;
E replacement of missing tissue/parts;
F repair to wounds/(damaged) tissue/cell replacement plus e.g. skin/hair/gut lining/blood cells;
G asexual reproduction plus e.g. bulbs tubers runners/used by bacteria/yeast;
H allows large numbers of offspring to be produced/ref. cloning/quick colonisation;
I ref. to cancer i.e. proliferation of cells;

6

- (ii) J haploid vs. diploid/mitosis maintains chromosome numbers, meiosis halves it; (not: just 23 vs 46)
K two divisions involved;
L chromosomes are different/crossing over occurs; (allow: ref. independent assortment)
M meiosis produces gametes; (allow: examples e.g. sperm and egg cells; not: sex cells))
N allows for variation;
O allows sexual reproduction to take place;

4

WJEC 2012

6. The photographs show chromosomes during the stages of mitosis.



H



I



J



K



L

- (a) (i) Place the stages into the correct sequence. The first box has been completed. [1]



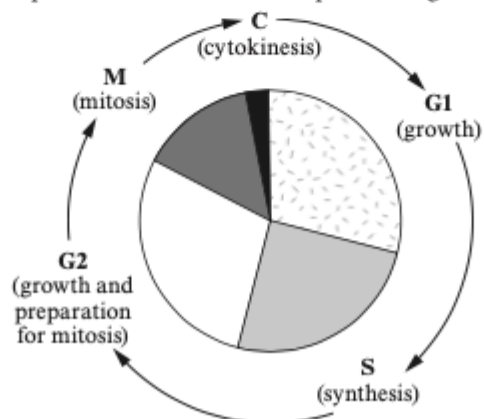
- (ii) Name stages:

[2]

I

L

- (b) The diagram represents the events that take place during the cell cycle.



The table below shows the DNA content of a cell measured during one cell cycle.

<i>Stage</i>	<i>DNA content of cell/arbitrary units</i>
G1	20
S	20 increasing to 40
G2	40
M	40
C	40 decreasing to 20

- (i) State the name of the period in the cell cycle that includes stages G1, S and G2.[1]

.....

.....

- (ii) State **two** events that occur during this period. [2]

.....

.....

- (c) Using information provided in the diagram and the table, explain why it is important that the DNA content of the cell increases during stage S and decreases during stage C. [2]

.....

.....

.....

- (d) Explain how mitosis maintains genetic stability. [2]

.....

.....

.....

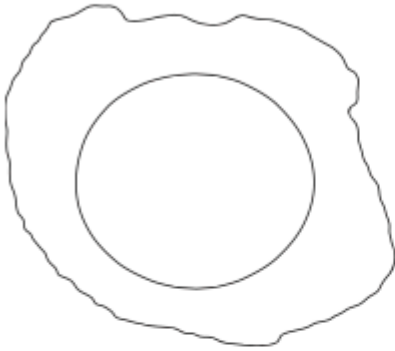
(Total 10 marks)

6. (a) (i)	J K L H I	1
(ii)	I = telophase	1
	L = metaphase	1
(b) (i)	interphase	1
(ii)	ATP production/ metabolically active; Replication of DNA; NOT synthesis/ doubling {Making/ replacing} new organelles/ replication of mitochondria/ chloroplasts NOT replication of organelles Protein synthesis; Cell increase in size (not growth) (any two)	2
(c)	DNA Doubled / DNA content increased from 20 to 40	1
	and then halved (to maintain DNA content) (in two daughter cells.) (ignore reference to chromosomes)	1
(d)	Two genetically identical daughter cells are produced; {Genetically identical/ clone} of parent cell.	2

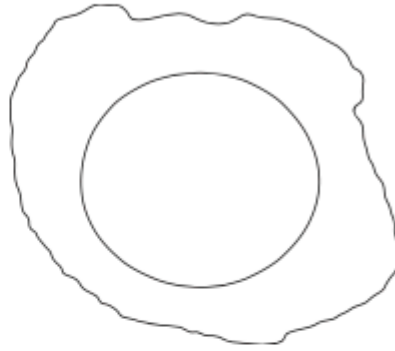
(Total 10 marks)

4. An Australian ant, *Myrmecia pilosula*, carries all its genetic information in a **single pair** of homologous chromosomes. Female worker ants are **diploid**, males are **haploid**.

- (a) Complete the drawings of the ant body (somatic) cells below, using drawings similar to structure **A** shown below to represent a single chromosome. [2]



Female worker ant body cell

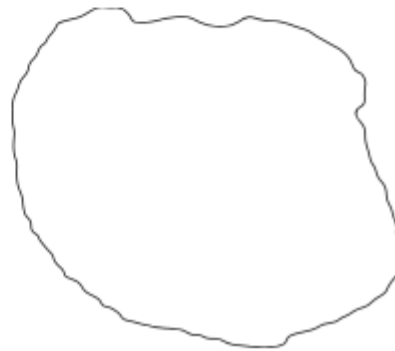
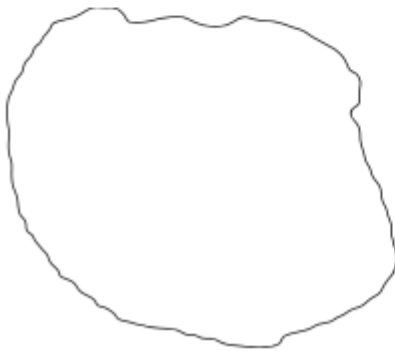


Male ant body cell

- (b) Using the cell outlines provided below, make **labelled** drawings to show the appearance of the **female** worker ant cells at the following stages. [4]

- (i) **metaphase** of mitosis,

- (ii) **anaphase** of mitosis.



- (iii) Adult ants emerge from pupae fully grown. Describe the purpose of mitosis in female worker ants. [2]

.....

.....

- (iv) Suggest the additional purpose of mitosis in fully grown male ants. [1]

.....

- (v) In ant colonies only some ants, called 'queens', produce egg cells and lay eggs. Name the type of cell division that the 'queens' use to produce haploid egg cells. [1]

.....

- (vi) What is the significance of the queens producing **haploid** egg cells? [1]

.....

.....

- (c) During a mating flight the queen collects sperm cells from male ants, which she stores in an organ called the spermatheca. The haploid egg cells from the queen's ovaries pass the spermatheca as they are laid. Some eggs are fertilised as they pass out, some eggs pass out unfertilised.

How will the ants that develop from fertilised egg cells differ from the ants that develop from unfertilised egg cells? [1]

.....

.....

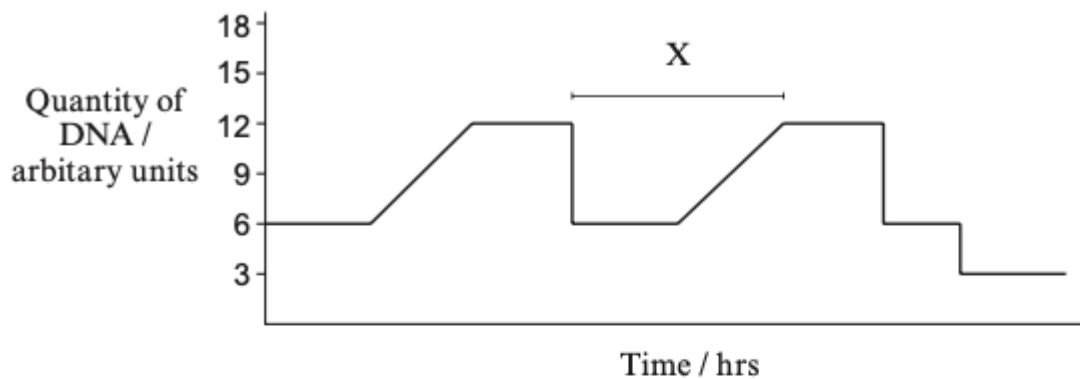
(Total 12 marks)

4. (a) 2 chromosomes in female cell; 2
1 chromosome in male cell;
Diagrams must match each other.
Accept 'chromatids' in each cell. Do not accept chromatid in male cell if chromosomes drawn in female cell or opposite.
- (b) (i) 2 Chromosomes arranged on equator of spindle; (ignore orientation) 2
2 V shaped {chromosomes / chromatids} with centrosomes towards each centriole/pole;
Ecf from one diagram to other.
- (ii) Labelling: chromatids, centromere, spindle, centrioles, equator, cell 2
membrane.
2 marks for 4 correct labels on either diagram;
1 mark for 3.
- (iii) To provide {genetically identical cells / clones}; 2
Repair / replacement {of cells / tissue} / regeneration qualified;
NOT growth.
- (iv) Making gametes / sperm cells / sex cells / produce haploid cells for 1
reproduction;
- (v) Meiosis / reduction division; 1
Spelling must be correct.
- (vi) Genetic variation (in the offspring) / restore diploid number (in 1
zygote) OWTTE;
- (c) Fertilised eggs will develop into females, unfertilised eggs into 1
males; (both for 1 mark);
Accept: fertilised will give genetically varied ants, unfertilised would give clones;
IGNORE haploid / diploid.

Question total 12

WJEC 2013

- (b) The graph below shows the relative quantity of DNA in an animal cell during two complete cell cycles.



- (i) Name the stage in the cell cycle which is represented by the time period labelled X on the graph above. [1]

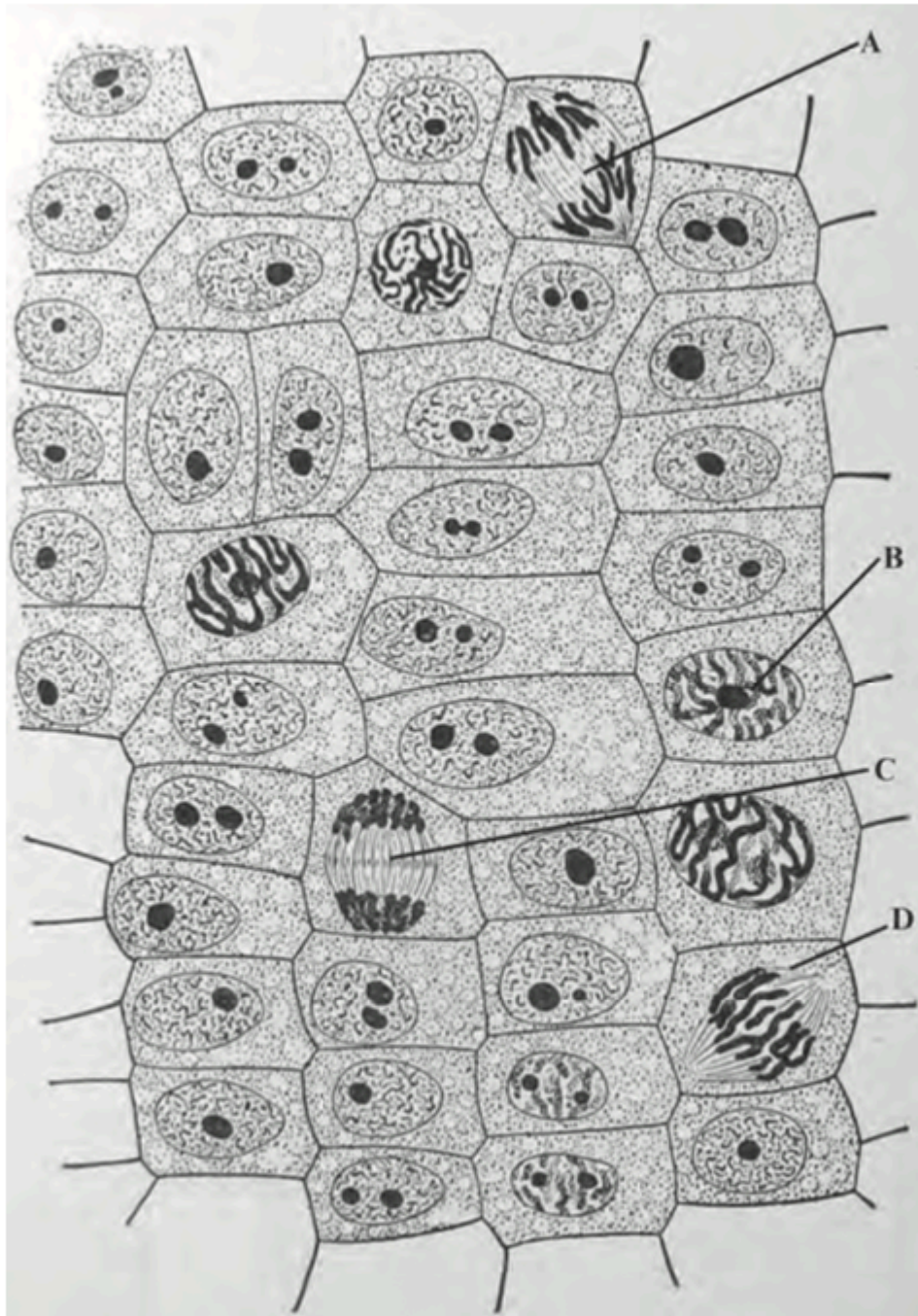
- (ii) Using evidence from the graph, give a reason for your answer in part (b)(i). [1]

- (iii) State the type of nuclear division that is shown in the second cell cycle and explain how the evidence in the graph led you to this conclusion. [2]

(Total 8 marks)

1.	(a)	(i) Stage A – telophase; Stage C – metaphase;	2
		(ii) Centromeres split/ divide; <u>Chromatids/ chromosomes</u> are being <u>pulled</u> to (opposite) poles; (due to) contraction/ shortening of the spindle (fibres);	2
	(b)	(i) Interphase;	1
		(ii) The (quantity of) DNA has <u>doubled</u> / (quantity of) DNA changes from 6 to 12; NOT increase	1
		(iii) Meiosis; (correct spelling) (At the end of the cell cycle) the (quantity) of DNA has been <u>halved</u> (and halved again) / can describe with numbers /involves 2 (consecutive) divisions; Ignore reference to chromosomes	2
Question 1 total			[8]

4. The drawing below is taken from plant tissue which shows cells undergoing mitosis.



(a) What plant tissue could be observed to produce this drawing?

[1]

(b) Identify from the diagram opposite the stages of mitosis labelled A to D. [4]

A

B

C

D

(c) One stage of the **cell cycle** shown on the diagram is present in greater numbers than the others. Name this stage and explain this observation. [2]

Stage:

Explanation

.....

(d) How would cells produced by meiosis differ from those produced by mitosis? [2]

.....

.....

.....

.....

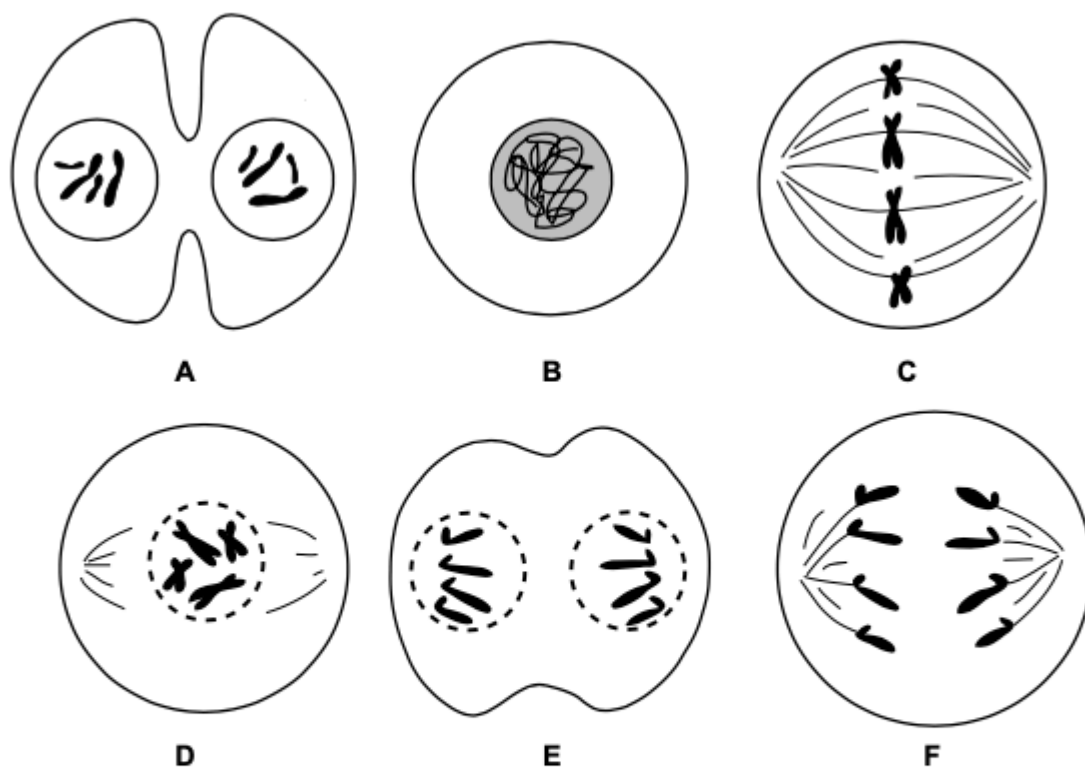
.....

(Total 9 marks)

4.	(a)	Root <u>tip</u> / shoot <u>tip</u> / meristem;	1
	(b)	A Anaphase; B Prophase; C Telophase; D Metaphase;	4
	(c)	Interphase; It is the longest phase;	2
	(d)	(All cells) would be {haploid/half the number of chromosomes}; NOT cells have fewer/ less chromosomes (All cells) would be genetically different;	2
Question 4 Total			[9]

WJEC 2014

2. The diagrams below show the different stages of the cell cycle in a body cell from an animal.



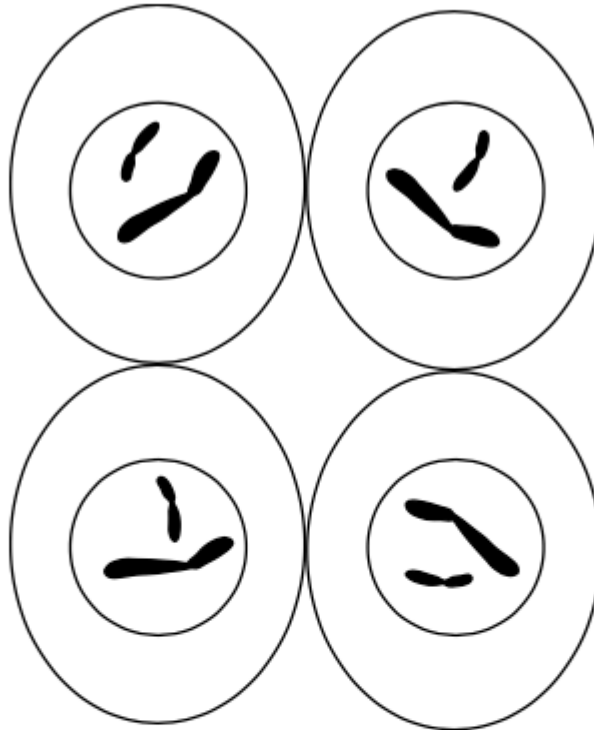
- (a) (i) Put the stages from the diagrams above in the correct sequence. The last stage has been done for you. [1]

					A
--	--	--	--	--	---

- (ii) Name the process represented in diagram A.

[1]

Cell division also occurs in the ovaries of animals. The diagram below shows the final stage of cell division in the ovary of the same animal.



- (b) (i) Using the diagrams above, describe and explain **one** difference between these cells and those produced in part (a) opposite. [2]

.....

.....

.....

.....

- (ii) Explain the importance of this type of cell division in the animal. [2]

.....

.....

.....

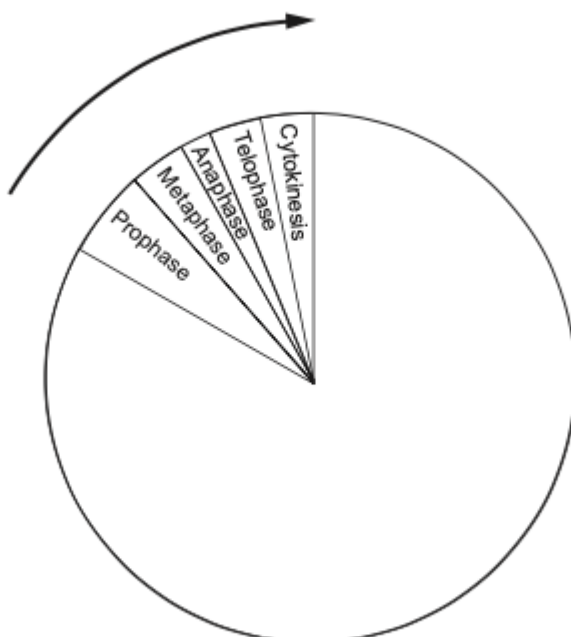
.....

2	(a)	(i)	B, D, C, F, E;	1
		(ii)	Cytokinesis;	1
	(b)	(i)	4 cells are produced compared with 2 / cells are haploid as oppose to diploid/only contain one set of chromosomes compared with two sets of chromosomes; NOT 2 chromosomes (can be neutral) As a result of two (consecutive) divisions;	2
		(ii)	(Meiosis produces haploid gametes which) allows the diploid state to be restored {at fertilisation/in the zygote} / prevents doubling of the chromosome number at fertilisation; Meiosis produces <u>genetically</u> different {gametes/cells} / results in <u>genetic</u> variation (in the offspring);	2
Question 2 Total				[6]

WJEC 2015

4. The diagram below shows the relative lengths of the cell cycle in actively dividing cells taken from the root tip of a garlic plant.

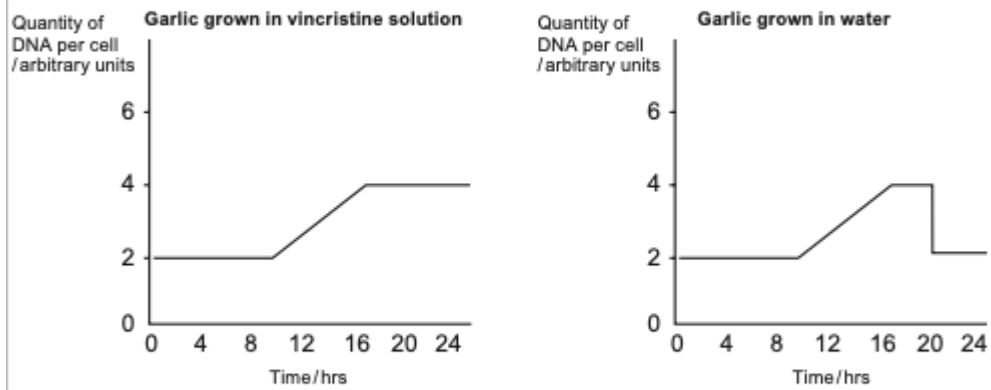
Mitosis



- (a) Describe the changes that occur to a plant cell during prophase. [4]
- (b) The percentage of cells in each stage of the cell cycle is proportional to the length of that stage. Using a microscope, a student observed 100 cells and found 5 undergoing prophase. If the total length of the cell cycle is 24 hours, calculate the length of prophase in minutes. Show your working. [2]

Answer = minutes

- (c) Vincristine is a chemical which inhibits mitosis by preventing the formation of the spindle fibres. Garlic bulbs were grown in a solution of vincristine and the quantity of DNA present in a cell from the root tip was measured over the 24 hour length of the cell cycle. The results are shown below together with the results from garlic bulbs grown in water.

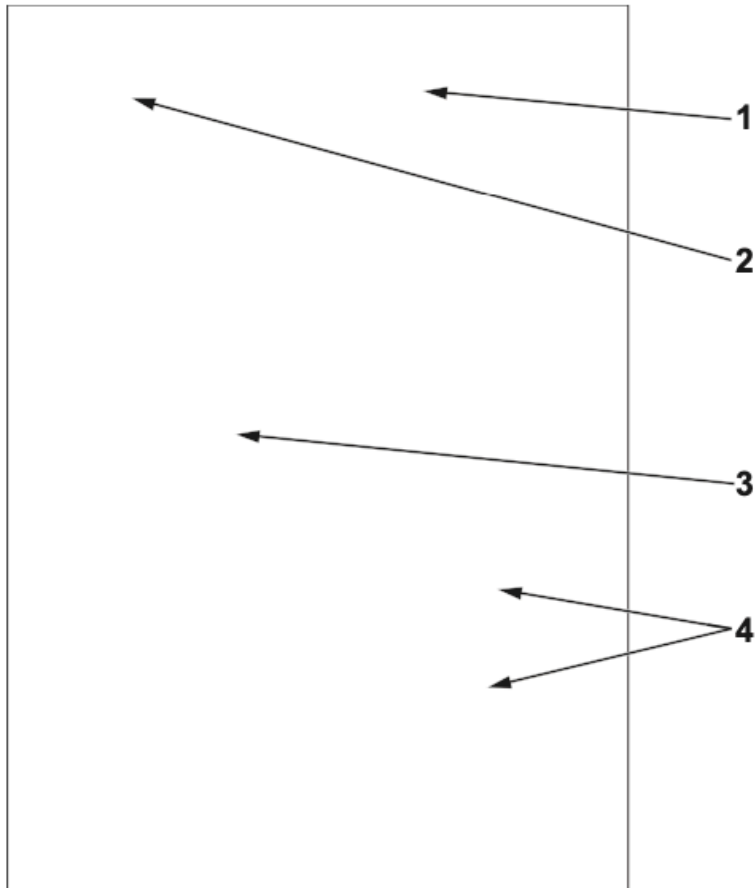


Using your knowledge of the cell cycle, explain how the results of this experiment show that vincristine inhibits mitosis. [3]

4	(a)	Any 4 from 1. {nuclear membrane / nucleolus} disappear; 2. <u>chromosomes</u> {shorten / thicken / contracts / condense} / chromatin condenses; 3. (appear as) {<u>two / pair / sister</u>} chromatids; NOT pairing up 4. chromatids joined at centromere; 5. spindle forms; Reject ref to centrioles making spindle fibres Accept labelled diagram for 3, 4 and 5	Max 4
	(b)	$5/100 \times (24 \times 60)$ or $(24 \times 60)/20 = 72$ Correct Answer = 2 marks correct calculation, incorrect answer = 1 mark	2
	(c)	Any 3 from 1. both show <u>interphase</u> because DNA {doubles / changes from 2 to 4 / replicates}; NOT DNA increases unqualified 2. (with vincristine) there is {no halving of DNA / DNA does not decrease from 4 to 2 arbitrary units}; 3. (lack of spindle fibres) prevents chromatids being pulled to the {poles / ends of spindle}(in anaphase); allow chromosomes 4. therefore no cytokinesis / no separation into two cells; 5. daughter cells not produced	Max 3
	(d)	meiosis produces <u>genetically</u> different cells AND mitosis produced {<u>genetically</u> identical cells / clones}; meiosis halves chromosome number AND mitosis maintains chromosome number / cells produced by meiosis are haploid AND those by mitosis are diploid ;	2
		Question 4 total	[11]

WJEC 2016

- (a) The photomicrograph below shows a root tip squash. With reference to the cells labelled **1-4**, describe and explain the sequence of events in mitosis. [10]



7	(a)	The photomicrograph below shows a root tip squash. With reference to the cells labelled 1- 4, describe and explain the sequence of events in mitosis. A Prophase and 3 ; B chromosomes appear as {two/sister/ a pair of } chromatids / (DNA/chromatin) condensation ; C Joined at centromere ; D Nuclear membrane disappears/ nucleolus disappears ; E spindle formation ; F metaphase and 4 ; G Chromosomes/ chromatids line up at equator ; H Chromosomes/ chromatids attach to spindle <u>by centromeres</u> ; I anaphase and 2 ; J Centromere divides; K Chromatids/chromosomes move to opposite poles; L Contraction/shortening of spindle fibres; M Telophase and 1 ; N Nuclear membrane/ nucleolus reforms; O chromosomes decondense ; Reject: chromatids Note: ref. to each event must take place in correct stage Question 7a total	[10]
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